Improving the displayed exponents in arXiv:2610.06783v1

A working derivation for review

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Abstract

We sharpen the binomial-tail analysis of the paper's box data structure and modify its encoding algorithm to omit unusable leaves. Under the same assumptions $N \geq D^{18}$ and $|W| \leq N^2/\sqrt{D}$, this gives deterministic time $O(N^2/D^{0.07195})$, improving the displayed 0.063 saving; with Majdoddin's five-term tensor the same regime gives $O(N^2/D^{0.0991})$ preprocessing and $O(D^{0.4009})$ queries. A separate reduction intersects independent modular filters before scanning candidate witness blocks, and folds the residue classes of each modulus into the outer dimensions of a single thin product per block, so that the band encodings are shared across residues. It gives a Las Vegas algorithm for integer all-edges Exact Triangle, min-plus product, and APSP in expected $O(n^{2.995943})$ time; batching the independent middle-block products into the tensor slots of one tile, with prefix-contracted queries, improves this to $O(n^{2.995535})$. A bucket-pair version of the same folding reduces integer 3SUM directly to one thin product per modulus and keeps the full saving, giving expected $O(n^{1.996001})$ time. A deterministic sequence of conditional modular moments gives deterministic integer 3SUM in $O(n^{1.996200})$; applied to each outer pair, it gives deterministic integer all-edges Exact Triangle, min-plus product, and APSP in $O(n^{2.996261})$. Combining this with batching, deterministic cyclic reconstruction of omitted encodings, and the SODA 2024 rectangular bounds improves the exponent to 2.9958892. A separate conditional statement, assuming the rectangular theorem of the unrefereed preprint arXiv:2610.10494v2, whose stated bound at $k = 1$ would give $\omega \leq 2.25$, would give 2.9958323; it is not counted among the bounds derived here. A separate construction using Majdoddin's five-term embedding and composite incidence certificates, with fixed-profile inner-basis encodings and balanced prefix queries, gives deterministic $O(n^{2.993676})$ time for integer all-edges Exact Triangle, min-plus product, and APSP; used as a black box in the real-input reduction, the same data structure gives real 3SUM and APSP expected times $O(n^{1.993805})$ and $O(n^{2.993805})$. Splitting each composite modulus between residue folding and a cyclic polynomial ring, and charging residual and stored-box queries separately, further improves this deterministic bound to $O(n^{2.993441})$. A partial cache of direct inner projections, partitioning each residual query by its first cached positions, improves it to $O(n^{2.993401})$. Changing the local identity, re-embedding the inputs with Walsh transforms, and caching the remaining residual terms as polynomials gives $O(n^{2.99294792})$ deterministic time. Deterministic middle translations then restrict the rare source symbols; grouped completion across middle blocks, asymmetric rectangular multiplication, and a separate jointly capped box encoder improve the bound to $O(n^{2.9926863})$. A shrinking-bias source construction and a small numerical refinement give $O(n^{2.9926862626})$; the original geometry/query bottleneck remains. A separate signed-window bit-refinement reduction gives deterministic exact all-edges witness counts, including min-plus multiplicities, in $O(n^{2.99294792})$ time with its earlier witness. Applied to fixed original 3SUM buckets with cyclic histogram products, it gives deterministic 3SUM and exact #3SUM in $O(n^{1.9931055})$ time. Rank-prefix hashing with complementary certification gives real 3SUM and APSP expected times $O(n^{1.996001})$ and $O(n^{2.996001})$ with the ten-term data structure. Rectangular products and isolated witness recovery sharpen convolution and weighted-clique bounds; asymmetric covers also give min-sum subset convolution in expected $O^*(2.997190^n)$ time. With square five-term tiles, integer min-plus convolution takes $O(n^{1.993927})$ time, both Las Vegas and deterministically, and min-sum subset convolution expected $O^*(2.995627^n)$ time. Read on lopsided instances, the deterministic product bound makes the source's reduction from directed APSP with polylogarithmic weights explicit: $O(n^{2.525754})$ against Zwick's $O(n^{2.5275})$, and $O(n^{2.5251975})$ with one further rectangular table row. Over $\mathbb{F}_2$ the

fixed-bucket construction gives deterministic 3XOR and exact #3XOR in $O(n^{1.9931055})$. We also derive deterministic and real-input corollaries, a wider thin-matrix regime, and the displayed bounds that the source's further reductions inherit. Exact rational interval certificates check every numerical inequality used in the stated bounds. These are mathematical derivations for independent review, not a claim of literature priority or proof-assistant certification.

1 Results and scope

The source is Alman–Vassilevska Williams, arXiv:2610.06783v1, by Josh Alman and Virginia Vassilevska Williams. We use its algebraic identity and batch embedding (Lemmas 6–9), its tiling, and its box construction (Lemmas 27–29). The changes proved here are the encoding restriction, sharper counting, modular filtering and residue folding, conditional modular moments, certified rank-prefix counts, isolated witness recovery, and rectangular product reductions. Except in the separately stated subset-convolution result, input integers and weights have polynomial magnitude, with the exponent treated as a fixed constant. The subset-convolution result explicitly charges arbitrary input bit length. The word RAM has $O(\log N)$ or $O(\log n)$-bit words. APSP assumes no negative cycles.

Problem/regime	Bound derived here	Guarantee
Wanted thin-product entries, $N \geq D^{18}$, $\lvert W\rvert \leq N^2/\sqrt{D}$	$O(N^2/D^{0.0991})$	Deterministic; query $O(D^{0.4009})$; the paper has 0.063 and 0.437
Wanted entries, $N \geq D^{14}$, $\lvert W\rvert \leq N^2/\sqrt{D}$	$O(N^2/D^{0.0853})$	Deterministic; query $O(D^{0.4147})$
Wanted entries, $D \leq N^{0.129}$, $\lvert W\rvert \leq N^2/\sqrt{D}$	$O(N^2/D^{0.0001})$	Deterministic; fixed query density
Integer 3SUM and exact #3SUM	$O(n^{1.9931055})$	Deterministic; fixed buckets and signed-window count refinement
Integer 3SUM	$O(n^{1.9931055})$	Las Vegas, inherited from the deterministic bound
Integer all-edges Exact Triangle, APSP/min-plus	$O(n^{2.9958892})$	Deterministic; per-pair moments, lazy encodings, SODA 2024 multiplication
Integer all-edges Exact Triangle, APSP/min-plus	$O(n^{2.9926862626})$	Deterministic; exact-half structured sources
Integer #Exact Triangle, min-plus witness counts	$O(n^{2.99294792})$	Deterministic exact counts
Integer APSP/min-plus	$O(n^{2.9926862626})$	Las Vegas, inherited from the deterministic bound
Real 3SUM	$O(n^{1.993805})$	Las Vegas expected time; certified comparison counts
Integer min-plus convolution	$O(n^{1.993927})$	Deterministic, hence also Las Vegas
Directed APSP, weights $(\log n)^{O(1)}$	$O(n^{2.525754})$	Deterministic; Zwick's $O(n^{2.5275})$; 2.5251975 with one extra table row
3XOR and exact #3XOR, $d = O(\log n)$	$O(n^{1.9931055})$	Deterministic
Min-sum subset convolution, β-bit tables	$O^*(2.995627^n)$	Las Vegas expected time; square five-term tiles
Real APSP/min-plus/Exact Triangle	$O(n^{2.993805})$	Las Vegas expected time

The integer randomized result uses a different guarantee from the paper's deterministic integer headline. The real-input bounds allow comparisons, additions, and subtractions on real numbers, as in the source reductions. These asymptotic bounds do not assert practical speedups. Section 7 lists the bounds inherited by weighted k-Clique, by the problems solved by one min-plus product or one APSP, by hinted OMv, by min-plus convolution and MonoConvolution, and by the counting versions.

2 Pruning the input encoding

Write $D = 4^m$, $L = \lceil cm \rceil$, $K = \binom{L}{m}$, $N_0 = 3^{L-m}$, and $M = KN_0^2 = \binom{L}{m}9^{L-m}$. The recursion has ten child types: nine P_{ij} and one P_0. Each left or right input array at depth k has 7^{L-k} entries. An output whose inner set has size m uses only leaves containing at most m occurrences of P_0.

Lemma 1 (Restricted encoding)**.** *All encoded scalars at leaves with at most m occurrences of P_0 can*

be computed in $O(S)$ arithmetic operations, where

$$S = S(L, m) = \sum_{j=0}^{m} \binom{L}{j} 9^{L-j}.$$

They can be stored in a trie of $O(S)$ vertices, with $O(L)$ lookup time.

Proof. Run only the input-encoding recursion. Do not create a P_0 child if its prefix already contains m copies of P_0. Each retained child array is the same linear combination of its parent's seven slices as in the original algorithm; no sibling's value is needed. Thus all retained leaf scalars are unchanged.

Let p_k be the number of retained prefixes at depth k. Every retained prefix has at least 9^{L-k} retained leaf descendants, by choosing only P_{ij} thereafter. These descendant sets are disjoint. Therefore

$$p_k 9^{L-k} \le S, \qquad \sum_{k=0}^{L} p_k 7^{L-k} \le S \sum_{h=0}^{L} (7/9)^h \le \frac{9}{2} S.$$

Forming the at most ten child arrays at a node costs $O(7^{L-k})$, proving the time bound. Similarly $\sum_k p_k \le S \sum_{h \ge 0} 9^{-h} = 9S/8$ bounds the trie size. We do not allocate the old 10^L-entry array. Dense root initialization and the paper's $O(L7^L)$ input formation are also $O(S)$ because $L7^L = O(9^L)$ and $9^L \le S$. □

Every low-order query leaf is retained. A stored box has at most $m - t$ positions marked P_0 or free; hence every leaf needed by its dynamic program is retained too. All scalar values are unchanged.

For fixed $c > 10$, put $\beta_d = \binom{L}{m-d} 9^{L-m+d}$. Then $S = \sum_{d=0}^{m} \beta_d$, $\beta_0 = M$, and

$$\frac{\beta_d}{\beta_{d-1}} = \frac{9(m-d+1)}{L-m+d} \le \frac{9m}{L-m+1} < \frac{9}{c-1} < 1.$$

Thus $S = O(M)$. There are $O(N/\sqrt{M})$ row and column bands, so all encodings now cost $\widetilde{O}(N\sqrt{M})$ instead of $\widetilde{O}(N10^L/\sqrt{M})$.

3 Exact box-tail exponents

Let $\mathcal{H}(x) = -x \ln x - (1-x)\ln(1-x)$, with natural logarithms, and define

$$T(c) = \frac{\frac{1}{2} c\mathcal{H}(1/c) + (c-1)\ln 3}{\ln 4}, \tag{1}$$

$$G(c, \theta) = \frac{c\mathcal{H}(1/c) - c\mathcal{H}((1-\theta)/c) - \theta \ln 9}{\ln 4}, \tag{2}$$

$$q(\theta) = \frac{\mathcal{H}(\theta) + \theta \ln 9}{\ln 4}, \tag{3}$$

$$E(c) = \begin{cases} T(c), & c \ge 10, \\ c \ln 10 / \ln 4 - T(c), & 1 < c < 10. \end{cases} \tag{4}$$

Theorem 2 (Refined data structure)**.** *Fix $c > 1$ and $0 < \theta < 0.9$ such that*

$$\frac{9(1-\theta)}{c-1+\theta} < 1, \qquad G(c, \theta) > 0.$$

For $L = \lceil cm \rceil$, $t = \lceil \theta m \rceil$, and $N \geq \sqrt{K} N_0$, the data structure has preprocessing time and space

$$\widetilde{O}\big(ND^{E(c)} + N^2 D^{-G(c,\theta)}\big)$$

and individual query time $\widetilde{O}(D^{q(\theta)})$. Here the suppressed factors are powers of $\log D$.

Proof. Lemma 29 of the paper bounds boxes per tile by $(m+1)\sum_{d=t}^{m} \beta_d$. For $d \geq t+1$, successive ratios are eventually bounded by a constant less than one under the displayed condition. Consequently the tail is $O(\beta_t)$, and the exact ratio is

$$\frac{\beta_t}{M} = 9^t \frac{\binom{L}{m-t}}{\binom{L}{m}} = D^{-G(c,\theta)} \operatorname{poly}(m).$$

Standard two-sided entropy estimates for binomial coefficients prove the last equality up to polynomial factors. Multiplying by $O(N^2/M)$ tiles gives the box term. Queries use $O(L\sum_{d=0}^{t}\binom{m}{d}9^d)$ values. Because $\theta < 0.9$, the summands are increasing up to the cutoff, and their sum is $D^{q(\theta)}\operatorname{poly}(m)$.

For $c > 10$, the encoding lemma gives $\sqrt{M} = D^{T(c)}\operatorname{poly}(m)$. For $c \leq 10$, the maximum term in S is at $j \approx L/10 \leq m$, so $S = 10^L \operatorname{poly}(m)$ and the encoding exponent is the second branch of E. Both branches agree at $c = 10$.

The input-layer count remains $\binom{L}{m}3^{L-m}4^m \leq 7^L$. The algebraic embedding and the box identities require $L \geq m$, not $L \geq 10m$. The latter condition in the paper was used for its global geometric bound. This proof therefore also covers the stated $c < 10$ range. For fixed $c > 1$, $N \geq N_0 = 3^{L-m}$ implies $L = O(\log N)$; the original intermediate-value bounds remain polynomial in N. Padding D to a power of four and rounding L, t changes constants or polylogarithmic factors only. Small bounded D is handled directly. □

For $D \leq N^e$ and $|W| \leq N^2/D^\kappa$, a sufficient strict certificate for $O(N^2/D^g)$ is

$$g < G, \qquad g < \kappa - q, \qquad g < 1/e - E, \qquad 1/e > T. \tag{5}$$

Every strict gap absorbs the suppressed logarithms.

Corollary 3 (Same regime; larger saving)**.** *Take $c = 21.775$, $\theta = 0.11121$, and $e = 1/18$. Then*

$$G = 0.0719572213\ldots, \quad q = 0.4280411447\ldots, \quad E + G = 17.9991675570\ldots.$$

Preprocessing takes $O(N^2/D^{0.07195})$, and queries take $O(D^{0.42805})$. Thus $|W| \leq N^2/\sqrt{D}$ can be answered in $O(N^2/D^{0.07195})$.

This increases the displayed saving exponent 0.063 by about 14.2%; it is not a claim of a 14.2% wall-clock speedup.

For a different tradeoff, $c = 9.33$, $\theta = 0.135$, $e = 0.129$ satisfy (5) with $\kappa = 1/2$ and $g = 0.0001$. This extends the supported inner dimension to $D \leq N^{0.129}$ at that density. It does *not* extend the paper's universal statement for every positive κ or every positive query exponent. The smaller saving is part of this tradeoff.

Batching independent products into the tensor slots

The tiling above uses all $K = \binom{L}{m}$ independent product slots of the source's Lemma 9 for a grid of blocks from one matrix product. When many independent products of the same shape are available, assigning one product to each slot gives a different encoding amortization and a weaker outer dimension requirement.

Lemma 4 (Direct-sum tiling). *Let $D = 4^m$, $L = \lceil cm \rceil$ for fixed $c > 10$, $K = \binom{L}{m}$, $N_0 = 3^{L-m}$, and $M = KN_0^2$. Consider B independent products of $N \times D$ by $D \times N$ integer matrices with $B \geq K$ and $N \geq N_0$, all dimensions and magnitudes polynomial in the ambient size. The B products can be preprocessed, and their Q_{tot} requested entries answered, in*

$$\widetilde{O}\left(BNN_0 + BN^2D^{-G} + Q_{\mathrm{tot}}D^q\right)$$

time, with the box exponent G of Theorem 2 and either the query exponent q of that theorem or the exponent q_{pc} of Lemma 5 below.

Proof. Partition the products into packs of K, padding the last pack with zero products, and fix for each pack a bijection between its slots and the subsets $Q \subseteq [L]$ of size m. Partition every input's rows and columns into common consecutive blocks of N_0 positions, padding the last blocks with zeros. For each row-block index i build one left array a_i: in the coordinates of inner set Q place row block i of the left matrix of the product assigned to Q; for each column-block index j build one right array b_j analogously. The left array depends on i and the pack only, and the right array on j and the pack only, so each retained encoding is computed once and shared by every tile (i, j).

The source's Lemma 9 permits completely independent matrices in the K slots: on the tile (i, j) the output with inner set Q is exactly the (i, j) block of the product assigned to Q, because an output with inner set Q forces both input inner sets to contain Q while every active input has exactly m inner positions. The assignment stays fixed across tiles, each requested entry has a unique pack, slot, tile, and output string, and its query runs unchanged with no sum over slots.

A pack has $\lceil N/N_0 \rceil$ row and column blocks and therefore $2\lceil N/N_0 \rceil$ encodings of cost $O(S) = O(M)$ each by the restricted encoding lemma, including input placement since $KN_0D \leq 7^L \leq S$. Over $\lceil B/K \rceil$ packs this is $\widetilde{O}(BNN_0)$ because $B \geq K$ and $N \geq N_0$ absorb the ceilings. Each pack has $\lceil N/N_0 \rceil^2$ tiles with the unchanged box preprocessing $\widetilde{O}(MD^{-G})$, which totals $\widetilde{O}(BN^2D^{-G})$. Finally $N \geq N_0$ and fixed c give $L = O(\log N)$, so the intermediate-value bounds are unchanged. □

The condition $N \geq N_0$ replaces $N \geq \sqrt{K}N_0$ only for this layout, and its price is the capacity condition $B \geq K$. Write

$$A_0(c) = \frac{(c-1)\ln 3}{\ln 4}, \qquad C_K(c) = \frac{c\mathcal{H}(1/c)}{\ln 4},$$

so that $N_0 = \widetilde{\Theta}(D^{A_0})$, $K = \widetilde{\Theta}(D^{C_K})$, and $T = A_0 + C_K/2$. The encoding exponent of a batched product is therefore A_0 in place of $E = T$. For lopsided instances with middle part H, the reductions below supply only $\widetilde{\Theta}(H/\sqrt{D})$ independent products per prime, so $B \geq K$ must be checked separately; the earlier condition $H \geq \sqrt{D}$ alone does not permit substituting A_0 for E.

Contracting a prefix of each query

The query of Theorem 2 reads α_t stored boxes and the low-order leaves one by one. Boxes with few leading stars can instead be aggregated on the fly, using only the stored encodings.

Lemma 5 (Prefix contraction). *Write $P = \varphi_{P_0}\psi_{P_0}$ and $I = \sum_{i,j\leq 2} p_{ij}q_{ij}$ for the one-level bilinear kernels. The identity of the source gives*

$$\sum_{i,j\leq 3} \varphi_{P_{ij}}\psi_{P_{ij}} = I - P, \qquad \sum_{\lambda} \varphi_\lambda\psi_\lambda = I,$$

the second sum over all ten terms. Moreover $p_{ij} = \varphi_{P_{ij}} - \varphi_{P_{i3}}$ and $q_{ij} = \psi_{P_{ij}} - \psi_{P_{3j}}$ for $i, j \leq 2$. Fix a query with inner set Q, the switching order t, and an integer $0 \leq \ell \leq m - t$; let F be the first ℓ

positions of Q in the order used to define boxes and $Q' = Q \setminus F$. Then the output equals the sum of the $9^t\binom{m-\ell}{t}$ stored boxes with at least ℓ stars plus a contraction computed from the retained encodings in $\mathrm{poly}(L)\,6^\ell \sum_{j<t}\binom{m-\ell}{j}6^j$ operations. For $t = \lceil \theta m \rceil$, $\ell = \lfloor rm \rfloor$, $0 \le r < 1-\theta$, and $\theta/(1-r) < 6/7$, the query time is $\widetilde{O}(D^{q_{\mathrm{pc}}})$ with

$$q_{\mathrm{pc}} = \max(q_1, q_2), \quad q_1 = \frac{(1-r)\mathcal{H}(\theta/(1-r)) + (\theta+r)\ln 6}{\ln 4}, \quad q_2 = \frac{(1-r)\mathcal{H}(\theta/(1-r)) + \theta \ln 9}{\ln 4},$$

For $0 < r \le \theta \ln(9/6)/\ln 6$, this is strictly below the exponent q of Theorem 2. For other admissible r, use the displayed maximum without asserting an improvement. No preprocessing cost or tile condition changes.

Proof. The boxes of the query with at least ℓ stars cover exactly the leaves whose restriction to Q' has at least t terms other than P_0: the stopping rule sees its t-th such term before reaching F exactly in this case. For fixed $e \ge \ell$ there are $\binom{m-e-1}{t-1}$ sets V, hence $9^t \sum_{e\ge\ell}\binom{m-e-1}{t-1} = 9^t\binom{m-\ell}{t}$ such boxes; read them from the trie. The remaining leaves have fewer than t terms other than P_0 on Q' and arbitrary terms on F. With the fixed terms outside Q suppressed, their sum is the bilinear kernel

$$I^F \sum_{d<t} \sum_{\substack{U\subseteq Q' \\ |U|=d}} (I-P)^U P^{Q'\setminus U} = I^F \sum_{\substack{A\subseteq Q' \\ |A|<t}} (-1)^{t-|A|-1}\binom{m-\ell-|A|-1}{t-|A|-1} I^A P^{Q'\setminus A},$$

by $\sum_{d=j}^{t-1}(-1)^{d-j}\binom{m'-j}{d-j} = (-1)^{t-j-1}\binom{m'-j-1}{t-j-1}$ with $m' = m-\ell$. A contraction with kernel I at the $j = \ell + |A|$ positions of $F \cup A$, P at $Q' \setminus A$, and the query's terms elsewhere is computed from 6^j retained encodings on each side: at each I position the six forms $\varphi_{P_{ij}}$, $i \le 2$, $j \le 3$, give the four forms p_{ij} by a 4×6 transform, and symmetrically on the right; applying these transforms one position at a time, each shrinking its dimension, costs $O(j6^j)$, after which the 4^j matching coordinates are multiplied and summed. Every fetched leaf has at most $m-\ell-|A| \le m$ copies of P_0, so it is retained. Summing over A gives the stated operation count; the binomial coefficients have $O(m)$ bits, so the intermediate-value bounds survive. With $t = \lceil \theta m \rceil$ and $\ell = \lfloor rm \rfloor$ the terms of the residual sum increase up to the cutoff when $\theta/(1-r) < 6/7$, and entropy estimates give the two exponents; $(1-r)\mathcal{H}(\theta/(1-r)) < \mathcal{H}(\theta)$ for $0 < r < 1-\theta$, which gives the strict comparison with q when also $r \ln 6 \le \theta \ln(9/6)$. The balance $q_1 = q_2$ occurs at $r = \theta\ln(9/6)/\ln 6$; explicit certificates use a nearby rational r and check both exponents. □

Lemma 6 (Prefix-compatible encoding budget)**.** *Under Lemma 5, suppose also $\ell \le t$ and use the prefix-contracted algorithm for every query. It suffices to retain the input encodings only at leaves with at most $m-\ell$ copies of P_0. For fixed $c > 10$ and $\ell = \lfloor rm \rfloor$, the encoding cost per array is $\widetilde{O}(MD^{-G(c,r)})$. In Lemma 4, the total encoding cost is consequently $\widetilde{O}(BNN_0D^{-G(c,r)})$; its geometry and capacity hypotheses, box costs, and prefix-query costs are unchanged.*

Proof. The residual contraction indexed by $A \subseteq Q'$ accesses P_0 only at $Q' \setminus A$: each of the six forms used by the shrinking transform at an I position is a P_{ij} form, and the fixed positions outside Q also use P_{ij}. Every fetched leaf therefore has $m-\ell-|A| \le m-\ell$ copies of P_0. Every stored box and every intermediate box in its dynamic program has at most $m-t$ symbols that are P_0 or stars. Its leaf values thus need at most $m-t \le m-\ell$ copies of P_0 as well.

Run the restricted input recursion with budget $m-\ell$ in place of m. The identical prefix argument bounds its entire internal work and compact trie size by a constant times

$$S_{m-\ell} = \sum_{j=0}^{m-\ell}\binom{L}{j}9^{L-j} = \sum_{d=\ell}^{m}\beta_d.$$

Dense root initialization, input placement, and subset-slot enumeration remain $O(S_{m-\ell})$, since $S_{m-\ell} \geq 9^L$ and their cost is at most $O(L7^L)$. For $c > 10$, the same geometric ratio bound gives $S_{m-\ell} = O(\beta_\ell)$, and the entropy estimate gives

$$\frac{\beta_\ell}{M} = 9^\ell \frac{\binom{L}{m-\ell}}{\binom{L}{m}} = D^{-G(c,r)} \operatorname{poly}(m).$$

The layout still uses $O((B/K)(N/N_0))$ encoded arrays under the original two packing hypotheses, giving the stated total. The reduction never uses the old direct low-order query after making this additional pruning. □

Deterministic cyclic-order lazy encoding

Lemma 7 (Cyclic-order lazy encoding)**.** *Fix $c > 10$ and $0 < \alpha < \theta$. Let $D = 4^m$, $L = \lceil cm \rceil$, $R_0 = \lfloor (1-\alpha)m \rfloor$, and $S_{R_0} = \sum_{j=0}^{R_0} \binom{L}{j} 9^{L-j}$. The encodings needed by Lemma 5 can be represented deterministically in $\widetilde{O}(S_{R_0})$ preprocessing time and space per input band. All box values are preprocessed as before. A scalar leaf with $z > R_0$ copies of P_0 can be retrieved in*

$$\operatorname{poly}(L) 3^{\lceil L(z-R_0)/z \rceil}$$

time; leaves with $z \leq R_0$ take $\operatorname{poly}(L)$ time.

Proof. For each of the L cyclic rotations of the tensor coordinates, permute both the input array and the lookup labels by that rotation. Build the restricted encoder with budget R_0, storing the whole remaining input array at every retained prefix, not only the final scalar. If p_d is the number of retained prefixes at depth d, every prefix has at least 9^{L-d} retained descendants, so

$$\sum_{d=0}^{L} p_d 7^{L-d} \leq \frac{9}{2} S_{R_0}.$$

This pays for every stored array and all recursion work. Input permutation and initialization take $O(L7^L) = \widetilde{O}(S_{R_0})$ per rotation. Coordinate permutations do not change the multilinear evaluation when applied to both the array and its query string.

Consider a target leaf with $z > R_0$ positions labeled P_0, put $s = z - R_0$, and set $R = \lceil Ls/z \rceil$. Among the L cyclic windows of length R, the average number of these positions is $zR/L \geq s$, because each position belongs to exactly R windows. A window therefore contains at least s positions. Rotate it to be the final R coordinates. The preceding coordinates contain at most $z - s = R_0$ copies of P_0, so the first forbidden child, if any, occurs within this final window. Its parent array is stored and has at most R coordinates remaining. Every one-level form has at most three nonzero coefficients, so directly contracting the remaining array along the target forms reads only the Cartesian product of their supports, at most 3^R entries, and costs $\operatorname{poly}(L)3^R$. This is a sparse contraction: no dense 7^R temporary array is formed. Finding the rotation and retained parent costs only $\operatorname{poly}(L)$. This is deterministic and works for every logical leaf, including adaptively chosen leaves.

The stored boxes need only leaves with at most $m - t$ copies of P_0. Since $\alpha < \theta$, for sufficiently large m these are below R_0; bounded exceptional sizes can be evaluated directly. Box preprocessing therefore uses its original recurrence and costs. The retained arrays and direct contractions have the same polynomial intermediate-value bounds as the original input encoder. □

The geometric-tail and entropy argument yields

$$S_{R_0} = \widetilde{\Theta}(MD^{-G(c,\alpha)}).$$

Thus single-product encoding exponent $E = T$ becomes $T - G(c,\alpha)$, and the direct-sum encoding exponent A_0 becomes $A_0 - G(c,\alpha)$. All tiling and batching capacity conditions remain unchanged.

For a prefix-contracted query let $\ell = \lfloor rm \rfloor$, with $0 < r < \alpha < \theta$ and $\theta/(1-r) < 6/7$. In the residual term indexed by $j = xm$, every fetched leaf has $z = m - \ell - j$ copies of P_0. It is already stored for $x \geq \alpha - r$, up to rounding. The remaining terms have total exponent bounded by the maximum of

$$\frac{(1-r)\mathcal{H}(x/(1-r)) + (r+x)\ln 6}{\ln 4} + \frac{c\ln 3}{\ln 4}\frac{\alpha - r - x}{1-r-x}, \qquad 0 \leq x \leq \alpha - r.$$

A convenient closed upper bound is

$$q_{\text{lazy}} = r\log_4 6 + c\log_4 3\frac{\alpha - r}{1-r} + (1-r)\log_4\left(1 + 6\,3^{-c(1-\alpha)/(1-r)^2}\right).$$

Indeed, with $h = 1-r$ and $\tau = \alpha - r$, concavity gives $(\tau - x)/(h-x) \leq \tau/h - (h-\tau)x/h^2$, and maximizing the resulting entropy-plus-linear expression over $0 \leq x \leq h$ gives the displayed bound. All rounding errors in the omitted-leaf range change the exponent of a lookup by $O(1/m)$, since $1 - r - x \geq 1 - \alpha > 0$; they therefore contribute only constant or polynomial factors in L. The complete query exponent is $\max(q_1, q_2, q_{\text{lazy}})$, where q_1, q_2 are the two original prefix exponents. In particular, if $q_{\text{lazy}} < \max(q_1, q_2)$, the old query bound holds with the strictly smaller encoding exponent above.

4 Independent modular filters remove the scan loss

The following reduction changes the algorithm, independently of the encoding modification. Write $S(a,b,c) = w(a,b) + w(a,c) + w(b,c)$. Its two ingredients are a folding of the residue classes of a modulus into the outer dimensions of one thin product, which shares the band encodings of the data structure across residues, and the intersection of independent modular filters before any exhaustive scan.

Lemma 8 (Residue folding)**.** *Fix a prime p and a block B of the third vertex part, and put $p_1 = \lceil \sqrt{p} \rceil$ and $p_2 = \lceil p/p_1 \rceil$, so that $p \leq p_1 p_2$ and $\max(p_1, p_2) \leq \sqrt{p} + 1$. Index rows by the pairs (a, ρ_1) with $0 \leq \rho_1 < p_1$, columns by the pairs (b, ρ_2) with $0 \leq \rho_2 < p_2$, and the middle part by $B \times \mathbb{Z}_p$. Define the Boolean matrices*

$$X[(a,\rho_1),(c,\sigma)] = 1 \iff \sigma \equiv w(a,c) + \rho_1 \pmod p,$$
$$Y[(c,\sigma),(b,\rho_2)] = 1 \iff \sigma \equiv -w(b,c) - p_1\rho_2 \pmod p.$$

For an outer pair (a,b), write $w(a,b) \bmod p = \rho_1 + p_1\rho_2$ with $0 \leq \rho_1 < p_1$; then $0 \leq \rho_2 < p_2$, and

$$(XY)[(a,\rho_1),(b,\rho_2)] = \#\{c \in B : \ S(a,b,c) \equiv 0 \pmod p\}.$$

The product has outer dimensions at most $n(\sqrt{p}+1)$ and inner dimension $|B|\,p$, and the n^2 outer pairs are queried at n^2 distinct positions.

Proof. The entry counts the $c \in B$ with $w(a,c) + \rho_1 \equiv -w(b,c) - p_1\rho_2$, that is, with $w(a,c) + w(b,c) + w(a,b) \equiv 0 \pmod p$, because $\rho_1 + p_1\rho_2 \equiv w(a,b)$. Since $0 \leq w(a,b) \bmod p < p \leq p_1 p_2$, the digits ρ_1, ρ_2 exist and are unique, and distinct pairs (a,b) occupy distinct positions. Finally $p_1 \leq \sqrt{p} + 1$ and $p_2 \leq p/\sqrt{p} + 1$. □

The lemma replaces the p residue-specific instances of a block, each an $n \times n$ product, by one product with outer dimension $N = O(n\sqrt{p})$. The box preprocessing $N^2 D^{-G}$ is unchanged in total, and the queries are unchanged, but the band encodings cost ND^E instead of pnD^E: they are shared across all residues. The same effect can be obtained with rows (a, ρ) for all $\rho \in \mathbb{Z}_p$ and unchanged columns, using rectangular tiles with $\sqrt{pK} \times \sqrt{K/p}$ block products; the square version above needs no change to the data structure.

Theorem 9 (Amplified reduction with folding)**.** *Suppose a thin-product data structure preprocesses $N \times D$ by $D \times N$ integer matrices in $\widetilde{O}(ND^E + N^2 D^{-G})$ time and answers an entry in $\widetilde{O}(D^q)$ time whenever $N \geq D^T \operatorname{poly}(\log D)$, where $G > 0$, $q < 1/2$, and $E, T > 0$ are constants. For a fixed $0 < e < 1$ with $D = n^e$ and $e(T - 1/4) < 1$, all-edges Exact Triangle has a Las Vegas algorithm with expected time*

$$\widetilde{O}\left(n^2 D^{E-1/4} + n^3 D^{-G} + n^3 D^{q-1/2} + n^2 D^{3/4} + n^2\sqrt{D}\right). \tag{6}$$

Without Lemma 8, the same proof with p residue-specific instances per block gives the encoding term $n^2 D^E$, the construction term $n^2 D$, and the condition $eT < 1$.

Proof. Let $\ell = \log n$, $P = \lceil \sqrt{D}\, \ell^2 \rceil$, $h = \lfloor D/(2P) \rfloor$, and $k = \lceil 4 \log_2 n \rceil$. Partition the third vertex part into blocks of at most h vertices. Sieve the primes in $[P, 2P]$ and sample k of them uniformly and independently with replacement. They and the blocks are fixed before queries.

For each sampled prime p and block B, construct the folded instance of Lemma 8. Its middle dimension is at most $2Ph \leq D$ and its outer dimensions are at most $n(\sqrt{2P} + 1)$; pad it with zeros to a square $N \times D$ by $D \times N$ product with $N = \Theta(n\sqrt{P})$. Since $N \geq n\sqrt{P} \geq nD^{1/4} \log n$ and $e(T - 1/4) < 1$, the hypothesis $N \geq D^T \operatorname{poly}(\log D)$ holds for all sufficiently large n. For every outer pair, compute $w(a, b) \bmod p$ once per sampled prime, and query the single position $((a, \rho_1), (b, \rho_2))$ of the lemma in every block. The residue classes need not be balanced: the data structure charges each entry query separately. Thus each pair is queried exactly once per prime and block. A pair/block passes a prime if some c in that block makes $S(a, b, c) = 0 \pmod p$. Scan a block using the original integer sums only if it passes *all* sampled primes. Stop scanning a given pair after its first exact zero witness is found.

Every genuine zero passes every prime, and marking a pair requires an explicit exact check. The algorithm is therefore always correct. Passing all filters alone is not a certificate: different primes may have different modular witnesses in the block.

Fix a zero-free pair/block. Each nonzero sum, of magnitude at most $3n^\nu$, has at most $\log(3n^\nu)/\log P$ prime divisors at least P. There are $\Omega(P/\log P)$ primes in the interval. A union bound over at most h sums gives

$$\Pr[\text{pass one prime}] \leq C_\nu h \log n / P = O_\nu((\log n)^{-3}).$$

For all sufficiently large n, this is at most $1/2$. Independence across prime choices gives survival probability at most $2^{-k} \leq n^{-4}$. There are $O(n^3/h)$ pair/blocks, so their total expected failed-scan cost is $O(n^3 n^{-4})$. At most n^2 successful scans occur, costing $O(n^2 h)$. No independence between different pair/blocks is assumed.

Let $b = \lceil n/h \rceil$ be the number of blocks. There are $kb = \widetilde{O}(n/\sqrt{D})$ folded instances, each with $N = \widetilde{O}(nD^{1/4})$, since $P = \widetilde{O}(\sqrt{D})$. Their band encodings cost $\widetilde{O}(kbND^E) = \widetilde{O}(n^2 D^{E-1/4})$, and their box preprocessing costs $\widetilde{O}(kbN^2 D^{-G}) = \widetilde{O}(n^3 (P/h) D^{-G}) = \widetilde{O}(n^3 D^{-G})$, since $P/h = \Theta(\log^4 n)$. The number of entry queries is $kbn^2 = \widetilde{O}(n^3/h) = \widetilde{O}(n^3 D^{-1/2})$, because each pair is queried once per prime and block. These give the first three terms of (6). Writing the two Boolean matrices of every instance explicitly costs $\widetilde{O}(kbND) = \widetilde{O}(n^2 D^{3/4})$. The sieve, the residues $w(a, b) \bmod p$, and combining the answers are smaller. Successful scans give the last term. Processing one block at a time suffices. Finitely many small sizes can use brute force. Randomness affects time only. This computes all answers

for the designated outer-edge class. If answers for all three edge classes are required, repeat with the three choices of middle part, changing the cost by a constant factor. □

Corollary 10 (Integer all-edges Exact Triangle)**.** *Take $c = 21.0477$, $\theta = 0.111971$, and $e = 0.05828$ in the refined data structure of Theorem 2, with the explicit rounding $D = 4^{\lfloor \log_4(n^e) \rfloor} = \Theta(n^e)$. Its tiling condition $N \geq \sqrt{K} N_0 = \sqrt{M}$ holds for the folded instances, whose outer dimension is $N = \Theta(n\sqrt{P}) \geq nD^{1/4}$, for all sufficiently large n, since*

$$e(T - 1/4) = 0.995905586384774\ldots < 1, \qquad \sqrt{M} = \widetilde{\Theta}(D^T).$$

This uses the modified parameterized construction: $e > 1/18$, so the convenient $N \geq D^{18}$ corollary alone does not cover these parameters. The strict savings are

$$eG = 0.004057096756\ldots, \qquad e(1/2-q) = 0.004057095599\ldots, \qquad 1-e(E-1/4) = 0.004094413615\ldots,$$

all above 0.004057*. All-edges Exact Triangle therefore has expected running time $O(n^{3-0.004057}) = O(n^{2.995943})$.*

The numerical supremum of this parameter family, about 0.0040572, is *not* claimed. The chosen strict gaps also absorb the extra logarithms introduced by amplification. Without Lemma 8, the parameters $c = 21.2733$, $\theta = 0.111731$, $e = 0.056846$ satisfy the unfolded certificate with saving 0.003999, giving $O(n^{2.996001})$; the companion script certifies both parameter sets.

For clarity, the complete expected bound is

$$\widetilde{O}\left(n^2 D^{E-1/4} + n^3 D^{-G} + n^3 D^{q-1/2} + n^2 D^{3/4} + n^2\sqrt{D}\right).$$

Its five exponents after setting $D = n^{0.05828}$ are:

Cost	Exponent of n, before logarithmic factors
Encodings, $n^2 D^{E-1/4}$	$2.995905586384774\ldots$
Boxes, $n^3 D^{-G}$	$2.995942903243984\ldots$
Queries, $n^3 D^{q-1/2}$	$2.995942904400685\ldots$
Matrix construction, $n^2 D^{3/4}$	2.04371
Successful scans, $n^2\sqrt{D}$	2.02914

Let τ denote the maximum of these exact exponents. Thus

$$\tau = 3 - e(1/2 - q) = 2.995942904400685\ldots, \quad 2.995943 - \tau = 0.000000095599315\ldots > 0.$$

We retain the stronger intermediate bound $\widetilde{O}(n^\tau)$ through the remaining reductions, and round the exponent only at the end.

Corollary 11 (Batched folded reduction)**.** *For each sampled prime, the $B = \lceil n/h \rceil = \widetilde{\Theta}(n/\sqrt{D})$ folded middle-block products of Theorem 9 are independent and have the common shape $N \times D$ by $D \times N$ with $N = \widetilde{\Theta}(nD^{1/4})$. Apply Lemma 4 to them, with the prefix-contracted queries of Lemma 5. The modular answers, the filter analysis, and the scans are unchanged, and the expected time becomes*

$$\widetilde{O}\left(n^2 D^{A_0-1/4} + n^3 D^{-G} + n^3 D^{q_{\mathrm{pc}}-1/2} + n^2 D^{3/4} + n^2\sqrt{D}\right),$$

valid when $nD^{1/4} \geq N_0$ and $n/\sqrt{D} \geq K$ up to polylogarithms, that is, $e(A_0-1/4) < 1$ and $e(C_K+1/2) < 1$ for $D = n^e$. Take

$$(c, \theta, r, e) = (20.0337,\ 0.11371328,\ 0.025732677,\ 0.06711237).$$

Then $A_0 = 15.083850\ldots$, $C_K = 2.865215\ldots$, $q_1 = 0.4334671995\ldots$, $q_2 = 0.4334671992\ldots$, *and the savings are*

$$eG = 0.004465176\ldots, \qquad e(1/2 - q_1) = 0.004465173\ldots,$$
$$e(1/2 - q_2) = 0.004465173\ldots, \qquad 1 - e(A_0 - 1/4) = 0.004465145\ldots,$$

all above 0.004465, *with* $1 - e(C_K + 1/2) = 0.774\ldots > 0$. *The maximal exact cost exponent is*

$$\tau_{\mathrm{b}} = 2 + e(A_0 - 1/4) = 2.995534854890842\ldots, \qquad 2.995535 - \tau_{\mathrm{b}} = 0.000000145109157\ldots > 0,$$

so all-edges Exact Triangle has expected running time $O(n^{2.995535})$.

The unbatched bound $O(n^{2.995943})$ remains the one to use when fewer than K independent products of a common shape are available, as in the lopsided corollaries of Section 7. The supremum of the batched family, about 0.0044652, is not claimed; using the encoder budget $m - \ell$ instead of m would add about $4 \cdot 10^{-6}$ and is also not claimed.

Keeping the full saving for integer APSP

Here is a direct proof of the tight transfer noted in the source's footnote 10. For each individual min-plus product, choose constants u, v making its finite integer matrices $\bar{A} = A + u$ and $\bar{B} = B + v$ entrywise nonnegative. Since each candidate sum uses one entry from each matrix,

$$\bar{A} \star \bar{B} = (A \star B) + (u + v).$$

The shifts are local to this matrix product, and are removed from its answer. Relabel the nonnegative matrices A, B and recursively compute

$$C' = \lfloor A/2 \rfloor \star \lfloor B/2 \rfloor,$$

where $\star$ denotes min-plus multiplication. For every candidate index k, $A_{ik} + B_{kj} \geq 2C'_{ij}$, and an index attaining C'_{ij} gives $A_{ik} + B_{kj} \leq 2C'_{ij} + 2$. Thus each integer entry of $C = A \star B$ lies in $\{2C'_{ij}, 2C'_{ij} + 1, 2C'_{ij} + 2\}$.

For $r = 0, 1$, an all-edges Exact Triangle call with edge weights A_{ik}, B_{kj}, and $-(2C'_{ij} + r)$ returns a Boolean answer $Z^{(r)}_{ij}$. Reconstruct the minimum with the explicit priority rule

$$C_{ij} = \begin{cases} 2C'_{ij}, & Z^{(0)}_{ij} = 1, \\ 2C'_{ij} + 1, & Z^{(0)}_{ij} = 0 \text{ and } Z^{(1)}_{ij} = 1, \\ 2C'_{ij} + 2, & \text{otherwise.} \end{cases}$$

Both tests can succeed through different candidates, so the first has priority. There is one recursive min-plus call per bit level, followed by two all-edges calls. The recursion ends at the all-zero matrices after $O(\log U)$ levels for $U = n^{O(1)}$. Thus only logarithmically many all-edges calls are needed, with polynomial integer weights. Fresh randomness in each call preserves the expected-time bound even though the calls are adaptive.

Repeated min-plus squaring with a zero diagonal computes APSP in at most $\lceil \log_2 n \rceil$ products. If finite input weights satisfy $|w| \leq W$, use an integer sentinel $B > 4nW + 1$ for absent edges. Every walk considered has at most $2n$ edges: a walk with no sentinel has magnitude at most $2nW$, and a walk containing a sentinel has value above $2nW$. Classify final values above $2nW$ as infinity. With

no negative cycles, shortest paths can be chosen simple. Bit scaling and repeated squaring add only logarithmic factors to the intermediate bounds $\widetilde{O}(n^{\tau})$ and $\widetilde{O}(n^{\tau_{\mathrm{b}}})$. For each fixed r, the strict gaps $\tau < 2.995943$ and $\tau_{\mathrm{b}} < 2.995535$ give

$$n^{\tau}(\log n)^r = O(n^{2.995943}), \qquad n^{\tau_{\mathrm{b}}}(\log n)^r = O(n^{2.995535}).$$

Consequently integer min-plus product and APSP inherit expected $O(n^{2.995535})$ time from the batched reduction, and $O(n^{2.995943})$ from the unbatched one, with the exponents rounded after both reductions.

Singleton decoding removes the successful-scan term

The remaining scan term can be removed in expectation without changing the thin-product data structure. The additional ingredient is randomized isolation of one genuine witness, followed by an exact verification and a fallback scan.

Lemma 12 (Isolated witness recovery)**.** *Let the outer parts have at most R vertices each and the middle part have H vertices, all polynomial in an ambient size n, with polynomially bounded integer weights. Suppose the thin-product data structure has preprocessing cost $\widetilde{O}(ND^E + N^2D^{-G})$, query cost $\widetilde{O}(D^q)$, and the tile requirement $N \geq D^T \operatorname{poly}(\log D)$, where $G > 0$, $0 \leq q < 1/2$, and $E, T > 0$ are fixed. Let $D = \Theta(n^d)$ for any fixed $d > 0$, rounded to a power of four, and suppose $H \geq \sqrt{D}$ and $RD^{1/4} \geq D^{T+\eta}$ for a fixed $\eta > 0$. Then all-edges Exact Triangle has an always-correct algorithm with expected running time*

$$\widetilde{O}\left(RHD^{E-1/4} + R^2HD^{-G} + R^2HD^{q-1/2} + RHD^{3/4}\right). \tag{5}$$

The same bound, with additional logarithmic factors, applies to the $R \times H$ by $H \times R$ integer min-plus product.

Proof. Use the original parameters $P = \lceil \sqrt{D} \log^2 n \rceil$ and $h = \lfloor D/(2P) \rfloor$. Choose a sufficiently large fixed L and retain $\lceil L \log_2 n \rceil$ independent original modular filters for every block of at most h middle vertices. In addition, for every block and each $j = 0, \ldots, \lceil \log_2 h \rceil$, perform $C \log n$ independent trials. A trial retains each middle vertex with probability 2^{-j} and chooses a fresh independent uniform prime from $[P, 2P]$. Use the usual folded instance, setting the entries of one input associated with unretained vertices to zero. Query both its integer product and a second product obtained by multiplying the other input's entries associated with vertex c by its distinct label $\lambda(c)$ in $\{1, \ldots, H\}$. These return the retained modular-witness count and the sum of the corresponding labels. If the count is one, the label sum identifies the sole modular witness. Check its original integer sum and mark the pair only when that sum is zero.

Fix a pair/block containing $z \geq 1$ genuine witnesses, and put $j = \lceil \log_2 z \rceil$ and $t = 2^{-j}$. The probability of retaining exactly one genuine witness is $zt(1-t)^{z-1} \geq 1/(2\exp(1))$; for $z = 1$ it is one. Conditional on any fixed mask with one genuine witness, at most $h - 1$ nonzero sums remain. The prime-divisor bound in Theorem 9 shows that a fresh independent prime avoids all of them with probability at least $1/2$ for sufficiently large n. Thus each trial at the indicated level decodes a genuine witness with probability at least $1/(4\exp(1))$. Choosing the constant C sufficiently large makes failure in every trial at most n^{-L} for any fixed desired L. No independence across distinct pairs or blocks is required.

After the trials, exhaustively scan every still-unsolved pair/block that passed all the original modular filters. A zero-free block reaches this fallback with probability at most n^{-L} by the original amplification. A block with a genuine witness reaches it only if all its isolation trials failed, also with probability at most n^{-L}. Summing the actual block sizes bounds the expected total fallback cost by $R^2Hn^{-L} = o(1)$ for sufficiently large fixed L. A genuine witness always passes the original filters, so it is either decoded

and exactly checked or found in a fallback scan. This proves correctness for every random outcome, including the outcome in which all isolation trials fail.

There are only polylogarithmically many products per block. The two products per trial have outer dimension $\widetilde{O}(RD^{1/4})$, inner dimension at most D, and one query per outer pair. Their weighted entries have polynomial magnitude, so the existing integer word-size bounds apply. With $H \geq \sqrt{D} \geq h$, there are $O(H/h) = \widetilde{O}(HD^{-1/2})$ blocks, giving exactly the four terms in (5). Mask generation, label decoding, and exact checks fit these costs. For $H < h$, one must instead retain $\lceil H/h \rceil$ in the number of instances; the hypothesis is not removed by isolation. The bit-scaling transfer applies without any change to the two outer dimensions or the middle dimension. Fresh randomness in each call preserves the expected bound under its adaptive calls. □

Integer 3SUM without the Exact Triangle detour

The source's Theorem 21(a) reduces 3SUM to Exact Triangle on $n^{1/2}$ vertices and halves the saving. Its footnote 10 notes that the direct randomized reduction of Kopelowitz, Pettie, and Porat [2] to offline set disjointness keeps all of it. That reduction splits a hash value into a high and a low digit placed on the two sides of a set-disjointness query, which is Lemma 8 applied to a middle part $\mathbb{Z}_p$. We state it with residues modulo primes, with the balance of the buckets handled by sub-buckets, so that it uses only the tools already in this note. Throughout, 3SUM is given sets A, B, C of at most n integers of absolute value at most n^ν and asks for $x \in A$, $y \in B$, $z \in C$ with $x + y = z$. The one-set form, which asks for three distinct elements of a set S with $x + y + z = 0$, is the instance $A = B = S$, $C = -S$ in which witnesses with a repeated element are skipped during scans. A witness with a repeated element is determined by one of its elements, so there are at most $3n$ of them; a sub-query whose only witnesses are of this kind is scanned without success, and at most $3n$ such scans occur in either algorithm below.

Lemma 13 (Bucket-pair folding)**.** *Let r and p be distinct primes, $p_1 = \lceil \sqrt{p} \rceil$, $p_2 = \lceil p/p_1 \rceil$, and $s \geq 1$. Partition A into the residue buckets $A_i = \{x \in A : x \equiv i \pmod r\}$ and cut every bucket into sub-buckets of at most s elements; do the same for B. Index rows by the pairs (α, ρ_1) with α a sub-bucket of A and $0 \leq \rho_1 < p_1$, columns by the pairs (β, ρ_2) with β a sub-bucket of B and $0 \leq \rho_2 < p_2$, and the middle part by $\mathbb{Z}_p$. Define the Boolean matrices*

$$X[(\alpha, \rho_1), \sigma] = 1 \iff \sigma \equiv x - \rho_1 \pmod p \text{ for some } x \in \alpha,$$
$$Y[\sigma, (\beta, \rho_2)] = 1 \iff \sigma \equiv p_1\rho_2 - y \pmod p \text{ for some } y \in \beta.$$

For $z \in C$ write $z \bmod p = \rho_1 + p_1\rho_2$ with $0 \leq \rho_1 < p_1$. Then $(XY)[(\alpha, \rho_1), (\beta, \rho_2)] > 0$ if and only if some $x \in \alpha$ and $y \in \beta$ satisfy $x + y \equiv z \pmod p$. If $x + y = z$ with $x \in A_i$, then $y \in B_{(z-i) \bmod r}$. The product has at most $(n/s + r)(\sqrt{p} + 1)$ rows, as many columns, and inner dimension p.

Proof. A common middle vertex σ exists if and only if $x - \rho_1 \equiv p_1\rho_2 - y$, that is, $x + y \equiv \rho_1 + p_1\rho_2 \equiv z \pmod p$. Reduction modulo r is additive, which gives the bucket statement. A bucket of $|A_i|$ elements has $\lceil |A_i|/s \rceil \leq |A_i|/s + 1$ sub-buckets, so there are at most $n/s + r$ of them, and $p_1, p_2 \leq \sqrt{p} + 1$. □

Theorem 14 (Las Vegas integer 3SUM)**.** *Let the data structure satisfy the hypothesis of Theorem 9, and fix $0 < e < 1$ with $D = n^e$ and $eT < 1$. Integer 3SUM has a Las Vegas algorithm with expected time*

$$\widetilde{O}\left(nD^E + n^2D^{-G} + n^2D^{q-1/2} + n^2D^{-1/2} + nD\right). \tag{8}$$

Proof. Let $P = \lfloor D/2 \rfloor$, $s = \lfloor \sqrt{P}/(C_\nu \log n) \rfloor$ for a constant C_ν fixed below, and $R = \lceil n/s \rceil$.

Bucket prime. For every prime $r \in [R, 2R]$ compute $\Phi(r) = \sum_i (|A_i|^2 + |B_i|^2)$ in $O(n)$ time and keep a minimizer. A nonzero difference of two inputs has at most $\log(2n^\nu)/\log R$ prime divisors in $[R, 2R]$, which contains $\Omega(R/\log R)$ primes, so the average of Φ over these primes is $2n + O(n^2 \log n/R) = O(ns \log n)$, and so is its minimum. This costs $O(nR) = \widetilde{O}(n^2 D^{-1/2})$.

Universe primes. Sieve $[P, 2P]$, exclude r from the candidates, and sample $k = \lceil 4 \log_2 n \rceil$ of the remaining primes uniformly and independently with replacement. For each sampled prime p build the instance of Lemma 13 and pad it to a square $N \times D$ by $D \times N$ product with $N = O((n/s)\sqrt{P}) = O(n \log n)$. Since $N \geq n$ and $eT < 1$, the tile condition holds for large n. For every $z \in C$, residue i, sub-bucket $\alpha \subseteq A_i$, and sub-bucket $\beta \subseteq B_{(z-i) \bmod r}$, query the position of the lemma in every instance. Scan the sub-query (z, α, β), by testing $z - x \in \beta$ for each $x \in \alpha$ in a sorted copy of β, only if its entry is positive in all k instances, and stop at the first exact solution.

Correctness. A solution $x + y = z$ with $x \in A_i$ has $y \in B_{z-i}$ by the lemma and satisfies the congruence modulo every prime, so its sub-query is positive in every instance and is scanned; a scan reports only exact solutions. The output is therefore always correct.

Queries. Using $|A_i||B_j| \leq (|A_i|^2 + |B_j|^2)/2$, the number of sub-queries is

$$\sum_{z \in C} \sum_i \left\lceil \frac{|A_i|}{s} \right\rceil \left\lceil \frac{|B_{z-i}|}{s} \right\rceil \leq \frac{n\Phi(r)}{2s^2} + \frac{2n^2}{s} + nr = O\Big(\frac{n^2 \log n}{s}\Big) = \widetilde{O}(n^2 D^{-1/2}),$$

each costing $\widetilde{O}(D^q)$ per instance.

Failed scans. A sub-query without an exact solution has at most s^2 nonzero values $x + y - z$, of magnitude at most $3n^\nu$, each with at most $\log(3n^\nu)/\log P$ prime divisors in $[P, 2P]$. By the union bound it is positive for a uniformly sampled prime with probability at most $s^2 C'_\nu \log n/P \leq 1/2$ for C_ν large enough, and independence over the k primes gives survival probability at most n^{-4}. With $\widetilde{O}(n^2/s)$ sub-queries, the expected cost of failed scans is $\widetilde{O}((n^2/s)\, n^{-4}\, s) = o(1)$.

Costs. Per instance, with $N = O(n \log n)$: band encodings $\widetilde{O}(nD^E)$, boxes $\widetilde{O}(n^2 D^{-G})$, and writing the two Boolean matrices $\widetilde{O}(nD)$. The queries cost $\widetilde{O}(n^2 D^{q-1/2})$, the bucket prime $\widetilde{O}(n^2 D^{-1/2})$, the sieve $\widetilde{O}(D)$, sorting $\widetilde{O}(n)$, and the single successful scan $\widetilde{O}(s)$. These are the terms of (8). □

Corollary 15 (Integer 3SUM)**.** *Take $c = 21.2733$, $\theta = 0.111731$, and $e = 0.056846$ in Theorem 2. Then*

$$eG = 0.003999294\ldots, \quad e(1/2 - q) = 0.003999284\ldots, \quad 1 - eE = 0.004000455\ldots,$$

all above 0.003999; the terms nD and $n^2 D^{-1/2}$ have savings $1 - e$ and $e/2$, and $1 - eT > 0$. Integer 3SUM therefore has a Las Vegas algorithm with expected time $O(n^{2-0.003999}) = O(n^{1.996001})$.

These are the unfolded parameters of Theorem 9, and the encoding relief $D^{-1/4}$ is not available here: a sub-query compares s^2 sums, so false-positive control forces $s^2 \leq p/\,\mathrm{polylog}$, the whole inner dimension is spent on residues, and the outer dimension is pinned to $\widetilde{\Theta}(n)$. The supremum of this constraint set, about 0.0039993, is not claimed. For integer inputs this improves the earlier real-input bound $O(n^{1.99632})$. The rank-prefix construction below brings real 3SUM to the same $O(n^{1.996001})$ expected bound.

5 Deterministic and real-input corollaries

The deterministic constructions and the real-input constructions use different guarantees. Their full reduction algebra is included below.

Deterministic integer reduction

Choose modulus scale $p = \Theta(n^a)$ and third-part block size $h = \Theta(n^b)$ independently. Put $e = a + b$, $z = a - b > 0$. As in Theorem 17 of the source, choose a prime minimizing modular triangle count, which also minimizes the false-positive count. It satisfies $F(p) = O(n^3 \log n/p)$. By Lemma 8 there are $O(n/h)$ folded instances, one per block, of outer dimension $O(n\sqrt{p})$ and actual inner dimension $O(ph)$, and $O(n^3/h)$ queries. No equal-sized chunking of the residue classes is needed because queries are charged individually. The relevant savings below exponent 3 are

$$eG - z, \qquad e(1/2 - q) - z/2, \qquad z, \tag{7}$$

from boxes, queries, and failed scans. The shared encoding term $\widetilde{O}((n/h)\, n\sqrt{p}\, D^E)$ additionally has saving $1 - e(E - 1/4) - 3z/4$, and the folded tiles fit when $n\sqrt{p} \geq \sqrt{M}$. Prime selection with Strassen costs $O(n^{\log_2 7+3a})$; explicit construction costs $O(n^{2+3a/2})$. At most one successful scan per pair adds $O(n^{2+b})$. All these terms must be checked, not only (7).

Take

$$(c, \theta, e, z) = (21.6844,\ 0.1163026,\ 0.0565835,\ 0.0021252).$$

The three quantities in (7) are respectively $0.002125821042\ldots$, $0.002125622860\ldots$, and 0.0021252, and the encoding saving is $0.002317406777\ldots$. The remaining costs have strict slack as certified by the companion script, including slack for the logarithmically many min-plus and APSP calls. Thus deterministic all-edges Exact Triangle and the preceding min-plus/APSP transfer take $O(n^{2.997875})$ time with one chosen prime; the per-pair conditional moments of Corollary 23 improve this to $O(n^{2.996261})$. The source's Theorem 21(a) gives 3SUM time $n^{2-0.002125/2+o(1)}$, hence the safely rounded $O(n^{1.998938})$. Without folding, the parameters (21.915, 0.116108, 0.0551975, 0.0020952) give $O(n^{2.997905})$ and $O(n^{1.998953})$, as also certified.

Deterministic integer 3SUM

The bucket-pair reduction also derandomizes, with one modulus chosen as in the source's Theorem 17 and the failed scans charged to false positives.

Theorem 16 (Deterministic integer 3SUM)**.** *With the data structure of Theorem 2, choose $D = \Theta(n^e)$, a sub-bucket size $s = \Theta(n^b)$ with $z := e - 2b > 0$, and $R = \lceil n/s \rceil$. Integer 3SUM has a deterministic algorithm with running time*

$$\widetilde{O}\left(n^{1+z/2}D^E + n^{2+z}D^{-G} + n^{2+z/2}D^{q-1/2} + n^{2-z} + n^{1+3e/2+z/2} + n^{2-b} + n^{1+z/2}D\right),$$

provided $n^{1+z/2} \geq D^T \operatorname{poly}(\log D)$.

Proof. Choose the bucket prime $r \in [R, 2R]$ as in Theorem 14, in $O(nR) = O(n^{2-b})$ time, so that $\Phi(r) = O(ns \log n)$. Let $P = \Theta(D)$ with $2P \leq D$. For every prime $p \in [P, 2P]$ compute

$$C(p) = \#\{(x, y, z) \in A \times B \times C :\ x + y \equiv z \pmod{rp}\}$$

by one exact cyclic convolution of length rp of the residue histograms of A and B modulo rp, followed by a sum over $z \in C$; with Kronecker substitution and Schönhage–Strassen multiplication this takes $\widetilde{O}(rp)$ per prime and $\widetilde{O}(Prp) = \widetilde{O}(n^{1-b+2e}) = \widetilde{O}(n^{1+3e/2+z/2})$ in all. Keep a minimizer p among the candidates other than r. Removing at most one prime preserves the asymptotic averaging bound. The exact solutions satisfy every congruence, so $C(p) = F(p) + Z_0$ with Z_0 independent of p and $F(p)$ the

number of false triples, which the minimizer therefore minimizes. A false triple has $x + y \equiv z \pmod r$, hence is one of the $T = \sum_z \sum_i |A_i||B_{z-i}| \le n\Phi(r)/2 = O(n^2 s \log n)$ bucket-compatible triples, and its nonzero value $x+y-z$ has at most $\log(3n^\nu)/\log P$ prime divisors in $[P, 2P]$, which contains $\Omega(P/\log P)$ primes. Averaging gives $F(p) = O(T \log n/P) = O(n^2 s \log^2 n/P)$.

Build the instance of Lemma 13 for (r, p, s), whose outer dimension is $N = O((n/s)\sqrt{p}) = O(n^{1+z/2})$, query the $\widetilde{O}(n^2/s) = \widetilde{O}(n^{2-b})$ compatible sub-queries once, scan a sub-query if its entry is positive, and stop at the first exact solution. Every solution lies in a compatible sub-query with a positive entry, and scans are exact, so the algorithm is correct. A failed scan (z, α, β) contains a false triple (x, y, z) with $x \in \alpha$ and $y \in \beta$, and distinct failed scans contain distinct triples because the sub-buckets partition A and B; so failed scans cost $O(F(p)\, s \log n) = \widetilde{O}(n^2 s^2/P) = \widetilde{O}(n^{2-z})$. The remaining terms are the band encodings ND^E, the boxes $N^2 D^{-G}$, the queries $n^{2-b} D^q = n^{2+z/2} D^{q-1/2}$, writing the matrices ND, and one successful scan. □

Writing δ for the saving below exponent 2, the certificate reads

$$eG - z > \delta, \qquad e(1/2 - q) - z/2 > \delta, \qquad z > \delta, \qquad 1 - eE - z/2 > \delta,$$
$$1 - 3e/2 - z/2 > \delta, \qquad (e - z)/2 > \delta, \qquad 1 - e - z/2 > \delta, \qquad 1 + z/2 - eT > 0.$$

The first three are the Exact Triangle quantities (7); the encoding constraint lies between the unfolded and the folded Exact Triangle versions.

Corollary 17 (Deterministic integer 3SUM)**.** *Take*

$$(c, \theta, e, z) = (21.915,\ 0.116108,\ 0.055255,\ 0.0020975).$$

The savings are $0.002097422\ldots$, $0.002097397\ldots$, 0.0020975, *and* $0.002122108\ldots$ *for boxes, queries, failed scans, and encodings, with the other terms far larger; all exceed* 0.002097. *Integer 3SUM therefore has a deterministic algorithm with running time* $O(n^{1.997903})$. *The source's Theorem 21(a) with the deterministic Exact Triangle bound gives only* $O(n^{1.998938})$.

The supremum of this constraint set, about 0.0020974, is not claimed. This single-modulus construction must absorb all failed scans. The next construction instead chooses several primes by an exact conditional objective, and may certify YES before any scanning.

Deterministic integer 3SUM by conditional modular moments

The preceding single-modulus choice can be strengthened deterministically. The new ingredient is an exact objective for the intersection of several modular filters, allowing a different witness at each prime. Its minimization either gives a bounded scan budget or certifies an affirmative answer.

Theorem 18 (Conditional modular moments)**.** *Fix an integer* $k \ge 1$. *With the data structure of Theorem 2, choose* $D = \Theta(n^e)$, $s = \Theta(n^b)$, *with* $0 < b < e/2 < 1/2$ *and* $z = e - 2b > 0$. *Three-set integer 3SUM has a deterministic decision algorithm with running time*

$$\widetilde{O}\left(n^{1+z/2} D^E + n^{2+z} D^{-G} + n^{2+z/2} D^{q-1/2} + n^{2-kz} + n^{1+(k+1/2)e+z/2} + n^{2-b} + n^{1+z/2} D\right),$$

provided $n^{1+z/2} \ge D^T \operatorname{poly}(\log D)$. *The distinct-element one-set form has the same bound, up to polylogarithmic factors. An actual witness can also be recovered with polylogarithmic overhead.*

Proof. Choose the bucket prime $r \in [R, 2R]$, where $R = \lceil n/s \rceil$, and partition into sub-buckets as in Theorem 16. Thus $\Phi(r) = O(ns \log n)$ and the number Q_0 of compatible sub-queries $u = (w, \alpha, \beta)$ is $\widetilde{O}(n^2/s)$. Let $P = \Theta(D)$ with $2P \le D$, and let $\mathcal{P}$ contain all primes in $[P, 2P]$ except r. Let $U \ge 2$ be an integer bounding every $|x + y - w|$ in the input, and put $\lambda = \lfloor \log_2 U \rfloor + 1$. Every nonzero difference has at most λ distinct prime divisors. These choices are computable from the input; no real-number comparison is needed in the selection rule below. Small sizes with fewer than k available primes may be handled directly.

For each sub-query, define its actual pair count

$$c_p(u) = \#\{(x, y) \in \alpha \times \beta : x + y \equiv w \pmod{p}\}.$$

This counts multiplicities; it need not equal the Boolean product entry of Lemma 13, but it is positive exactly when that entry is positive. For distinct primes $p_1, \dots, p_t$, put

$$J_t = \sum_u \prod_{j=1}^{t} c_{p_j}(u), \qquad J_0 = Q_0.$$

Exact objective computation. Put $M_t = r \prod_{j=1}^{t} p_j$. For each sub-bucket α of A, enumerate every ordered tuple $(x_1, \dots, x_t) \in \alpha^t$, allowing repeated coordinates, and add one to a histogram at the unique CRT residue v mod M_t satisfying

$$v \equiv \mathrm{bucket}(\alpha) \pmod{r}, \qquad v \equiv x_j \pmod{p_j} \quad (1 \le j \le t).$$

Build the analogous histogram for B. The exact cyclic convolution of these two histograms, summed at w mod M_t for $w \in C$, equals J_t. Indeed, the congruence modulo r imposes bucket compatibility, while the other coordinates impose the t modular equations independently. Expanding the product of pair counts makes exactly these independent choices of witnesses inside the same two sub-buckets. This is not a test that a single witness passes all primes.

The number of tuples is at most ns^{t-1} per side. CRT coefficients can be precomputed, and $t \le k$ is fixed. Histogram construction therefore costs $\widetilde{O}(ns^{t-1} + M_t)$. Exact cyclic convolution costs $\widetilde{O}(M_t)$ using Kronecker substitution and integer multiplication, as in the preceding reduction. Its coefficients count at most $n^2 s^{2t-2}$ tuple pairs, and the final objective is at most $n^3 s^{2t-2}$, so their bit lengths are $O_k(\log n)$. Also $M_t = n^{O_k(1)}$; indices, residues, and arithmetic intermediates use $O_k(\log n)$ bits, which require only a constant number of RAM words. Thus the exact arithmetic does not introduce an uncharged power of n.

Greedy choice with an affirmative certificate. Suppose $p_1, \dots, p_{t-1}$ are fixed. Exclude them from $\mathcal{P}$, and let m_t be the number of remaining candidates. Compute $J_t(p)$ for each candidate p and choose a minimizer. If

$$m_t J_t(p) > \lambda s^2 J_{t-1}. \tag{6}$$

return YES. Otherwise retain this prime and continue, for at most k stages. The comparison in (6) is an exact integer comparison.

To justify the affirmative answer, assume that the instance has no exact solution. Every difference in every sub-query is then nonzero, so

$$\sum_{p \text{ remaining}} c_p(u) \le \lambda |\alpha||\beta| \le \lambda s^2.$$

Multiply by the nonnegative weight $\prod_{j<t} c_{p_j}(u)$ and sum over u. The average candidate objective is at most $\lambda s^2 J_{t-1}/m_t$, so its minimum cannot satisfy (6). Hence (6) is a valid certificate that a solution exists. No knowledge of the number of genuine witnesses is assumed.

If all k stages complete, then $m_t = \Omega(P/\log P)$ and $\lambda = O(\log n)$ give

$$J_k \le Q_0 \prod_{t=1}^{k} \frac{\lambda s^2}{m_t} = \widetilde{O}\left(\frac{n^2}{s}\left(\frac{s^2}{P}\right)^k\right).$$

Build the k thin products of Lemma 13 and query every compatible sub-query in each product. Scan a sub-query only if every answer is positive. Each such sub-query contributes a positive integer to J_k, so there are at most J_k scans. Each scan costs $O(s \log n)$, and their total cost is $\widetilde{O}(n^2(s^2/P)^k) = \widetilde{O}(n^{2-kz})$. Every exact solution passes all filters, and every scanned answer is checked in the original integers. Thus the procedure is correct whether it returns through (6) or through scanning. If $J_t = 0$, it may return NO immediately, since a genuine solution would contribute at least one.

Complete cost. Stage t tries $O(P)$ candidates while the earlier primes are fixed. Its cost is

$$\widetilde{O}\left(Pns^{t-1} + rP^{t+1}\right).$$

Since $s \le P$ and $r = \Theta(n/s)$, the second term dominates the first. For fixed k all stages therefore cost $\widetilde{O}((n/s)P^{k+1}) = \widetilde{O}(n^{1+(k+1/2)e+z/2})$. The other costs are exactly the $k = O(1)$ copies of the folded products and queries in Theorem 16, together with bucket selection. Enumeration of the sub-queries and sorting are covered by those terms.

Distinct elements and witnesses. In the one-set form, forbidden repeated-element zero triples would invalidate the NO-instance premise used above. Instead use a deterministic separating family. Give the elements distinct binary indices. Any three distinct indices have three distinct patterns on some two bit positions: choose a bit separating one index from the other two, then a bit separating the latter pair. Enumerate the $O(\log^2 n)$ pairs of positions and all ordered selections of three distinct patterns among their four patterns. Use the corresponding three disjoint classes for $A, B, -C$. Every legal triple occurs in one such instance, and no instance permits a repeated element. If an actual witness is wanted, repeatedly bisect one set and invoke this decision algorithm on the restricted instance known to contain a solution. There are $O(\log n)$ such calls, after which the three singletons give a witness. These overheads are polylogarithmic. □

Corollary 19 (Improved deterministic integer 3SUM)**.** *Take*

$$(k, c, \theta, e, z) = (18,\ 22.4828,\ 0.111021,\ 0.0538428,\ 0.00021113).$$

For the boxes, queries, scans, encodings, and greedy selection, the savings below exponent 2 *are, respectively,*

$$\begin{aligned} eG - z &= 0.003800286893920\ldots, \\ e(1/2 - q) - z/2 &= 0.003800294398088\ldots, \\ kz &= 0.00380034, \\ 1 - eE - z/2 &= 0.003806152261295\ldots, \\ 1 - (k + 1/2)e - z/2 &= 0.003802635. \end{aligned}$$

The other costs and tile condition have strict slack, as certified by exact rational intervals. Consequently integer 3SUM, including its distinct-element one-set form and witness recovery, has deterministic time $O(n^{1.996200})$*. The strict gap absorbs the polylogarithmic factors of the convolutions, the separating family, and the decision self-reduction.*

This new affirmative certificate is a decision argument. It does not by itself give an all-edges Exact Triangle algorithm or improve the stated deterministic APSP exponent. It does, however, apply to Exact Triangle itself.

Deterministic decision Exact Triangle by conditional modular moments

For a pair/block $u = (a, b, B)$ and a prime p write $N_p(u) = \#\{c \in B : \ p \mid S(a, b, c)\}$, the modular-witness count returned by the folded product of Lemma 8. The objective $J_t = \sum_u \prod_{j \le t} N_{p_j}(u)$ counts tuples of witnesses, one per prime, inside the same pair/block, and plays the role of the 3SUM objective.

Theorem 20 (Conditional modular moments for Exact Triangle)**.** *Fix an integer $k \ge 1$ and rationals $\alpha > 0$, $\mu > 0$. Let the tripartite graph have two outer parts of $R = n^\alpha$ vertices and a middle part of $H = n^{\alpha\mu}$ vertices, with integer weights of magnitude $n^{O(1)}$. With the data structure of Theorem 2 put $D = \Theta(n^{\alpha e})$, $P = \Theta(n^{\alpha a})$, $h = \lfloor D/(2P) \rfloor = \Theta(n^{\alpha b})$, $e = a + b$, $z = a - b > 0$, and $b < \mu$. Exact Triangle (decision) has a deterministic algorithm with running time $\widetilde{O}(n^{\alpha(2+\mu)-\alpha\delta})$ for every δ with*

$$eG - z > \delta, \qquad e(1/2 - q) - z/2 > \delta, \qquad kz > \delta, \qquad 1 - e(E - 1/4) - 3z/4 > \delta,$$
$$1 - 3e/4 - 3z/4 > \delta, \qquad 1 - a > \delta, \qquad 1 + a/2 > eT,$$

together with the selection condition

$$0.628\,\mu - 0.372\,(k-1)b - (k+1)a > \delta \ \text{ if } \mu + (k-1)b \le 1, \qquad 0.628 - (k-1)b - (k+1)a > \delta \ \text{ otherwise,}$$

where 0.628 and 0.372 stand for $3 - \omega$ and $\omega - 2$ with the source's bound $\omega < 2.372$.

Proof. Partition the middle part into $\lceil H/h \rceil$ blocks of at most h vertices; there are $Q_0 = R^2 \lceil H/h \rceil$ pair/blocks. Let $\mathcal{P}$ be the primes in $[P, 2P]$, m_t the number still available at stage t, and $\lambda = \lfloor \log_2(3n^{O(1)}) \rfloor + 1$ a bound on the number of distinct prime divisors of any nonzero sum $S(a, b, c)$.

Exact objective. For fixed $p_1, \dots, p_{t-1}$ and a candidate p, $J_t(p) = \sum_u \prod_{j<t} N_{p_j}(u)\, N_p(u)$ is the number of tuples $(a, b, c_1, \dots, c_t)$ with all c_j in one block and $p_j \mid S(a, b, c_j)$ for every j (with $p_t = p$). Over the ring $\mathcal{R}_t = \mathbb{Z}[x_1, \dots, x_t]/(x_1^{p_1} - 1, \dots, x_t^{p_t} - 1)$, let $\Pi[a, (B, c_1, \dots, c_t)] = \prod_j x_j^{w(a,c_j)}$ and $\Sigma[(B, c_1, \dots, c_t), b] = \prod_j x_j^{w(c_j,b)}$, indexed by tuples inside a block; then $J_t(p)$ is the sum over (a, b) of the coefficient of $\prod_j x_j^{-w(a,b) \bmod p_j}$ in $(\Pi\Sigma)[a, b]$, exactly as in the source's Theorem 17 for one prime. The middle dimension is $M_t = \lceil H/h \rceil h^t = \widetilde{\Theta}(Hh^{t-1})$, a ring multiplication costs $\widetilde{O}(\prod_j p_j)$ by Kronecker substitution to one integer product, and the matrix product is blocked into square products of order $\min(R, M_t)$: $R^2 M_t^{\omega-2}$ ring operations when $M_t \le R$ and $M_t R^{\omega-1}$ otherwise. With $O(P)$ candidates, stage t costs $\widetilde{O}(P^{t+1} R^2 M_t^{\omega-2})$ or $\widetilde{O}(P^{t+1} M_t R^{\omega-1})$, dominated by $t = k$; in exponents of n^α this is $2 + \mu + 0.372(\mu + (k-1)b) + (k+1)a - \mu$ in the first regime and $\mu + (k-1)b + 1.372 + (k+1)a$ in the second, giving the two selection conditions against the budget $2 + \mu$. Coefficients count at most $R^2 H^k h^{k-1}$ tuples and all moduli are $n^{O(1)}$, so every integer has $O_k(\log n)$ bits.

Greedy choice and certificate. At stage t choose a minimizer p_t of J_t. If the instance has no zero triangle, every u has $\sum_{p \in \mathcal{P} \text{ remaining}} N_p(u) \le \lambda h$, so summing with the nonnegative weights $\prod_{j<t} N_{p_j}(u)$ gives $\sum_p J_t(p) \le \lambda h J_{t-1}$ and $m_t J_t(p_t) \le \lambda h J_{t-1}$. If instead $m_t J_t(p_t) > \lambda h J_{t-1}$, return YES; this is a valid certificate. If $J_t = 0$, return NO. Otherwise, after k stages, $J_k \le Q_0 \prod_t (\lambda h / m_t) = \widetilde{O}(Q_0 (h/P)^k)$.

Filtering. For each chosen prime and block build the folded instance of Lemma 8, with outer dimension $R\lceil \sqrt{p} \rceil$ and inner dimension at most $2Ph \le D$; the tile condition is $R\sqrt{P} \ge \sqrt{M}$, that is, $1 + a/2 > eT$. Query every pair once per prime and block, scan a pair/block with the original integer sums only if all k answers are positive, and stop at the first zero. A genuine zero triangle survives every filter, so the output is correct. Each scanned pair/block contributes at least one to J_k, so the scans cost $O(hJ_k) = \widetilde{O}(R^2 H (h/P)^k) = \widetilde{O}(n^{\alpha(2+\mu)-\alpha k z})$.

Remaining costs. The k data structures per block cost

$$\widetilde{O}\Big(\frac{H}{h}(R\sqrt{P})^2 D^{-G}\Big), \quad \widetilde{O}\Big(\frac{H}{h} R\sqrt{P}\, D^E\Big), \quad \widetilde{O}\Big(\frac{H}{h} R^2 D^q\Big), \quad \widetilde{O}\Big(\frac{H}{h} R\sqrt{P}\, D\Big)$$

for boxes, encodings, queries, and writing the matrices; computing the residues $w(a,b) \bmod p$ costs $O(kR^2)$. In exponents of n^α these give the stated savings $eG-z$, $1-e(E-1/4)-3z/4$, $e(1/2-q)-z/2$, $1-3e/4-3z/4$, and $1-a$, as in Theorem 16. The condition $b<\mu$ ensures that blocks fit in the middle part. □

Corollary 21 (Deterministic Exact Triangle)**.** *Take $\alpha=\mu=1$, $k=11$, and*

$$(c,\theta,e,z) = (21.62,\ 0.11216,\ 0.05671,\ 0.00034).$$

The savings for boxes, queries, failed scans, encodings, and selection (second regime) are

$$0.003743746\ldots,\quad 0.003744811\ldots,\quad 0.00374,\quad 0.004385942\ldots,\quad 0.00385,$$

with the other terms and the tile condition far from binding; all exceed 0.003739*. Integer Exact Triangle is therefore decided deterministically in $O(n^{2.996261})$ time, and a zero triangle is found by $O(\log n)$ decision calls on halves of one part, also in $O(n^{2.996261})$ time. The source's Theorem 19 gives $O(n^{3-0.00175}\log n) \le O(n^{2.9983})$, and the single-prime deterministic all-edges bound above gives $O(n^{2.997875})$.*

Numerical exploration found savings near 0.003746 at $k=11$ in this restricted family; this is not a proved supremum. Selection costs grow with P^{k+1}. The aggregate certificate decides Exact Triangle only. Applying the same threshold to each outer pair separately localizes zeros to pairs at no extra cost.

Theorem 22 (Per-pair conditional moments)**.** *Under the hypotheses of Theorem 20, all-edges Exact Triangle, which asks for every outer pair (a,b) whether some middle vertex c has $S(a,b,c)=0$, has a deterministic algorithm with the same running time $\widetilde{O}(n^{\alpha(2+\mu)-\alpha\delta})$.*

Proof. Write $\pi=(a,b)$ for an outer pair, $Z(\pi,B)=\#\{c\in B:\ S(a,b,c)=0\}$, $\rho_t=\lambda h/m_t$, and

$$J_t(\pi;p)=\sum_B \prod_{j<t} N_{p_j}(\pi,B)\, N_p(\pi,B), \qquad J_{t-1}(\pi)=J_{t-1}(\pi;p_{t-1}), \qquad J_0(\pi)=\lceil H/h\rceil,$$

so that $J_t(p)=\sum_\pi J_t(\pi;p)$. The product in the proof of Theorem 20 extracts the coefficient $J_t(\pi;p)$ of every pair separately before summing, for every remaining candidate p, so the whole table $(J_t(\pi;p))_{\pi,p}$ is available at the stated selection cost.

Maintain a set U_t of undecided pairs, with U_0 all R^2 pairs. At stage t declare YES every $\pi\in U_{t-1}$ with

$$\min_p J_t(\pi;p) > \rho_t\, J_{t-1}(\pi),$$

let U_t be the other pairs of U_{t-1}, and choose p_t minimizing $\sum_{\pi\in U_t} J_t(\pi;p)$ over the remaining candidates.

Soundness. If $Z(\pi,B)=0$ for every block, then $\sum_p N_p(\pi,B)\le \lambda h$ for every block, so $\sum_p J_t(\pi;p)\le \lambda h J_{t-1}(\pi)$ and $\min_p J_t(\pi;p)\le \rho_t J_{t-1}(\pi)$. Every declared pair therefore has a zero triangle.

Leak control. Put $\Gamma_{t-1}(\pi)=\sum_B Z(\pi,B)\prod_{j<t} N_{p_j}(\pi,B)$. Since $N_p(\pi,B)\ge Z(\pi,B)$ term by term, $\Gamma_{t-1}(\pi)\le J_t(\pi;p)$ for every candidate p; hence $\Gamma_{t-1}(\pi)\le \rho_t J_{t-1}(\pi)$ for every $\pi\in U_t$.

Contraction. A zero is divisible by all m_t candidates and a nonzero sum by at most λ of them, so $\sum_p N_p(\pi,B)\le m_t Z(\pi,B)+\lambda h$, and for $\pi\in U_t$

$$\frac{1}{m_t}\sum_p J_t(\pi;p) \le \Gamma_{t-1}(\pi)+\rho_t J_{t-1}(\pi) \le 2\rho_t J_{t-1}(\pi).$$

The minimizer is at most the candidate average, so $\sum_{\pi\in U_t} J_t(\pi;p_t)\le 2\rho_t \sum_{\pi\in U_{t-1}} J_{t-1}(\pi)$, and after k stages $\sum_{\pi\in U_k} J_k(\pi)\le 2^k Q_0\prod_t \rho_t = \widetilde{O}(Q_0(h/P)^k)$ for fixed k.

Filtering. Build the k folded data structures per block as before, query every pair/block with $\pi \in U_k$, scan a pair/block with the original sums only if all k counts are positive, and stop at the first zero of each pair. A pair of U_k whose scans find no zero is NO, since a genuine zero passes every filter; so all R^2 answers are exact. Each scanned pair/block contributes at least one to $\sum_{\pi \in U_k} J_k(\pi)$, so the scans cost $O(h\, 2^k Q_0 (h/P)^k \operatorname{polylog}) = \widetilde{O}(n^{\alpha(2+\mu)-\alpha k z})$; the constant 2^k is absorbed by the strict gap $kz > \delta$. No other cost changes. □

The decision theorem certifies the aggregate; the per-pair thresholds use the same table and the same λ, h, and m_t, and the uncertified mass contracts by $2\rho_t$ instead of ρ_t.

Corollary 23 (Deterministic all-edges Exact Triangle, min-plus product, and APSP)**.** *With $\alpha = \mu = 1$ and the $k = 11$ witness of the preceding corollary, integer all-edges Exact Triangle takes $O(n^{2.996261})$ deterministic time. The bit-scaling transfer of the subsection on integer APSP makes two all-edges calls per bit level and $O(\log n)$ min-plus squarings, all deterministic, and the strict gap $0.00374 > 0.003739$ absorbs the logarithmic factors. Hence integer min-plus product and APSP take $O(n^{2.996261})$ deterministic time, improving $O(n^{2.997875})$. For the lopsided splits of Corollary 28 below the same transfer computes the rectangular min-plus product, so Min- and Max-Weight k-Clique with polynomially bounded integer weights take $O(n^{k-0.003739\lfloor k/3 \rfloor})$ deterministic time for every fixed $k \geq 3$, and $O(n^{4.99276})$ for $k = 5$ through the split $(2, 2, 1)$.*

Sharper deterministic square products

Lemma 24 (Rectangular conditional-moment selection)**.** *Let $w(\kappa)$ be a certified upper bound on $\omega(1, \kappa, 1)$ over $\mathbb{Q}$. In the square outer-parts case of Theorem 22, the selection condition can be replaced by*

$$3 - w(1 + (k-1)b) - (k+1)a > \delta.$$

Here an arbitrarily small fixed exponent allowance for matrix multiplication must fit within the displayed strict gap. Neither the per-pair certificate nor its contraction changes.

Proof. At stage t, a candidate prime requires multiplication of an $n \times M_t$ matrix by an $M_t \times n$ matrix over the existing cyclic polynomial ring, where $M_t = \widetilde{O}(nh^{t-1})$. Pad every middle dimension to $\widetilde{O}(nh^{k-1})$. There are $O(P)$ candidates, and each ring operation costs $\widetilde{O}(P^k)$ bit operations. Thus all fixed k stages cost

$$\widetilde{O}\big(P^{k+1} n^{w(1+(k-1)b)+\varepsilon}\big).$$

The monomial-matrix construction is covered by this bound since the matrix multiplication exponent covers reading both inputs. Coefficient extraction and all per-pair comparisons are covered as well.

For completeness the field algorithm can be used over this ring without assuming divisions by ring elements. Fix an exact bilinear matrix multiplication algorithm over $\mathbb{Q}$ at a sufficiently large but constant base size to realize the chosen exponent allowance, then recurse. Its only divisions are by its fixed nonzero rational constants, which are invertible in the rational cyclic-polynomial algebra. Clearing those fixed denominators level by level gives common denominators of size $n^{O(1)}$. The linear encodings and recombinations at each of $O(\log n)$ levels amplify coefficient magnitudes by only a fixed constant per level; the bilinear multiplication adds at most a polynomial number of products. Consequently intermediate numerators and denominators have $O(\log n)$ bits for fixed k and fixed base algorithm. The exact Kronecker substitution bound for a ring multiplication therefore remains $\widetilde{O}(P^k)$, including bit complexity. The final extracted coefficients are the same nonnegative integers as in the original selector. □

Published rectangular bound and interpolation. Table 1 of Vassilevska Williams, Xu, Xu, and Zhou, *New Bounds for Matrix Multiplication: from Alpha to Omega* (SODA 2024; arXiv:2307.07970v2), gives

$$\omega(1, 1.2, 1) \le 2.535063, \qquad \omega(1, 1.5, 1) \le 2.794941.$$

These are external matrix-multiplication bounds. Our local rational certificate checks their subsequent use, not their source tensor analysis. For $1.2 \le \kappa \le 1.5$, tensor products of the two algorithms give

$$w(\kappa) = 2.535063 + \frac{2.794941 - 2.535063}{1.5 - 1.2}(\kappa - 1.2) = 2.535063 + 0.86626(\kappa - 1.2).$$

Indeed, tensoring rational proportions $1-\lambda$ and λ of fixed base algorithms gives outer dimensions with exponent one and middle exponent $(1-\lambda)1.2 + \lambda 1.5$, while the operation exponent is the same convex combination. Rational proportions suffice for our rational κ; fixing the base large enough absorbs each algorithm's exponent allowance. This is an upper bound by interpolation, not interpolation of unproved numerical optima.

Theorem 25 (Batched moments with deterministic lazy encodings)**.** *Let $k \ge 1$ be fixed, $c > 10$, $0 < r < \alpha < \theta$, $r < 1 - \theta$, and $\theta/(1-r) < 6/7$. Here α is the encoding-pruning fraction, not the outer-size parameter of Theorem 20; both outer parts now have size n, as does the middle part. Put*

$$q_* = \max(q_1, q_2, q_{\text{lazy}}), \quad A_* = A_0(c) - G(c, \alpha), \quad 0 < z < e, \quad a = (e+z)/2, \quad b = (e-z)/2 < 1.$$

Let w be a valid rectangular multiplication bound as in Lemma 24. Integer all-edges Exact Triangle, min-plus product, and APSP have deterministic time $O(n^{3-\delta})$ whenever, for some fixed $\varepsilon > 0$,

$$\begin{gathered} eG(c,\theta) - z > \delta, \qquad e(1/2 - q_*) - z/2 > \delta, \qquad kz > \delta, \\ 1 - e(A_* - 1/4) - 3z/4 > \delta, \qquad 3 - w(1 + (k-1)b) - (k+1)a - \varepsilon > \delta, \\ 1 - 3e/4 - 3z/4 > \delta, \qquad 1 - a > \delta, \\ 1 + a/2 > eA_0(c), \qquad 1 - b > eC_K(c). \end{gathered}$$

APSP retains the no-negative-cycle assumption and integer weights have polynomial magnitude.

Proof. Run the per-pair certificates and common-prime minimization of Theorem 22. Lemma 24 bounds the selection work by the fifth inequality. All k primes are chosen deterministically, and the final scans cost $\widetilde{O}(n^{3-kz})$ by the same $2\rho_t$ contraction.

For each chosen prime $p \in [P, 2P]$, residue folding gives $B = \lceil n/h \rceil = \Theta(n^{1-b})$ products, each with outer dimension $N = n\lceil\sqrt{p}\rceil = \Theta(n^{1+a/2})$ and common inner dimension $D = \Theta(n^e)$, padded to a power of four. Pad missing matrix entries with zero indicators. Apply Lemma 4 to these B products, with the deterministic representation of Lemma 7. The final two inequalities ensure $N \ge N_0$ and $B \ge K$, including rounding. Reducing the encoding budget does not reduce N_0 or K. The encodings, boxes, and all prefix-contracted queries cost, respectively,

$$\widetilde{O}(BND^{A_*}), \qquad \widetilde{O}(BN^2D^{-G(c,\theta)}), \qquad \widetilde{O}(Bn^2D^{q_*}).$$

Their exponents are $2-b+a/2+eA_*$, $3-b+a-eG(c,\theta)$, and $3-b+eq_*$. Subtracting from 3 gives the asserted savings. Each pair is queried once per block and prime at its folded position. Matrix formation costs $\widetilde{O}(BND)$, with saving $1 - 3e/4 - 3z/4$; the earlier sufficient residue allowance $1-a$ also holds. All input formation, stored intermediate arrays, sparse retrievals, and cyclic rotations are included in these terms. Every gap is strict, so polylogarithmic overhead, the min-plus bit-scaling calls, and APSP squarings fit the claimed ordinary $O(n^{3-\delta})$ bound. □

Corollary 26 (Deterministic APSP with the SODA 2024 bounds)**.** *Use the displayed interpolation from arXiv:2307.07970v2 and take*

$$k = 11, \quad c = 21.8429984, \quad \theta = 0.1125375917,$$
$$r = 0.0254666251, \quad \alpha = 0.0478165400,$$
$$e = 0.0613141114, \quad z = 0.0003737179, \quad \varepsilon = 10^{-8}.$$

Then $a = 0.03084391465$, $b = 0.03047019675$, *and* $\kappa = 1 + (k-1)b = 1.3047019675$. *The exact interval certificates give*

cost term	*saving below exponent* 3
boxes	0.004110896862224...
residual queries	0.004110896885301...
stored-box queries	0.004110896880317...
lazy retrievals (certified lower bound)	0.004110898310961...
scans	0.004110896900000
encodings	0.004110899160297...
selection, including 10^{-8}	0.004110887833450

The outer-band and product-capacity slacks are, respectively,

$$0.002654752597761\ldots \qquad \textit{and} \qquad 0.789932983443956\ldots.$$

All exceed their required thresholds, so $\delta = 0.0041108$ *yields*

$$\boxed{\textit{deterministic integer APSP/min-plus/all-edges Exact Triangle in } O(n^{2.9958892}).}$$

The lazy-query certificate bounds its small exponential tail by $3^{-X} < 3^{-21}$, *where* $X = c(1-\alpha)/(1-r)^2 > 21$, *before applying rational interval logarithms. This improves* 2.996261 *without using the October 2026 rectangular theorem discussed next.*

Corollary 27 (A conditional matrix-multiplication dependency)**.** *Assume the rectangular matrix-multiplication bound of Uznański [5], Corollary 1.2:*

$$\omega(1, \kappa, 1) \le w(\kappa) := 1 + \kappa + \frac{1}{4\kappa} \qquad (\kappa \ge 1/2).$$

Then the algorithm of Theorem 25, with Lemma 7, gives deterministic all-edges Exact Triangle, min-plus product, and APSP on polynomially bounded integer weights in $O(n^{2.9958323})$ *time. This additional bound is recorded separately from the preceding result because it uses a different, recent external theorem.*

Proof. Only the moment-selector exponent changes. Take

$$k = 13, \quad c = 21.4287303, \quad \theta = 0.1128443719, \quad r = 0.0255360478,$$
$$\alpha = 0.0483646500, \quad e = 0.0625769172, \quad z = 0.00032059682, \quad \delta = 0.0041677.$$

Thus $a = (e+z)/2 = 0.03144875701$, $b = (e-z)/2 = 0.03112816019$, and $\kappa = 1 + (k-1)b = 1.37353792228$. Reserve $\varepsilon = 10^{-8}$ in the matrix-multiplication exponent. The selector saving is

$$3 - w(\kappa) - (k+1)a - \varepsilon = 0.004167749375717\ldots > \delta.$$

The other cost and geometry inequalities are unchanged. In particular, $X = c(1-\alpha)/(1-r)^2 > 21$, so the lazy-query tail is bounded exactly using $3^{-X} < 3^{-21}$ and rational interval logarithms. Both

prefix-query terms, this lazy-query upper bound, boxes, scans, encodings, matrix formation, and both layout conditions have strict rational certificates at the displayed parameters.

The external paper uses arithmetic exponents with an arbitrarily small positive allowance; its Lemma 6.1 fixes a finite exact bilinear identity before recursing, and it states characteristic-zero validity. The exact polynomial-algebra and bit-complexity transfer in Theorem 25 therefore applies. Equivalently, a fixed complex rank identity for a rational tensor has an algebraic solution to its coefficient equations; all constants then lie in one fixed number field. Representing that field in a rational basis gives fixed-size coefficient operations, with $O(\log n)$-bit numerators and denominators through the recursion. No variable polynomial-ring element is inverted. The strict gaps absorb the remaining logarithmic factors in filtering, bit scaling, and repeated min-plus squaring. □

Corollary 28 (Deterministic zero-weight k-Clique decision)**.** *Split the k parts into groups of $\alpha = \lfloor k/3 \rfloor$, α, and $m = k - 2\alpha \geq \alpha$ parts. Then $\mu = m/\alpha \geq 1$, the second selection regime applies, and every saving of Corollary 23 scales by α. Zero-Weight k-Clique, decision and search, with polynomially bounded integer weights therefore takes $O(n^{k-0.003739\lfloor k/3 \rfloor})$ deterministic time for every fixed $k \geq 3$, which improves the deterministic zero-weight bound of Corollary 54 for every $k \geq 3$. For $k = 5$ the split $(2, 2, 1)$, with $\mu = 1/2$, $k' = 8$ stages, and*

$$(c, \theta, e, z) = (23.04,\ 0.11108,\ 0.053230,\ 0.0004529),$$

has savings per unit of α equal to $0.003624049\ldots$, $0.003625265\ldots$, 0.0036232, $0.004230855\ldots$, and $0.003711165\ldots$ for boxes, queries, failed scans, encodings, and selection in the first regime, with the graph construction $O(R^2 + RH)$ saving μ; all exceed 0.00362, so Zero-Weight 5-Clique is decided in $O(n^{4.99276})$ time. The minimum- and maximum-weight versions follow from the per-pair theorem above.

Earlier real-input comparison bound

Use the source's Lemmas 36–37. Set comparison-list length $d = n^u$ and bucket size $s = n^{1+u-e}$, so the matrix middle dimension is $O(n^e)$ when $u < e$. One comparison count costs, up to polylogarithms,

$$n^{2-eG} + |\mathcal{P}|n^{eq} + n^{2+2u-e}.$$

The last term comes from $nds + ds^2$; its second summand is smaller since $u < e$. The explicit input construction and sorting terms are smaller at the parameters below.

For real 3SUM, the source reduction has $|\mathcal{P}| = O(n^{2-u})$ and polylogarithmically many such calls. For real min-plus/APSP, multiplying the per-count costs by the source's list and call counts gives the same saving below exponent 3 rather than 2. In both cases the saving is

$$\min\{eG,\ u - eq,\ e - 2u\}.$$

Take $c = 20.276$, $\theta = 0.104566$, $e = 0.059612$, and $u = e(1+q)/3$. The first saving is $0.003680563566\ldots$ and the other two are $0.003680624519\ldots$, all strictly above 0.003680. The encoding saving $1 - eE$ also exceeds this value. The Las Vegas guarantees of the source reductions are preserved, yielding expected $O(n^{1.99632})$ real 3SUM and $O(n^{2.99632})$ real APSP/min-plus.

Real inputs: rank-prefix hashing and complementary certification

The same-block enumeration in the comparison-count reduction can be removed by changing its algorithm. The replacement below returns exact counts, with randomness affecting only its running time. It uses only comparisons on the real inputs: all subsequent hashing acts on their polynomially bounded integer ranks.

Lemma 29 (Certified comparison counts)**.** *There are n row lists and n column lists, each of length at most $1 \le d \le n$, and a prescribed set $\mathcal{P} \subseteq [n]^2$ of list pairs. Suppose either (a) every list has at most one occurrence of each of at most d colors, or (b) all occurrences have a single color. Put $\ell = \lceil \log_2(2nd+1) \rceil$ and let D be the least power of four at least $2d^2\ell^5$. Subject to the tiling condition of Theorem 2, the number of same-color pairs $x < y$ for every $(r, c) \in \mathcal{P}$ can be computed in Las Vegas expected time*

$$\widetilde{O}\left(nD^E + n^2D^{-G} + |\mathcal{P}|D^q + nD + nd\right).$$

The same holds for $x \le y$. Here $D = \widetilde{O}(d^2)$ and also $D \ge d^2$.

Proof. Sort all occurrences of each color separately, assigning distinct ranks from 0 to one less than the number of occurrences of that color. For strict comparisons, break ties by putting column occurrences before row occurrences; ties within one side are arbitrary. Thus a row occurrence x and column occurrence y of the same color satisfy $x < y$ exactly when their ranks satisfy $r(x) < r(y)$. Every rank is less than $2nd$ and has a representation in ℓ bits, padded with leading zeros.

For bit position j, counted from the most significant end, write $h_j(x)$ for the prefix preceding that bit and $b_j(x)$ for that bit. Since distinct occurrences have distinct ranks, exactly one first differing bit exists for every row–column occurrence pair. Therefore

$$[r(x) < r(y)] = \sum_{j=1}^{\ell} [h_j(x) = h_j(y),\ b_j(x) = 0,\ b_j(y) = 1],$$

and the reversed bit orientation gives $[r(x) > r(y)]$, corresponding to $x \ge y$. Write $C_<(r, c)$ and $C_\ge(r, c)$ for the two exact counts. Their sum is the number $T(r, c)$ of same-color occurrence pairs.

In case (b), $T(r, c) = |\mathcal{R}_r|\,|\mathcal{C}_c|$. In case (a), compute T at the prescribed positions by multiplying the two color-incidence matrices. Their inner dimension is at most d; zero-pad it to the full D before applying Theorem 2. This costs one copy of the claimed preprocessing and query bound. Padding to D, rather than charging the box term at d, is important here.

Set $H = d$ in case (a), and $H = d^2$ in case (b), and sieve the primes in $[P, 2P]$, where $P = H\ell^4$. For a sampled prime p, replace prefix equality by equality modulo p. More explicitly, for the first orientation, index the middle coordinate by (k, j, σ), where k is the color, $1 \le j \le \ell$, and $0 \le \sigma < p$, and set

$$X_<[r, (k, j, \sigma)] = \#\{x \in \mathcal{R}_r : \chi(x) = k,\ b_j(x) = 0,\ h_j(x) \equiv \sigma \pmod p\},$$
$$Y_<[(k, j, \sigma), c] = \#\{y \in \mathcal{C}_c : \chi(y) = k,\ b_j(y) = 1,\ h_j(y) \equiv \sigma \pmod p\}.$$

Reverse the bit choices to form $X_\ge, Y_\ge$. The inner dimension is at most $d\ell p \le 2d^2\ell^5$ in case (a), and at most $\ell p \le 2d^2\ell^5$ in case (b); pad to D. Each matrix entry is at most d. The two products count the modular matches, so their answers $U_{<,p}$ and $U_{\ge,p}$ never underestimate $C_<$ and $C_\ge$. Genuine prefix equalities always survive; a false modular equality contributes only a nonnegative extra term. Each requested list pair is queried once in each product, with no factor of p.

For a fixed list pair, there are at most H same-color occurrence pairs, hence at most $H\ell$ possible false prefix equalities over both orientations. Every nonzero prefix difference has magnitude less than $2nd$, and is divisible by at most $\log(2nd)/\log P$ primes at least P. Since $[P, 2P]$ contains $\Omega(P/\log P)$ primes, the probability that any false contribution occurs for this list pair is

$$O\left(\frac{H\ell \log(2nd)}{P}\right) = O(\ell^{-2}) \le \frac{1}{2}$$

for all sufficiently large n. Thus one independently sampled prime makes both directional answers exact with probability at least $1/2$. This bound is uniform over the input and the prescribed pairs.

Keep, separately for each pair and direction, the minimum upper count seen so far. Certify a pair as soon as its two running minima sum to $T(r, c)$. Both minima are upper bounds on nonnegative exact counts whose sum is $T(r, c)$, so this equality proves that both answers are exact. The two minima need not come from the same prime. Continue with independent primes until every pair certifies. After t rounds, a fixed pair remains uncertified with probability at most 2^{-t}; a union bound and a geometric-tail sum give $O(\log(|\mathcal{P}| + 2))$ expected rounds, without requiring independence between pairs. The algorithm terminates almost surely and never returns an incorrect count.

Sorting costs $\widetilde{O}(nd)$, sieving costs $\widetilde{O}(P) \le \widetilde{O}(D)$, and explicitly forming both matrices costs $\widetilde{O}(nD)$ per round. Theorem 2 gives the remaining terms. Multiplication, hashing, counters, and ranks use polynomially bounded integers. For $x \le y$, put row occurrences before column occurrences when values tie. The identical construction then counts $\le$ and its complement $>$, and the same total T certifies them. Finitely many small input sizes can be handled directly. □

Apply the source's Lemma 36 with

$$(c, \theta, e) = (21.2733,\ 0.111731,\ 0.056846), \qquad d = \max\{1, \lfloor n^{e/2} \rfloor\}.$$

The actual middle dimension is the rounded D of Lemma 29, so $D = \Theta(n^e \log^5 n)$, not literally n^e. The strict tiling inequality $eT < 1$ absorbs this polylogarithmic factor and all rounding. All oracle calls use fresh prime samples, including when their lists and pair sets depend on earlier calls.

For real 3SUM, the source gives polylogarithmically many comparison instances of type (b), with $|\mathcal{P}| = O(n^2/d)$ and $\widetilde{O}(n^2/d + nd)$ further work. Their total expected cost is

$$\widetilde{O}\left(nD^E + n^2D^{-G} + n^2D^{q-1/2} + nD + n^2D^{-1/2} + nd\right).$$

For real min-plus, APSP, and Exact Triangle, there are $\widetilde{O}(n/d)$ counting calls. Each has d comparison instances of type (a), whose prescribed pair-set sizes sum to n^2. List formation costs $O(nd^2)$ per counting call. Hence the complete expected bound is

$$\widetilde{O}\left(n^2D^E + n^3D^{-G} + n^3D^{q-1/2} + n^2D + n^2d + n^3/d\right).$$

The existing unfolded parameter witness gives

$$eG = 0.003999294616\ldots, \quad e(1/2 - q) = 0.003999284085\ldots, \quad 1 - eE = 0.004000455769\ldots.$$

All three are strictly greater than 0.003999; the other displayed terms have larger savings, and $1 - eT = 0.004000455769\ldots > 0$. These strict gaps absorb all logarithms, including the expected number of certification rounds. The source's reductions and the replacement oracle are both Las Vegas. Consequently real 3SUM takes expected $O(n^{1.996001})$ time, and real min-plus, APSP, and Exact Triangle take expected $O(n^{2.996001})$ time, in the same model allowing only comparisons, additions, and subtractions on real inputs.

6 Five-term encoding and composite incidence certificates

This section uses an additional source, Majdoddin [6], Sections 3–7: the five-term partial-product identity, deterministic re-indexing maps, and squared embedding of Theorem 5.1. We modify its encoding and query algorithms and use its composite-modulus incidence certification from Section 15.4. The dependency is separate from the Alman–Vassilevska Williams construction used above. In particular, the independent-product batching lemma for the ten-term tensor is not assumed for this five-term construction. We reproduce the changed cost arguments and the certification argument below.

Put $D = 4^m$, $L = \lceil cm \rceil$, $c > 5$, $K = \binom{L}{m}$, $N_0 = 4^{L-m}$, and

$$T_5 = c - 1 + \frac{c\mathcal{H}(1/c)}{\ln 4}, \qquad G_5(c,x) = \frac{c\mathcal{H}(1/c) - c\mathcal{H}((1-x)/c) - x\ln 4}{\ln 2}.$$

The source's squared embedding has two halves of L coordinates, m inner coordinates in each half, $K_{\mathrm{g}} = \Theta(K)$ good sets, and $2m+1$ degree-split runs. A tile computes the cross products of K_{g} left and K_{g} right blocks, each of outer size N_0. Its fit condition is $N \geq KN_0 = D^{T_5}\,\mathrm{poly}(m)$. It does not provide K_{g}^2 independent products. The source's re-indexing maps cost $K(2^m + 2^{L-m})\,\mathrm{poly}(L)$ once; this is lower order under the stated fit condition.

Lemma 30 (Compact five-term prefix data structure)**.** *Fix $0 < r \leq \theta$, $r + \theta < 1$, and $\theta/(1-r) < 3/4$. With $t = \lceil 2\theta m \rceil$ and $\ell = \lfloor rm \rfloor$, the squared five-term construction has preprocessing time and space*

$$\widetilde{O}\big(ND^{E_5} + N^2 D^{-G_5(c,\theta)}\big), \qquad E_5 = T_5 - G_5(c,r),$$

and individual exact-integer query time $\widetilde{O}(D^{q_5})$, where

$$\begin{aligned} q_5 &= \max(q_{51}, q_{52}), \\ q_{51} &= \frac{2(1-r)\mathcal{H}(\theta/(1-r)) + 2\theta\ln 3 + r\ln 8}{\ln 4}, \\ q_{52} &= \frac{2(1-r)\mathcal{H}(\theta/(1-r)) + 2\theta\ln 4}{\ln 4}. \end{aligned}$$

The original fit condition $N \geq KN_0$ remains necessary.

Proof. Label the four nonprivate terms $11, 12, 21, 22$ and the private term 0. In the first half their left forms are

$$f_{11} = x_1 + p_1, \quad f_{12} = x_1 - p_2, \quad f_{21} = x_2, \quad f_{22} = x_2 + p_2, \quad f_0 = -x_1 - x_2;$$

the right forms are $g_{11} = y_1$, $g_{12} = y_2$, $g_{21} = y_1 - q$, $g_{22} = y_2 + q$, $g_0 = y_1 + y_2$. The second half exchanges left and right. Their sum is the inner kernel $I = p_1 y_1 + p_2 q$, and the private product is $P = f_0 g_0$. The identities

$$p_1 = f_{11} - f_{12} - f_{22} + f_{21}, \quad p_2 = f_{22} - f_{21}, \qquad y_1 = g_{11}, \quad q = g_{11} - g_{21}$$

compute an I contraction from four nonprivate left forms and two nonprivate right forms. Successive local transforms shrink each dimension to two; their work is polynomial in L times the number of source entries.

Choose the first ℓ inner coordinates of each half as a balanced prefix F. To make it an initial segment in the box ordering, preprocess boxes for all $(L+1)^2$ orders determined by two physical cuts: coordinates up to the respective cuts come first, and all others follow. A query uses the cuts at its ℓ-th inner coordinates. This changes cost by only a polynomial factor. The source's balanced boxes have at most m private or star symbols per half, at most $2m - t$ in total, and all stars before private symbols in the chosen order. Replacing the last star by any of five terms preserves these conditions, so the full dynamic program is closed. A leaf has at most $2m+1$ preimages for each order.

The prefix identity proved in Lemma 5, with $2m$ inner coordinates and four nonprivate labels, reads

$$I^F \sum_{\substack{A \subseteq Q \setminus F \\ |A| < t}} (-1)^{t-|A|-1} \binom{2m - 2\ell - |A| - 1}{t - |A| - 1} I^A P^{(Q \setminus F) \setminus A}$$

for the residual, plus $4^t\binom{2m-2\ell}{t}$ stored boxes. If A has a_1, a_2 positions in the two halves, its input access costs are bounded by $\mathrm{poly}(L)8^\ell(4^{a_1}2^{a_2} + 2^{a_1}4^{a_2})$. Since $(1+4x)(1+2x)$ is coefficientwise bounded by $(1+3x)^2$, all residuals together cost at most

$$\mathrm{poly}(L)8^\ell \sum_{j<t} \binom{2(m-\ell)}{j} 3^j.$$

The cutoff assumption and entropy estimates give q_{51} and q_{52}.

Retain only encoded leaves with private counts j_1, j_2 satisfying

$$j_1 \le m, \qquad j_2 \le m, \qquad j_1 + j_2 \le 2(m-\ell).$$

Every residual fetch is nonprivate at all 2ℓ prefix positions; every box and intermediate box has at most $2m - t \le 2(m-\ell)$ private or star positions. Thus all required leaves are retained. A separate bound $j_i \le m - \ell$ would be insufficient for the boxes. For the retained count S, each retained depth-d prefix has 4^{2L-d} retained all-nonprivate extensions. Hence, if there are p_d such prefixes, $p_d 4^{2L-d} \le S$. Every remaining input array has size at most 4^{2L-d}, so all internal arrays, their formation, and compact storage cost $O(LS)$ per run. Dense root initialization costs $12^L \le S$; no 25^L leaf array is allocated.

Writing $\beta_a = \binom{L}{m-a}4^{L-m+a}$, the geometric ratio is at most $4m/(L-m+1) < 1$. The box count, including its full DP closure, is at most $\mathrm{poly}(L)\sum_{a_1+a_2\ge t}\beta_{a_1}\beta_{a_2}$. Concavity of entropy makes its exponential maximum balanced between halves, so relative to $(KN_0)^2$ its saving is $G_5(c,\theta)$. The same argument with total deficit 2ℓ gives $S = (KN_0)^2 D^{-G_5(c,r)}\,\mathrm{poly}(L)$. Sharing each band's encodings across tiles yields the claimed E_5. The degree-split runs, box orders, lookups, and exact integer operations contribute polynomial logarithmic factors. Local coefficients are $0, \pm 1$; encoded sums and the binomial coefficients have polynomial magnitude in N, so arithmetic uses $O(\log N)$ bits. □

Lemma 31 (Fixed-profile inner-basis prefix queries)**.** *Put $D = 4^m$, $L = cm + O(1)$, $c > 5$, and*

$$C_K = \frac{c\mathcal{H}(1/c)}{\ln 4}, \quad T = c - 1 + C_K, \quad G = \frac{c\mathcal{H}(1/c) - c\mathcal{H}((1-\theta)/c) - \theta\ln 4}{\ln 2}.$$

Fix $0 < r < 1$, $0 < \theta < 1 - r$, and a rational $0 < y < 1$. Assume $\theta/(1-r) < 4/5$. The squared five-term embedding, restricted to the fixed-profile good sets described below, gives preprocessing time and space

$$\widetilde{O}\left(ND^{E_{\mathrm{aug}}} + ND^{E_{\mathrm{box}}} + N^2D^{-G}\right), \qquad E_{\mathrm{aug}} = T - r(C_K - 3/2), \quad E_{\mathrm{box}} = T - G,$$

and individual query time $\widetilde{O}(D^{\max(q_1,q_2)})$, where

$$q_1 = r + \frac{(1-r)\ln((1+3y)(1+2y)) - 2\theta\ln y}{\ln 4}, \qquad q_2 = \frac{2(1-r)\mathcal{H}(\theta/(1-r)) + 2\theta\ln 4}{\ln 4}.$$

The original tile condition $N \ge KN_0$, with $K = \binom{L}{m}$ and $N_0 = 4^{L-m}$, remains sufficient; it has exponent T. There is no independent-product batching assumption.

Proof. Let $\ell = rm + O(1)$ and $L_{\mathrm{pre}} = rL + O(1)$, rounded to integers. Keep the inner sets Q satisfying

$$|Q \cap [L_{\mathrm{pre}}]| = \ell, \qquad |Q| = m.$$

Their number is

$$K' = \binom{L_{\mathrm{pre}}}{\ell}\binom{L - L_{\mathrm{pre}}}{m-\ell} = \binom{L}{m}/\,\mathrm{poly}(m).$$

The two regional densities equal $1/c + O(1/m)$. Stirling's inequalities give the displayed comparison, including any bounded rounding changes. For every fixed set in this family, the source's independent binary re-indexing maps make it good with probability $\mu_m \mu_{L-m} > \mu_\infty^2$. The conditional-expectation algorithm may sum over this explicit family instead of all m-sets. Each summand and each conditional-rank computation is unchanged, so it deterministically finds $\Theta(K')$ good sets in $K'(2^m + 2^{L-m})\,\mathrm{poly}(L)$ time. The source's squared embedding and its degree-split runs accept this list of good sets. Thus the outer tile size is $\Theta(K')N_0$, where $N_0 = 4^{L-m}$; replacing K' by $K = \binom{L}{m}$ costs only polynomial factors.

Use the source's local forms f_{ij}, g_{ij}, f_0, g_0, with the second half transposed. Set $I = p_1 y_1 + p_2 q$ and $P = f_0 g_0$. The query prefix F is exactly the inner positions in the fixed physical prefix of each half, hence $|F| = 2\ell$. Add two normal-side input forms p_1, p_2 to the encoder. On the opposite side retain the ordinary forms g_{11}, g_{21} and use $y_1 = g_{11}$, $q = g_{11} - g_{21}$. Each input side therefore requires only 4^ℓ prefix source combinations. These are actual extra stored forms; their entire construction is charged below.

For the suffix use

$$p_1 = f_{11} + f_0 + f_{21}, \qquad p_2 = f_{22} - f_{21}, \qquad y_1 = g_{11}, \qquad q = g_{11} - g_{21}.$$

At a normal-side coordinate the sources are the private form and three ordinary forms; at an opposite-side coordinate they are the private form and two ordinary forms. The outputs are the private form and two inner coordinates. Private outputs depend only on private inputs, while an inner output can depend on private or ordinary inputs. Thus every predecessor of an output with at most j nonprivate coordinates also has at most j nonprivate coordinates. Apply the tensor of these fixed local transforms on the truncated grid, generating only grid states of degree below $t = 2\theta m + O(1)$. Its source count, including both input sides, is at most a polynomial in L times

$$4^\ell \sum_{a+b<t} \binom{m-\ell}{a}\binom{m-\ell}{b} 3^a 2^b.$$

The predecessor-closure property bounds every intermediate transform grid by the same source count up to a polynomial factor. In particular, one must not enumerate the entire untruncated Cartesian grid first. Multiply matching transformed output entries and weight every degree-j entry by

$$(-1)^{t-j-1}\binom{2(m-\ell)-j-1}{t-j-1}.$$

The prefix partition identity then gives the exact residual. Outputs with different inner subsets are multiplied only with matching outputs; no cross-subset terms are introduced. Since $0 < y < 1$, the positive generating polynomial bounds this source count by

$$\mathrm{poly}(L) 4^\ell y^{-t} ((1+3y)(1+2y))^{m-\ell},$$

which has exponent q_1.

The complementary part of the partition uses $4^t\binom{2(m-\ell)}{t}$ stored boxes, giving q_2. Order the physical prefixes of the two halves first, then their suffixes. The stars in F precede every remaining inner position in this one fixed order. The usual balanced box family, with at most m private-or-star positions per half, at most $2m - t$ in total, and all stars before private positions, contains every required box. Replacing the last star by any ordinary or private term preserves the family, including its intermediate boxes. A completed leaf has at most $2m+1$ preimages in this family. The canonical balanced-tail bound therefore gives box cost $\widetilde{O}((KN_0)^2 D^{-G})$ per tile. This uses the full-family bound, so it requires no assertion that restricting to profile sets removes boxes.

Two distinct encoders are charged. The ordinary one supplies box base cases and retains private counts at most m in each half and at most $2m-t$ in total. Its cost is $\widetilde{O}((KN_0)^2 D^{-G})$ per band. The augmented

query-source encoder has, on its normal half, four ordinary labels and at most ℓ projection labels in the physical prefix, with no private label there. Its suffix has four ordinary labels and at most $m-\ell$ private labels. On its opposite half the prefix allows all four ordinary labels but no private label; the suffix again allows at most $m-\ell$ private labels. This includes every source just used, including sources of smaller intermediate degree.

Define $U(n,b,w)=\sum_{j\le b}\binom{n}{j}w^j4^{n-j}$. The normal and opposite half counts are respectively

$$A=U(L_{\mathrm{pre}},\ell,2)U(L-L_{\mathrm{pre}},m-\ell,1),\qquad B=4^{L_{\mathrm{pre}}}U(L-L_{\mathrm{pre}},m-\ell,1).$$

The ratios before their terminal summands are bounded away from one: asymptotically they are at most $2/(c-1)$ and $4/(c-1)$. Consequently

$$A=O(2^\ell K'N_0),\qquad \frac{AB}{(K'N_0)^2}=O\left(\frac{8^\ell}{\binom{L_{\mathrm{pre}}}{\ell}}\right)=\widetilde{O}(D^{-r(C_K-3/2)}).$$

Incomplete projection prefixes are retained. Every retained prefix in each encoder has every all-ordinary completion, so its full remaining array of dimension at most 4^{2L-d} is charged to that many distinct retained descendants. Summing over depths charges every internal array, its formation, and compact storage. Both encoders include all ordinary leaves; dense root initialization 12^L is smaller than 16^L and is therefore included. The two input sides use transposed block orders.

Sharing each band's encodings across the ordinary grid of tiles converts these two per-band costs into $ND^{E_{\mathrm{box}}}$ and $ND^{E_{\mathrm{aug}}}$, up to polynomial factors in L. The full box count becomes N^2D^{-G}. The independent-product layout is never used. The fixed degree-split runs, source re-indexing maps, integer bit costs, and rounding contribute polynomial logarithmic factors under the original tile condition. □

Lemma 32 (Rectangular five-term products)**.** *Under the hypotheses and notation of Lemma 31, let $X\in\mathbb{Z}^{R\times D}$ and $Y\in\mathbb{Z}^{D\times C}$ have entries of magnitude polynomial in $R+C+D$, and suppose $\min\{R,C\}\ge KN_0\,\mathrm{poly}(m)=D^T\,\mathrm{poly}(\log D)$. Then the fixed-profile five-term data structure preprocesses XY in time and space*

$$\widetilde{O}\big((R+C)(D^{E_{\mathrm{aug}}}+D^{E_{\mathrm{box}}})+RC\,D^{-G}\big)$$

and returns any single entry in $\widetilde{O}(D^{\max(q_1,q_2)})$ time.

Proof. Pad R and C upward to multiples of $K_{\mathrm{g}}N_0$; the hypothesis makes this at most a doubling. Partition the rows of X into $R/(K_{\mathrm{g}}N_0)$ row bands and the columns of Y into $C/(K_{\mathrm{g}}N_0)$ column bands, each consisting of K_{g} blocks of N_0 rows or columns. A tile is one row band together with one column band; it is exactly the $K_{\mathrm{g}}\times K_{\mathrm{g}}$ grid of N_0-block products to which the squared embedding, the fixed-profile good sets, the degree-split runs, the box family, and the prefix partition identity of Lemma 31 apply verbatim, since that lemma's tile is the same grid with $N=K_{\mathrm{g}}N_0$. A left encoding depends only on its row band and a right encoding only on its column band. Both encoder families are therefore computed once per band and shared across the tiles of that band, with transposed block orders on the two sides: the ordinary encoder costs $\widetilde{O}((KN_0)^2D^{-G})$ per band and the augmented encoder $\widetilde{O}((KN_0)^2D^{-r(C_K-3/2)})$ per band, by the bounds S and AB in that proof, which charge every internal array and its compact storage. Multiplying by the $(R+C)/(KN_0)$ bands gives $(R+C)D^{E_{\mathrm{box}}}$ and $(R+C)D^{E_{\mathrm{aug}}}$. The balanced box family of one tile costs $\widetilde{O}((KN_0)^2D^{-G})$ and there are $RC/(KN_0)^2$ tiles, giving RCD^{-G}. A query for the entry (i,j) lies in one tile; its residual multiplies the retained truncated sources of the row band of i and the column band of j, and its complementary part reads stored boxes of that tile, so the two query exponents are unchanged. The re-indexing maps cost $K(2^m+2^{L-m})\,\mathrm{poly}(L)\le\min\{R,C\}\,\mathrm{poly}(L)$ once. No independent-product batching is used, and no $K_1\times K_2$ grid with $K_1\ne K_2$ arises: restricting the good sets of one side is equivalent to padding that side. □

Lemma 33 (Composite incidence certification)**.** *Suppose a deterministic counting oracle computes, for any modulus $d \le h\,\mathrm{polylog}(n)$ and any chosen set of incidences $x = (a, b, B)$ with $|B| \le h = \Theta(n^b)$, where $0 < b < 1$, the exact number $C_d(x)$ of vertices $v \in B$ for which d divides the original triangle sum. If every such call, including writing and reading its data, costs $\widetilde{O}(n^{3-s})$, then for every fixed integer $R \ge 1$ satisfying the parameter choices below, all-edges Exact Triangle has deterministic time $\widetilde{O}(n^{3-s+b/R} + n^2)$. The polylogarithmic power may depend on R and the polynomial input-magnitude bound.*

Proof. This is the per-incidence construction of [6, Section 15.4]. We give its proof to specify the oracle interface and account for true witnesses. Let $Z(x)$ count genuine zero sums and $F_d(x) = C_d(x) - Z(x)$. If every nonzero triangle sum has magnitude at most $U' \ge 1$, put $\zeta = \max(1, \ln U')$ and choose $P \ge 4\zeta(4h)^{1/R}$, also $P \ge 8\zeta$ and sufficiently large. Enumerate primes in $[P, 2P)$ deterministically. After removing fewer than R previously chosen primes, let M denote the actual number of remaining candidates; then $M \ge P/(4\ln P)$ for all sufficiently large n. Each nonzero sum has at most $\lambda = \lfloor \ln U'/\ln P \rfloor$ candidate prime divisors. Thus $\lambda/M \le 4\zeta/P =: a$. The finitely many smaller n are handled by brute force.

Fix the active incidence set $\mathcal{I}$ for one stage. Start at $d = 1$. For $R - 1$ steps compute all $C_{dp}(x)$ for every unused candidate prime, choose p minimizing their sum over $\mathcal{I}$, and replace d by dp. The true mass $\sum Z(x)$ is independent of p, so the false mass contracts by at most a at each step. Consequently $\sum_{x\in\mathcal{I}} F_d(x) \le h|\mathcal{I}|a^{R-1}$. Retain the selected modulus's individual counts, not just their total. For $R = 1$, initialize $C_1(x) = |B|$ without an oracle call.

In the last round compute $\mu(x) = \min_p C_{dp}(x)$. Discard an incidence as false if $\mu(x) = 0$; certify its pair YES if

$$M\mu(x) > \lambda C_d(x).$$

The first rule is sound since $\mu \ge Z$. The second is sound since a zero-free incidence has $M\mu \le \sum_p F_{dp} \le \lambda F_d = \lambda C_d$. For an uncertified incidence, $\mu \ge 1$ and $M\mu \le \lambda(Z + F_d)$. If $Z = 0$ this implies $\lambda F_d \ge M$; if $Z \ge 1$, using $\mu \ge Z$ implies $\lambda F_d \ge (M - \lambda)Z \ge M - \lambda$. Therefore the next active set, after also removing certified pairs, has size at most

$$\frac{\lambda}{M - \lambda} h|\mathcal{I}|a^{R-1} \le 2h|\mathcal{I}|a^R \le |\mathcal{I}|/2.$$

Here $a \le 1/2$ and $ha^R \le 1/4$ by the choice of P. A genuine witness is never discarded unless its pair is certified YES.

Restart with $d = 1$ on the new active set and repeat $\lceil \log_2(2n) \rceil$ stages. Initially there are at most $2n^3/h$ incidences. Scanning every remaining block then costs $O(n^2)$ and answers all remaining pairs exactly. Each candidate modulus is a product of at most R distinct primes, hence at most $(2P)^R = h\,\mathrm{polylog}(n)$ for fixed R; choose P within a constant factor of its stated threshold. There are $\widetilde{O}(n^{b/R})$ candidate primes and $O(R\log n)$ rounds. A sieve, comparisons, minima, and survivor bookkeeping are charged, along with every candidate count call. Moduli and counts have $O(\log n)$ bits. No CRT tuples of different witnesses or randomized tests are used. □

Lemma 34 (Lopsided composite incidence certification)**.** *Let the two outer parts have R and C vertices and the middle part H vertices, all polynomial in an ambient size n, with integer weights of magnitude $n^{O(1)}$, and let $h = \Theta(n^b)$ with $b > 0$ and $h \le H$. Suppose a deterministic counting oracle computes, for any modulus $d \le h\,\mathrm{polylog}(n)$ and any set $\mathcal{I}$ of incidences $x = (a, b, B)$ with $|B| \le h$, the exact numbers $C_d(x)$ of $v \in B$ with $d \mid S(a, b, v)$, every call costing at most $\widetilde{O}(\Gamma)$ including writing and reading its data. Then for every fixed integer $\Lambda \ge 1$, all-edges Exact Triangle on this instance has deterministic time*

$$\widetilde{O}\big(n^{b/\Lambda}(\Gamma + RCH/h) + RC\big),$$

where the polylogarithmic power depends on Λ and the magnitude bound.

Proof. Repeat the proof of Lemma 33 with the following replacements. The initial active set has at most $RC\lceil H/h\rceil \le 2RCH/h$ incidences. Each stage halves the active set, so after $\lceil \log_2(2H)\rceil$ stages at most RC/h incidences remain, and scanning their blocks with the original integer sums costs $O(RC)$. Every stage performs Λ rounds, each evaluating all $\widetilde{O}(n^{b/\Lambda})$ candidate primes with one oracle call per candidate and $O(|\mathcal{I}|)$ bookkeeping per candidate for the sums, minima, comparisons, and survivor lists. The prime pool in $[P, 2P)$ with $P \ge 4\zeta(4h)^{1/\Lambda}$ and $P \ge 8\zeta$, the bound λ on the prime divisors of a nonzero sum, the contraction $ha^{\Lambda} \le 1/4$, the soundness of both rules, and the leak bound are statements about single incidences and do not depend on the three part sizes. Every modulus is a product of at most Λ distinct primes below $2P$, hence at most $(2P)^{\Lambda} = h\,\mathrm{polylog}(n)$. A genuine witness is never discarded unless its pair is certified YES, so all RC answers are exact. Finitely many small n use brute force. □

Theorem 38 below inherits the same lopsided reading, since its oracle costs depend only on (N, D, q) with $N = R\lceil\sqrt{d}\rceil$ on both sides; Corollary 56 uses it for cliques.

The residue-folding construction above extends from primes to every integer modulus $d \ge 2$: it uses only congruence and the digit representation $\rho_1 + \lceil\sqrt{d}\rceil\rho_2$, without division or inversion in a field. It computes the exact modular count by an ordinary integer product. For a block of size h the folded dimensions are $D = hd$, $N = n\lceil\sqrt{d}\rceil$, and each active incidence requests one entry. Pad D to the next power of four and N to complete tiles; strict fit slack absorbs the resulting constants. In particular no fixed query density is needed: use preprocessing plus the actual number of queries.

Corollary 35 (An explicit five-term deterministic APSP exponent)**.** *Using the five-term embedding of [6], integer all-edges Exact Triangle, min-plus product, and APSP admit deterministic $O(n^{2.993676})$ algorithms. Inputs have polynomially bounded integer weights and APSP has no negative cycles.*

Proof. Apply Lemma 31 and the composite incidence certification of Lemma 33 with the exact rational parameters

$$(c, \theta, r, e, y) = (10.74132662,\ 0.056740653,\ 0.038571087,\ 0.083812911,\ 0.025146880),$$
$$b = e/2 = 0.0419064555.$$

Restrict every slot to exactly $rm + O(1)$ inner coordinates among the first $rcm + O(1)$ physical coordinates. The slot count differs from K only by a polynomial factor. Put

$$C_K = \frac{c\mathcal{H}(1/c)}{\ln 4}, \qquad T = C_K + c - 1, \qquad G = \frac{c\mathcal{H}(1/c) - c\mathcal{H}((1-\theta)/c) - \theta\ln 4}{\ln 2}.$$

Charge the two encoder families separately:

$$E_{\mathrm{aug}} = T - r(C_K - 3/2), \qquad E_{\mathrm{box}} = T - G.$$

The direct-prefix residual and stored-box queries have respective exponents

$$q_1 = r + \frac{(1-r)\ln((1+3y)(1+2y)) - 2\theta\ln y}{\ln 4}, \qquad q_2 = \frac{2(1-r)\mathcal{H}(\theta/(1-r)) + 2\theta\ln 4}{\ln 4}.$$

The exact-rational verifier gives

$$\begin{aligned} T &= 12.140552723405380\ldots, & G &= 0.075467246110185\ldots,\\ E_{\mathrm{aug}} &= 12.105868595138261\ldots, & E_{\mathrm{box}} &= 12.065085477295195\ldots,\\ q_1 &= 0.424532754092257\ldots, & q_2 &= 0.424532753969640\ldots. \end{aligned}$$

Use $R = 100000$ distinct factors per composite. Candidate enumeration costs exponent $\eta = b/R = 0.000000419064555$. At the final modulus scale n^b, the savings below exponent three are

boxes	$eG - \eta$	> 0.0063247105170,
residual queries	$e(1/2 - q_1) - \eta$	> 0.0063247105001,
stored queries	$e(1/2 - q_2) - \eta$	> 0.0063247105104,
augmented encodings	$1 - e(E_{\text{aug}} - 1/4) - \eta$	> 0.0063247215434,
ordinary box encodings	$1 - e(E_{\text{box}} - 1/4) - \eta$	> 0.0097428733695,
matrix formation	$1 - 3e/4 - \eta$	> 0.9371398976854,
incidence bookkeeping	$b - \eta$	> 0.0419060364354.

The unchanged geometric margin is $1 - e(T - 1/4) > 0.0034181628524$. For intermediate modulus $d = n^u$, $0 \le u \le b$ up to polylogarithms, partition the middle vertices into $O(n/h)$ blocks of size $h = \Theta(n^b)$. There are $O(n/h)$ folded products with dimensions $D = hd$ and $N = n\lceil\sqrt{d}\rceil$, and at most $O(n^3/h)$ queries across them. Multiplying the preprocessing and query costs in Lemma 31 by these counts gives savings $b - u + (b+u)G$ for boxes, $b - (b+u)q_i$ for queries, $1 + b - u/2 - (b+u)E_j$ for each encoder, and $1 - 3u/2$ for formation, before subtracting η. All decrease with u, as does the fit margin $1 + u/2 - (b+u)T$, so the displayed final-scale checks cover every call.

Every runtime saving exceeds $\delta = 0.006324$ with a strict gap greater than $7.1 \cdot 10^{-7}$. This absorbs all fixed-R logarithmic factors, profile rounding, integer bit costs, bit scaling, and APSP squaring. The data-structure lemma already charges both encoder families and all intermediate arrays. The exact all-edges interface supplies the previously proved min-plus and APSP transfers. This proves $O(n^{2.993676})$. The result does not change the separately analyzed clique or counting claims, and does not establish a bound below 2.99. □

Splitting the modulus between geometry and coefficients

Lemma 36 (Zero-prefix five-term queries over cyclic coefficient rings)**.** *Put $D = 4^m$, $L = cm + O(1)$, $c > 5$, and*

$$C_K = \frac{c\mathcal{H}(1/c)}{\ln 4}, \qquad T = c - 1 + C_K, \qquad G = \frac{c\mathcal{H}(1/c) - c\mathcal{H}((1-\theta)/c) - \theta \ln 4}{\ln 2}.$$

Fix $0 < \theta < 4/5$ and a rational $0 < y < 1$, and set

$$q_1 = \frac{\ln((1+3y)(1+2y)) - 2\theta \ln y}{\ln 4}, \qquad q_2 = \frac{2\mathcal{H}(\theta) + 2\theta \ln 4}{\ln 4}.$$

Let $N, D, q \le n^{O(1)}$, $q \ge 1$, and assume every input polynomial has coefficient ℓ_1 norm at most $n^{O(1)}$. Over $\mathcal{R}_q = \mathbb{Z}[x]/(x^q - 1)$, the source's full-grid five-term embedding preprocesses $N \times D$ by $D \times N$ matrices over $\mathcal{R}_q$ in

$$\widetilde{O}\left(qND^T + qN^2D^{-G} + qND\right)$$

time and space, under the original layout condition $N \ge KN_0$, where $K = \binom{L}{m}$, $N_0 = 4^{L-m}$; the sufficient exponent condition is $N \ge D^T \operatorname{poly}(\log D)$. Given a product position and a specified residue $j \bmod q$, its coefficient can be queried in

$$\widetilde{O}\left(qD^{q_1} + D^{q_2}\right).$$

For the monomial inputs below, all stored coefficients and exact arithmetic intermediates have $O(\log n)$ bits. There is no independent-product batching hypothesis.

Proof. This is the zero-prefix endpoint of the sparse-Yates query construction, with its two query terms charged separately. Use all good inner sets; there is no profile restriction and no augmented projection encoder. Write $t = 2\theta m + O(1)$. The local identities

$$p_1 = f_{11} + f_0 + f_{21}, \qquad p_2 = f_{22} - f_{21}, \qquad y_1 = g_{11}, \qquad q_{\mathrm{loc}} = g_{11} - g_{21}$$

give the same suffix transform as before, on all $2m$ inner coordinates. Its truncated source count is bounded by

$$\mathrm{poly}(L) \sum_{a+b<t} \binom{m}{a}\binom{m}{b} 3^a 2^b \le \mathrm{poly}(L) y^{-t}((1+3y)(1+2y))^m.$$

The private output uses only the private source, and an inner output uses private or nonprivate sources. Consequently every predecessor of a retained output has no greater nonprivate degree; all intermediate transform arrays lie in the same truncated family. No full Cartesian array is formed first. Multiply only matching transformed outputs; a degree-j output receives weight $(-1)^{t-j-1}\binom{2m-j-1}{t-j-1}$, as in Lemma 31. The partition identity gives the exact residual, with exponent q_1. The complementary part uses $4^t\binom{2m}{t}$ stored boxes, giving q_2. This argument permits prefix length zero directly; it does not invoke a limiting positive prefix whose parameters depend on n.

The ordinary encoder retains at most m private labels in each half. Every retained depth-k prefix has all 4^{2L-k} ordinary completions, which pay for its entire remaining input array. Summing over depths charges formation and compact storage; the dense root size 12^L is below the retained all-ordinary count 16^L. Its leaf and internal-array bound is $\widetilde{O}((KN_0)^2)$ per band, where $K = \binom{L}{m}$ and $N_0 = 4^{L-m}$. The full balanced box family, including its dynamic-program closure, has $\widetilde{O}((KN_0)^2 D^{-G})$ states per tile. Sharing encodings across the ordinary grid of tiles therefore gives scalar costs ND^T and N^2D^{-G}. The re-indexing maps and fixed degree-split runs have the same polynomial-logarithmic overhead as in the scalar construction.

Every local identity, encoder transform, box recurrence, and query identity is an integer linear or bilinear identity. It therefore remains valid over the commutative ring $\mathcal{R}_q$. Store a ring element as its q coefficients. Linear operations cost $O(q)$ word operations; cyclic products cost $\widetilde{O}(q)$ by exact integer polynomial arithmetic. For example, split signs and use Kronecker substitution with an $O(\log n)$-bit radix exceeding the convolution coefficient bound, multiply the resulting $O(q \log n)$-bit integers, and fold degrees modulo q. This proves the preprocessing bound, including the conservative charge qND for forming and initializing the ring-valued input arrays.

The output coefficient map is linear. In the stored-box portion, read only coefficient j of each requested box and apply its scalar query weight. This costs one coefficient access per box, with no factor q. For the residual, perform its truncated linear transforms on all coefficients, costing $O(q)$ per retained source or intermediate entry. For each matching pair of transformed entries, compute only coefficient j of their cyclic product, by a length-q dot product. The residual therefore costs $\widetilde{O}(qD^{q_1})$; it does not require extracting the whole output polynomial.

For completeness, the ring extension does not multiply bit lengths at every recursion level. Input coefficient norms are polynomially bounded; in the application below each entry is zero or a monomial of coefficient one. A fixed local linear transform increases coefficient ℓ_1 norm by a constant factor, and there are $O(L) = O(\log n)$ levels. Each encoded input thus has polynomial norm. Each leaf performs one bilinear cyclic product; cyclic reduction cannot increase ℓ_1 norm, and the product norm is at most the product of input norms. Subsequent box recurrences are linear, with polynomially many summands. Query transforms and binomial query weights likewise have polynomial norm. Formal degree-split vectors are only polynomially long and preserve these bounds. Thus coefficients and intermediates have polynomial magnitude, hence $O(\log n)$ bits. This charges the internal encodings as well as the retained leaves. □

Lemma 37 (A folded modulus with an unfurled cyclic factor)**.** *Let $d, q \geq 1$ be coprime. Partition the middle vertices into blocks of size at most h. Fold only the modulus d into the outer indices, so $d_1 = \lceil \sqrt{d} \rceil$, $d_2 = \lceil d/d_1 \rceil$ and the middle indices are $(v, \sigma) \in B \times \mathbb{Z}_d$. Over $\mathcal{R}_q$ use the entries*

$$X[(a, \rho_1), (v, \sigma)] = \mathbf{1}_{\sigma \equiv w(a,v) + \rho_1 \pmod d} \, x^{w(a,v) \bmod q},$$
$$Y[(v, \sigma), (b, \rho_2)] = \mathbf{1}_{\sigma \equiv -w(v,b) - d_1 \rho_2 \pmod d} \, x^{w(v,b) \bmod q}.$$

For $w(a, b) \bmod d = \rho_1 + d_1 \rho_2$, coefficient $-w(a, b) \bmod q$ at this product position is exactly the number of vertices $v \in B$ for which dq divides the original triangle sum. The dimensions are $N = n\lceil \sqrt{d} \rceil$ and $D = hd$, with one coefficient query per requested incidence.

Proof. For a fixed v, the two indicator entries have exactly one common σ if and only if the original sum is zero modulo d. Its monomial product has exponent $w(a, v) + w(v, b)$ modulo q; the designated coefficient selects exactly those sums that are also zero modulo q. Coprimality makes these two tests equivalent to divisibility by dq. Negative input weights are reduced to their usual residues. Padded middle vertices are absent, represented by zero entries. The special cases $d = 1$ or $q = 1$ follow from the same formulas. □

Theorem 38 (Composite incidence certification with cyclic factors)**.** *Fix integers $R > S \geq 1$, a rational $0 < b < 1$, and parameters of Lemma 36. Put*

$$a = b/R, \qquad u = Sb/R, \qquad v = b - u, \qquad e = b + u.$$

Suppose $0 < G < 1$, $q_1, q_2 > 0$, $T > 1/2$, and the following strict inequalities hold for some $\delta > 0$:

$$eG - a > \delta, \qquad u - eq_1 - a > \delta, \qquad b - eq_2 - a > \delta,$$
$$1 + u/2 - eT - a > \delta, \qquad 1 - b - u/2 - a > \delta, \qquad b - a > \delta, \qquad 1 + u/2 - eT > 0.$$

Then integer all-edges Exact Triangle has deterministic time $O(n^{3-\delta})$. The established deterministic transfers give the same exponent for integer min-plus product and APSP without negative cycles.

Proof. Use blocks of size $h = \Theta(n^b)$ and exactly the candidate-prime pool and certification rule of Lemma 33, with R factors. Its primes have scale $P = n^{b/R} \operatorname{polylog}(n)$, with the logarithmic power fixed. Within each halving stage the active incidence set stays fixed while the greedy candidates are evaluated. For the first S selected factors, use ordinary scalar folded calls. Let their product be d. Freeze this folded modulus for the rest of the stage. Start $q = 1$ and accumulate the remaining selected primes into q, evaluating each candidate qp through Lemma 37. The final certification round evaluates R-factor moduli dqp, without needing to select its last prime. Since primes are distinct within a stage, d and every candidate cyclic factor remain coprime.

The exponent of the frozen modulus is exactly $u = Sb/R$, up to polylogarithmic factors; the largest cyclic factor has exponent $v = (R - S)b/R$. This equality is part of the parameter choice, not a continuous relaxation of the number of factors. Rebuilding the folded or ring-valued encodings separately for every candidate is fully charged; no reuse of a candidate's input arrays or boxes is assumed. The original certification argument applies unchanged because every returned number is the exact count for the full candidate modulus. It halves all active incidences, including true incidences, and $O(\log n)$ restarts leave $O(n^2)$ final scanning work.

There are $O(n/h)$ folded products and at most $O(n^3/h)$ requested incidences in any call, even before any contraction occurs. During the first phase let the current scalar folded modulus have exponent s,

where $0 \le s \le u$. Before candidate enumeration the savings below exponent three are

boxes	$b - s + (b+s)G$,
residual and stored-box queries	$b - (b+s)q_i \quad (i = 1, 2)$,
encodings	$1 + b - s/2 - (b+s)T$,
matrix formation	$1 - 3s/2$,
incidence bookkeeping	b.

The layout margin is $1 + s/2 - (b+s)T$. All these expressions are minimized at $s = u$, using the sign hypotheses.

During the second phase $N = \widetilde{\Theta}(n^{1+u/2})$ and $D = \widetilde{\Theta}(n^e)$ are fixed. For a cyclic factor of exponent w, $0 \le w \le v$, the corresponding savings are

boxes	$b - u + eG - w$,
residual queries	$b - eq_1 - w$,
stored-box coefficient reads	$b - eq_2$,
encodings	$1 + b - u/2 - eT - w$,
matrix formation	$1 - 3u/2 - w$,
incidence bookkeeping	b.

The formation line includes the entire qND ring-input allocation for every block. These expressions are minimized at $w = v$, except the two constant lines. Substituting $v = b - u$ yields precisely the six runtime inequalities in the theorem. The first phase's endpoint is the second phase's $w = 0$ endpoint, so no earlier large call is omitted. The original full-grid layout is checked separately by $1 + u/2 - eT > 0$.

There are $\widetilde{O}(n^a)$ candidate primes per round and $O(R \log n)$ rounds. Subtracting a from every runtime saving charges all candidate calls, including formation, extraction, comparisons, and survivor lists. A sieve costs $\widetilde{O}(n^a)$ and is smaller than these bounds. The prime products, ring dimensions, residues, and actual witness counts have $O(\log n)$ bits; Lemma 36 additionally charges encoded arithmetic. All polynomial logarithmic factors have fixed powers because R, S, c, θ, y, b are fixed constants. Strict slack absorbs these factors, rounding of block and tensor sizes, and the min-plus and APSP transfers. No density assumption about the active queries or folded residue classes is used. □

Corollary 39 (An explicit cyclic-factor deterministic exponent)**.** *Use*

$$(c, \theta, y, b) = (10.67333763,\ 0.0590858825,\ 0.025178142,\ 0.04368276), \qquad (R, S) = (100000, 92277).$$

Then

$$u = 0.0403091404452, \quad v = 0.0033736195548, \quad e = 0.0839919004452, \quad a = 0.0000004368276.$$

The exact-rational certificate verifies every first- and second-phase endpoint and the parameter conditions. Its smallest runtime saving is greater than 0.0065597670136*, while the original layout margin is greater than* 0.0065608839*. Consequently all-edges Exact Triangle, integer min-plus product, and integer APSP admit deterministic* $O(n^{2.993441})$ *algorithms under the source five-term embedding and the stated integer-weight assumptions. The gap above the target saving* 0.006559 *exceeds* $7.67 \cdot 10^{-7}$*. This does not establish a bound below* 2.99*.*

Caching a partial set of inner projections

Lemma 40 (Global partial inner-projection cache)**.** *Keep the five-term embedding and the exponents T and G of Lemma 36. Fix $0 < \theta < 2/3$, $0 < \rho < \min\{2\theta, 2/3\}$, and $0 < y < 1$. Define*

$$C_\rho = \frac{\mathcal{H}(\rho) + \rho \ln 2}{\ln 4}, \qquad E = T + C_\rho,$$
$$q_{\text{in}} = \frac{2\mathcal{H}(\theta) + 2\theta \ln 2}{\ln 4}, \qquad q_{\text{old}} = \frac{\ln((1+3y)(1+2y)) - 2\theta \ln y}{\ln 4},$$
$$q_{\text{hi}} = q_{\text{old}} - \frac{\rho \ln(3/2)}{\ln 4}, \qquad q_1 = \max\{q_{\text{in}}, q_{\text{hi}}\},$$
$$q_2 = \frac{2\mathcal{H}(\theta) + 2\theta \ln 4}{\ln 4}.$$

Under the original condition $N \geq KN_0$, the modified data structure over $\mathcal{R}_q = \mathbb{Z}[x]/(x^q - 1)$ has preprocessing and storage

$$\widetilde{O}\left(qND^E + qN^2D^{-G} + qND\right)$$

and a specified coefficient query costs $\widetilde{O}(qD^{q_1} + D^{q_2})$. The input-size and coefficient-norm assumptions are those of Lemma 36. This cache is allowed at arbitrary physical positions; it does not restrict the good inner sets to a fixed physical prefix.

Proof. Let $t = \lceil 2\theta m \rceil$ and $\ell = \lfloor \rho m \rfloor$. In an input's normal half, augment the four ordinary forms and the private form $P = f_0$ with the two direct projections p_1, p_2. Retain words with at most m private-or-projection positions and at most ℓ projection positions. In the opposite half retain the old words with at most m private positions. The other input exchanges its normal and opposite halves. The projections are actual extra stored values.

Put

$$U = \sum_{k=0}^{m} \binom{L}{k} 4^{L-k}, \qquad C_\ell = \sum_{a=0}^{\ell} \binom{m}{a} 2^a.$$

The normal-half leaf count is

$$A_\ell = \sum_{k=0}^{m} \binom{L}{k} 4^{L-k} \sum_{a=0}^{\min\{k,\ell\}} \binom{k}{a} 2^a.$$

Here k specifies all private-or-projection positions and a specifies the projected ones. Thus each input's leaf count is $A_\ell U \leq U^2 C_\ell$. Since $c > 5$, the ordinary tail is bounded by a geometric series ending at $k = m$, so $U = \widetilde{O}(KN_0)$. Since $\rho < 2/3$, $C_\ell = \widetilde{O}(D^{C_\rho})$. Every retained depth-$d$ prefix has all 4^{2L-d} ordinary completions. These completion sets are disjoint for different prefixes at a fixed depth and pay for each entire remaining input array, whose local dimensions are at most four. There are at most seven constant-size child forms. Therefore all internal arithmetic, arrays and leaf writes cost $O(LA_\ell U)$ per band and degree run. This also pays for the 12^L-entry dense source array. Sharing band encodings gives $\widetilde{O}(ND^E)$ scalar work, without a full augmented Cartesian allocation.

We now change only the residual query. Its desired output words are on the $2m$ queried inner positions, with symbols private, first inner factor, or second inner factor, and with total nonprivate degree below t. Process the two input sides separately. On one side order the m normal positions increasingly. Write a and b for the normal and opposite nonprivate counts of an output word.

If $a < \ell$, read its normal inner factors directly from the cache, fix all other normal positions to private, and perform the old two-source transform on the opposite positions with degree below $t - a$. If $a \geq \ell$, let J_0 be the first ℓ normal nonprivate positions, and let $p = \max J_0$ in their order. Group outputs by J_0

and the 2^ℓ choices of their direct inner factors. Every normal position before p outside J_0 is private. On normal positions after p use the three-source transform, and on all opposite positions use the two-source transform, with free degree below $t-\ell$. These are exactly the identities

$$p_1 = f_{11} + P + f_{21}, \quad p_2 = f_{22} - f_{21}, \qquad y_1 = g_{11}, \quad q_{\mathrm{loc}} = g_{11} - g_{21}.$$

Every transform predecessor has no greater nonprivate degree than its output. The private predecessor in the expression for p_1 occurs only after the fixed pivot, so it cannot change J_0. Consequently each group is generated on a predecessor-closed truncated grid. At every intermediate step a normal axis has at most three nonprivate choices and an opposite axis at most two; the group's source count pays for all intermediate arrays up to polynomial factors. Every source has at most ℓ direct projections, private plus direct count at most m in the normal half, and at most m private positions in the opposite half, so all sources belong to the stored cache.

The low groups use

$$C_{\mathrm{low}} = \sum_{\substack{a+b<t \\ a<\ell}} \binom{m}{a}\binom{m}{b} 2^{a+b} \le \sum_{j<t} \binom{2m}{j} 2^j = \widetilde{O}(D^{q_{\mathrm{in}}})$$

sources. To count the high groups, fix the pivot and apply

$$\sum_p \binom{p-1}{\ell-1}\binom{m-p}{a-\ell} = \binom{m}{a}.$$

This identity assigns every a-set to its unique first ℓ positions. It gives the exact high-source count

$$\begin{aligned} C_{\mathrm{high}} &= \sum_{\substack{a+b<t \\ a\ge\ell}} \binom{m}{a}\binom{m}{b} 2^\ell 3^{a-\ell} 2^b \\ &\le (2/3)^\ell y^{-t}((1+3y)(1+2y))^m = \widetilde{O}(D^{q_{\mathrm{hi}}}). \end{aligned}$$

For bounded small m with $\ell = 0$, use one empty high group and the old transform; this does not affect the asymptotic bounds.

The left and right partitions generally differ, because their normal halves differ. Each side therefore writes its transformed values to a table indexed by the same full word on the $2m$ physical inner positions; we do not pair lexicographic groups. On each side these groups are disjoint and cover all output words of degree below t. The common table has $\sum_{j<t}\binom{2m}{j}2^j$ entries and is already charged. Multiply equal full keys, weight degree j by $(-1)^{t-j-1}\binom{2m-j-1}{t-j-1}$, and sum the source degree runs. This is the unchanged exact residual, including words which lie in a low group on one side and a high group on the other.

The high-box recurrence is unchanged. Its base leaves have no direct projections and are contained in the cache, so it costs $\widetilde{O}(N^2 D^{-G})$ and requires $\widetilde{O}(D^{q_2})$ query reads. Over $\mathcal{R}_q$, cache formation, all transform arrays and residual coefficient dot products retain their q factor. Stored boxes contain all coefficients, permitting a single scalar access for a requested coefficient. The new forms are projections with coefficients zero or one. The polynomial coefficient-norm and exact bit-complexity bounds of Lemma 36 therefore apply unchanged. □

Corollary 41 (Deterministic APSP with partial inner caches)**.** *Use the same composite certification and fixed prime-stage construction, with*

$$\begin{aligned} &c = 10.767898969, \quad \theta = 0.060343925615, \quad \rho = 0.066059242397, \\ &y = 0.025749961, \quad b = 0.04337385, \qquad (R,S) = (100000, 88618). \end{aligned}$$

Then integer all-edges Exact Triangle, min-plus product, and APSP without negative cycles have deterministic time $O(n^{2.993401})$, under the same source embedding and polynomially bounded integer-weight assumptions.

Proof. Here

$$a = b/R = 0.0000004337385, \qquad u = Sb/R = 0.038437038393,$$
$$v = b - u = 0.004936811607, \qquad e = b + u = 0.081810888393.$$

Apply Lemma 40 to each scalar or cyclic call in Theorem 38. The geometry still uses T; only the query-source encoding cost changes from T to E. In the first phase, at folded exponent $s \in [0, u]$, this cost's saving before candidate enumeration is $1 + b - s/2 - (b + s)E$. In the second phase, at cyclic exponent $w \in [0, v]$, it is $1 + b - u/2 - w - eE$. All other lines in the two phase tables are unchanged, using $q_1 = \max\{q_{\text{in}}, q_{\text{hi}}\}$ for residual queries. The exact certificate checks both endpoints of every affine expression, both residual branches separately, and every parameter condition.

At the last endpoint, after charging all $\widetilde{O}(n^a)$ candidate primes, the savings are bounded below by

stored boxes $eG - a$	0.0065996962020
direct residual $u - eq_{\text{in}} - a$	0.0065996854801
high residual $u - eq_{\text{hi}} - a$	0.0065996854801
stored-box queries $b - eq_2 - a$	0.0065997069234
cached encodings $1 + u/2 - eE - a$	0.0066000053788.

Matrix formation and incidence bookkeeping also exceed the target saving 0.006599. The original tile margin $1 + u/2 - eT$ exceeds 0.0236621656710; the larger cache exponent must not be substituted for T in that geometric condition. The minimum runtime slack above 0.006599 exceeds $6.85 \cdot 10^{-7}$. Thus all fixed powers of logarithms, rounding, exact coefficient arithmetic, bit scaling and APSP squaring are covered, giving the stated exponent. The previous 2.993441 and fixed-profile 2.993676 witnesses remain valid. This still does not establish an exponent below 2.99. □

A shared private factor and polynomial query caches

The next construction changes the local five-term identity and its matrix embedding. Its private and inner kernels share an opposite-side factor. A Walsh transform of the input's outer coordinates preserves the full good-set layout while making that shared factor useful for queries.

Lemma 42 (Walsh embedding with a shared private factor)**.** *Let $D = 4^m$, $L = cm + O(1)$, $c > 5$, $K = \binom{L}{m}$, and $N_0 = 4^{L-m}$. The following five-term identity admits a full-grid embedding with a constant fraction of the original K good inner sets, the same N_0 and D, and $2m + 1$ degree runs:*

λ	φ_λ	ψ_λ
11	$x_1 + p_1 + p_2$	$s + v$
12	$x_1 + p_1$	$s - v$
21	$x_2 - p_2$	$s + v - w$
22	x_2	$s - v + w$
0	$-x_1 - x_2$	$2s$

$$\sum_\lambda \varphi_\lambda \psi_\lambda = 2p_1 s + p_2 w =: I. \tag{A1}$$

The second physical half exchanges the normal and opposite inputs. For integer input matrices, the complete embedded output is 4^L times the corresponding matrix-product entry. The Walsh preprocessing of the original inputs costs $\widetilde{O}(ND)$ per matrix; formation of the source arrays is charged by the compact encoder below. The complete output is recovered by exact division by 4^L. The same statements hold coefficientwise over $\mathcal{R}_q = \mathbb{Z}[x]/(x^q - 1)$, with a factor $\widetilde{O}(q)$ in work. The tile condition remains $N \geq KN_0$.

Proof. Identity (A1) follows by expansion. Put $k = L - m$ and retain the source's binary maps $\Gamma : \mathbb{F}_2^L \to \mathbb{F}_2^m$ and $\Delta : \mathbb{F}_2^L \to \mathbb{F}_2^k$. For every good m-set Q, Γ_Q and $\Delta_{\overline{Q}}$ are invertible. These are the same combinatorial conditions and constant-density good-set construction as before; no new independence condition is imposed.

Index a left row within a good-set block by $(i, a) \in \mathbb{F}_2^k \times \mathbb{F}_2^k$, and a right column by (a, i). For each block Q, apply the unnormalized Walsh transform only to its opposite outer index:

$$\begin{aligned} \widehat{X}_Q[(i, z), h] &= \sum_{a \in \mathbb{F}_2^k} (-1)^{z \cdot a} X_Q[(i, a), h], \\ \widehat{Y}_Q[h, (z, i)] &= \sum_{a \in \mathbb{F}_2^k} (-1)^{z \cdot a} Y_Q[h, (a, i)]. \end{aligned} \tag{A2}$$

Fast Walsh transforms use $O(k)$ additions per original entry. They are independent of the queried good set in the opposite physical half.

For a normal word u, write $Q(u)$ for its p_1, p_2 positions, $P_2(u)$ for its p_2 positions, and $i(u)$ for its x_1, x_2 bits outside $Q(u)$. For an opposite word v, let $V(v)$ and $W(v)$ be its v and w positions. In degree run $d \in \{-m, \ldots, m\}$, define the left source array by

$$A_d[u'u''] = 2^{|P_2(u')|} \widehat{X}_{Q(u')} \big[(i(u'), \Delta[V(u'')]), (\Gamma[P_2(u')], \Gamma[W(u'')])\big] \tag{A3}$$

when $Q(u')$ is good and $|W(u'')| - |P_2(u')| = d$, and zero otherwise. Define the right source array symmetrically by

$$B_d[v'v''] = 2^{|P_2(v'')|} \widehat{Y}_{Q(v'')} \big[(\Gamma[W(v')], \Gamma[P_2(v'')]), (\Delta[V(v')], i(v''))\big] \tag{A4}$$

when $Q(v'')$ is good and $|P_2(v'')| - |W(v')| = d$. Each input has local dimensions four and three, hence root-array volume 12^L per degree run; the compact encoder below charges this volume and its formation.

For an original row (Q_L, i_L, a_L) and column (Q_R, a_R, i_R), place inner symbols at Q_L in the first half and Q_R in the second. Use normal outside bits i_L, i_R and opposite outside bits

$$j_L = \Delta_{\overline{Q_L}}^{\mathsf{T}} a_R, \qquad j_R = \Delta_{\overline{Q_R}}^{\mathsf{T}} a_L. \tag{A5}$$

These transpose character maps replace the earlier inverse locator; they are bijections and are computable in polynomial time in L.

Every inner symbol forces a normal p letter. Since a source word has exactly m such letters, its normal inner set is exactly the query's good set. Its outside normal factors consequently reduce to the indicated x_i with coefficient one. Inner p_2 pairs only with w, and inner p_1 only with s. If the two normal p_2 sets have sizes a, b, and there are $e_L, e_R \geq 0$ additional outside w letters, the two run values are $b + e_L - a$ and $b - a - e_R$. Equality forces $e_L = e_R = 0$. Thus every outside opposite factor is $s + (-1)^j v$.

The local $p_1 s$ coefficient and the source weights in (A3)–(A4) together supply the uniform factor 2^{2m}, independently of the middle-bit choices. The good Γ restrictions enumerate all 4^m middle indices exactly once. In the second physical half, the outside v subsets $V \subseteq \overline{Q_R}$ have signs $(-1)^{a_L \cdot \Delta[V]}$ by (A5). The map $V \mapsto \Delta[V]$ is bijective, so Walsh inversion of (A2) gives 2^k times the desired left input entry. The first half similarly gives 2^k times the desired right entry. Hence the degree-summed output equals

$$2^{2m} 2^{2k} (XY)_{ij} = 4^L (XY)_{ij}. \tag{A6}$$

Only the *complete* query numerator is divided by 4^L; individual residual terms or boxes need not be divisible. Over $\mathcal{R}_q$ every complete coefficient is exactly divisible, so this uses integer division, never inversion modulo q. The scaled identity holds over any commutative ring, but decoding over a general ring would require division by 4^L to be justified separately. □

Lemma 43 (Two-sided polynomial cache)**.** *Use Lemma 42, the original five-term box geometry, and*

$$T = c - 1 + \frac{c\mathcal{H}(1/c)}{\ln 4}, \qquad G = \frac{c\mathcal{H}(1/c) - c\mathcal{H}((1-\theta)/c) - \theta \ln 4}{\ln 2}.$$

For fixed $0 < \theta < 1/2$, set

$$q_1 = \frac{2\mathcal{H}(\theta)}{\ln 4}, \qquad E = T + q_1, \qquad q_2 = \frac{2\mathcal{H}(\theta) + 2\theta \ln 4}{\ln 4}.$$

Under $N \geq KN_0$, integer matrices over $\mathcal{R}_q$ can be preprocessed in time and space

$$\widetilde{O}\left(qND^E + qN^2D^{-G} + qND\right).$$

A specified coefficient of a selected product entry is then obtained in $\widetilde{O}(qD^{q_1} + D^{q_2})$ time. Input-size and polynomial coefficient-norm assumptions are as in the previous cyclic-ring lemma.

Proof. Write $P_N = -x_1 - x_2$, $P_O = 2s$, and $P = P_N P_O$. Introduce a separate formal variable z, truncate modulo z^t for $t = \lceil 2\theta m \rceil$, and put $H(z) = P_N + zp_1$. Then

$$P + zI = H(z)P_O + zp_2 w. \tag{A7}$$

This identity uses a single alternative channel at each coordinate.

In a normal half retain four ordinary forms and two distinguished forms $H(z), p_2$; in an opposite half retain four ordinary forms and two distinguished forms P_O, w. Retain at most m distinguished positions per half and at most t channel labels p_2 or w across both halves. The channel forms themselves carry no z factor in the cache. The other input exchanges the two halves.

Put $U = \sum_{j \leq m} \binom{L}{j} 4^{L-j}$ and $B_t = \sum_{h \leq t} \binom{2m}{h}$. The exact retained count is

$$S_t = \sum_{j,k \leq m} \binom{L}{j}\binom{L}{k} 4^{2L-j-k} \sum_{h \leq \min\{t, j+k\}} \binom{j+k}{h}. \tag{A8}$$

Indeed j, k choose the distinguished supports, and choosing h positions in their union determines all channel labels. Every other distinguished position has its unique baseline H or P_O. There is no factor 2^h. Since $j + k \leq 2m$,

$$(KN_0)^2 B_t \leq S_t \leq U^2 B_t = \widetilde{O}\big(D^{2T+q_1}\big). \tag{A9}$$

The lower bound is the $j = k = m$ summand. The upper bound uses $c > 5$ for the geometric ordinary tail and $\theta < 1/2$ for the increasing unweighted binomial tail. Integer ceilings contribute polynomial factors; the entropy expression is not extended beyond that regime.

All internal arrays are charged. Every retained depth-d prefix has all 4^{2L-d} ordinary completions. They preserve both distinguished caps and the joint channel cap, and their sets are disjoint across prefixes at fixed depth. The remaining input array has local dimensions at most four, so the completions pay for its entire volume. Thus each layer has at most S_t entries. Every node has at most six forms of constant support; its polynomial operations are additions, fixed integer scaling, and the one-place shift in $H(z)$. With $t = O(m)$ coefficients per entry, full initialization, intermediate arithmetic and storage, and leaf writes cost $O(\mathrm{poly}(m)S_t)$. In particular $12^L \leq 16^L \leq S_t$ pays for each dense source array. Generate the capped recursion directly, without a full 36^L leaf allocation. Sharing these encodings across the tiles of a band gives $\widetilde{O}(ND^E)$ scalar preprocessing.

For a query let Q be its $2m$ inner physical positions. For every channel set $C \subseteq Q$ with $|C| < t$, read one polynomial from each cache: use p_2 or w at channel positions according to the input's role, $H(z)$ or P_O

at other inner positions, and the fixed ordinary forms outside Q. Each word has exactly m distinguished positions per half and fewer than t channels, so it is retained. Both inputs use the same full physical set C. Denote the values by $L_C(z), R_C(z)$. Tensoring (A7) gives, modulo z^t,

$$F(z) = \mathrm{Contr}\left(\prod_{i\in Q}(P_i + zI_i)\right) = \sum_{\substack{C\subseteq Q\\ |C|<t}} z^{|C|} L_C(z) R_C(z), \tag{A10}$$

where the contraction includes the fixed outside ordinary kernels. All polynomial degrees are nonnegative, so truncation discards no needed coefficient. The standard exact low residual is

$$\sum_{j<t} \sigma_j [z^j] F(z), \qquad \sigma_j = (-1)^{t-j-1} \binom{2m-j-1}{t-j-1}. \tag{A11}$$

Equivalently, a channel set contributes $\sum_{j=|C|}^{t-1} \sigma_j [z^{j-|C|}](L_C R_C)$. This applies one z shift per physical channel, not one per input. The exact number of keys is $\sum_{h<t} \binom{2m}{h}$, so enumeration, lookups, polynomial products and weighting cost $\widetilde{O}(D^{q_1})$.

The high-box recurrence still expands a star using the four ordinary terms and P of (A1). Retain the usual at-most-m private-or-star positions per half, at most $2m - t$ in total, and stars before private positions. Its full state count is unchanged. A base leaf has no channel label and is present in the cache. At normal private positions take the constant coefficient $H(0) = P_N$; its base product is $L_{\mathrm{leaf}}(0) R_{\mathrm{leaf}}(0)$, not the product of the full cached polynomials. Thus no separate ordinary input encoder is required. The box preprocessing costs $\widetilde{O}(N^2 D^{-G})$ and the query reads $4^t \binom{2m}{t} = \widetilde{O}(D^{q_2})$ stored values. Adding these values to (A11), summing degree runs, and then dividing the complete numerator by 4^L yields the original product entry by (A6).

Over $\mathcal{R}_q$, z is separate from the cyclic variable x. Cache entries have $O(m)$ z coefficients, each of width q. Linear formation and exact cyclic box products cost their usual $\widetilde{O}(q)$ factor. A specified cyclic coefficient of a product $L_C R_C$ is obtained by polynomially many length-q coefficient dot products, so the residual costs $\widetilde{O}(qD^{q_1})$. Stored boxes contain every cyclic coefficient, and their requested coefficient is read in constant time.

The Walsh transforms increase coefficient ℓ_1 norms by at most 2^{L-m}, source weights by at most 2^m, and every subsequent constant-support linear circuit by at most a fixed constant to the $O(L)$ power. Bilinear products, the coefficients in (A11), and the polynomial number of degree runs preserve a polynomial norm bound. Thus all scalar coefficients have $O(\log n)$ bits under the existing fixed-parameter assumptions. Exact cyclic arithmetic, auxiliary polynomial operations, Walsh transforms, and final exact division contribute only the stated polynomial-logarithmic factors. □

Corollary 44 (Deterministic APSP with shared-factor polynomial caches)**.** *For polynomially bounded integer weights, all-edges Exact Triangle, min-plus product, and APSP without negative cycles admit deterministic $O(n^{2.99294792})$ algorithms.*

Proof. Use the existing composite-incidence certification and fixed prime-stage construction with

$$c = 10.66550171, \quad \theta = 0.06529782140, \quad b = 0.04604634600,$$
$$(R, S) = (1000000000, 768979045).$$

Put

$$\eta = b/R = 0.000000000046046346,$$
$$u = S\eta = 0.03540867517281957,$$
$$v = (R - S)\eta = 0.01063767082718043,$$
$$e = b + u = 0.08145502117281957.$$

These are fixed constants independent of n. The earlier increase in the fixed stage count gave 2.992948 from 2.992949. The present refinement to 2.99294792 retains exactly those parameters and tightens only the reported target; it introduces no new oracle. For sufficiently large n, the candidate pool contains more than $2R$ primes; with $R = 10^9$ this threshold, like the absorption of the $(\log n)^{\Theta(R)}$ factors by the strict slack, holds only for astronomically large n, so the bound is asymptotic. The factor $(\log n)^R$ in the completed modulus remains a fixed polylogarithmic factor, and its bit length is $b \log n + O(R \log\log n) = O(\log n)$. The $O(R \log n)$ stage work therefore fits the same asymptotic proof; no practical speedup is asserted. Replace the selected-entry oracle in Theorem 38 by Lemma 43. All normalization occurs inside that oracle, before its exact nonnegative modular counts are used by the incidence certificate. In particular, signed Walsh and cache intermediates do not alter the certificate's count interpretation.

In the scalar phase, for folded exponent $s \in [0, u]$, the savings after charging all $\widetilde{O}(n^\eta)$ candidate primes are

$$\begin{array}{ll} \text{boxes:} & b - s + (b+s)G - \eta, \\ \text{residual queries:} & b - (b+s)q_1 - \eta, \\ \text{stored queries:} & b - (b+s)q_2 - \eta, \\ \text{polynomial cache:} & 1 + b - s/2 - (b+s)E - \eta, \\ \text{matrix formation:} & 1 - 3s/2 - \eta. \end{array} \tag{A12}$$

The tile margin is $1 + s/2 - (b+s)T$. In the cyclic phase, for $w \in [0, v]$, the respective savings are

$$\begin{array}{ll} \text{boxes:} & b - u - w + eG - \eta, \\ \text{residual queries:} & b - w - eq_1 - \eta, \\ \text{stored queries:} & b - eq_2 - \eta, \\ \text{polynomial cache:} & 1 + b - u/2 - w - eE - \eta, \\ \text{matrix formation:} & 1 - 3u/2 - w - \eta. \end{array} \tag{A13}$$

The tile margin is $1 + u/2 - eT$, and incidence bookkeeping has saving $b - \eta$ in both phases. These expressions charge the full cache exponent $E = T + q_1$; geometry still uses T.

The exact-rational certificate checks 47 strict inequalities, including both endpoints of every affine expression in (A12)–(A13), the stage identities, and every tail and parameter condition. At the final cyclic endpoint the five savings exceed, respectively,

$$0.00705208124172, \quad 0.00705208125745, \quad 0.00705208123527,$$
$$0.00705208151941, \quad 0.93624931636754.$$

The original tile margin exceeds 0.03540867543478. The minimum runtime slack above the target saving 0.00705208 is greater than $1.23527 \cdot 10^{-9}$. Thus polynomial logarithmic factors from the Walsh transforms, source runs, auxiliary polynomial caches, exact coefficient arithmetic, integer rounding, bit scaling, and APSP squaring fit within the strict gap. This proves the stated exponent. The earlier certified witnesses remain valid; a bound below 2.99 has not been established. □

Light sources and shared completion of residual queries

The following construction improves Corollary 44 by changing the source encoder and completing requested residual values jointly across middle blocks. It retains the full-grid geometry, the complete-

output normalization, and the composite-incidence reduction. In this subsection write

$$D = 4^m, \quad L = \lceil cm \rceil, \quad k = L - m, \quad K = \binom{L}{m}, \quad N_0 = 4^k,$$
$$C_K = \frac{c\mathcal{H}(1/c)}{\ln 4}, \qquad T = c - 1 + C_K, \qquad Q = \frac{2\mathcal{H}(\theta)}{\ln 4},$$
$$G = \frac{c\mathcal{H}(1/c) - c\mathcal{H}((1-\theta)/c) - \theta \ln 4}{\ln 2}, \qquad q_2 = Q + 2\theta, \qquad t = \lceil 2\theta m \rceil.$$

Here Q without a subscript is an entropy exponent; physical good sets will have subscripts. Formal polynomials use z, cyclic coefficients use x in $\mathcal{R}_q$, and the cache dimension exponent is ζ.

Lemma 45 (Shifted private factor and light source decomposition)**.** *The embedding of Lemma 42 also applies to the four ordinary pairs*

$$(x_1 + p_2, s + v), \quad (x_1, s - v), \quad (x_2 - p_2, s + v - w), \quad (x_2, s - v + w)$$

and private pair $(-x_1 - x_2 + p_1, 2s)$*. Their sum is* $I = 2p_1 s + p_2 w$*. For* $\rho = 51/100$*, each original product decomposes deterministically into* $\mathrm{poly}(m)$ *products whose source arrays have at most* $\lfloor \rho m \rfloor$ *normal* p_2 *letters. In each product one may additionally set to zero every source entry with more than* $\lfloor \rho m \rfloor$ *opposite* w *letters, before both the residual and high-box encoders. The complete degree-summed output still equals* 4^L *times that product entry.*

Proof. Expansion proves the shifted identity. The source definitions (A2)–(A5) are unchanged: every complete contraction has I at the m normal distinguished positions, which force all m normal p_1, p_2 letters. Thus outside normal forms reduce to the same x_1, x_2 as before. The proof of (A6), including its weights, degree filter, and transpose locator, applies.

Here is a deterministic construction of the required translations. Put $\epsilon = 1/50$, $J = 10000m$, $r = 101/100$, and $C = (r + r^{-1})/2 = 20201/20200$. For a list $\delta_1, \ldots, \delta_J \in \mathbb{F}_2^m$, allowing repetitions, set

$$S_a = \sum_{j=1}^{J} (-1)^{a \cdot \delta_j}, \qquad \Phi = \sum_{a \neq 0} (r^{S_a} + r^{-S_a}).$$

A uniform independent list has expectation $2(2^m - 1)C^J$. The exact rational inequality $r^{200} > 4C^{10000}$ gives $\mathbb{E}\Phi < r^{\epsilon J}$. Reveal the Jm bits one at a time, choosing a bit that does not increase the conditional expectation. For each character its factor for the current list entry is r, r^{-1}, or C, according as its sign is determined positive, determined negative, or still unbiased. Enumerating all $2^m - 1$ characters computes both choices exactly. Common denominators and numerators have $O(m)$ bits, with fixed constants, so construction costs $2^m \,\mathrm{poly}(m)$ bit operations. The resulting $\Phi < r^{\epsilon J}$ forces $|S_a| < \epsilon J$ for every $a \neq 0$.

For every good set Q_n and middle half-index h, invertibility of Γ_{Q_n} and this bias bound imply

$$J^{-1} \sum_j \mathrm{wt}\big(\Gamma_{Q_n}^{-1}(h + \delta_j)\big) \leq (1 + \epsilon)m/2 = \rho m.$$

Let $\iota(Q_n, h)$ be the first list index attaining weight at most $\lfloor \rho m \rfloor$. Indices resolve repetitions in the list. For a left row with normal block Q_L and right column with normal block Q_R, define, for each pair of list indices (r, s),

$$X^{rs}_{(Q_L,i),(h_1',h_2')} = X_{(Q_L,i),(h_1'+\delta_r,h_2'+\delta_s)} \mathbf{1}\{\iota(Q_L, h_1' + \delta_r) = r\},$$
$$Y^{rs}_{(h_1',h_2'),(Q_R,j)} = Y_{(h_1'+\delta_r,h_2'+\delta_s),(Q_R,j)} \mathbf{1}\{\iota(Q_R, h_2' + \delta_s) = s\}.$$

Both inputs use the same global middle translation. Masks are evaluated at the original middle indices and depend only on the input's own normal good set. Each original summand belongs to exactly one pair, so $XY = \sum_{r,s} X^{rs}Y^{rs}$, coefficientwise over $\mathcal{R}_q$. The masks commute with the opposite-outer-index Walsh transform. In (A3)–(A4), a surviving normal p_2 set therefore has size at most $\lfloor \rho m \rfloor$.

Let the two normal p_2 counts in a complete contraction be a, b, and the additional outside opposite w counts be $e_L, e_R \geq 0$. The two degree runs are $b + e_L - a$ and $b - a - e_R$. Their equality forces $e_L = e_R = 0$; hence the opposite w counts are b, a, both at most $\lfloor \rho m \rfloor$. Every discarded coordinate vanishes termwise in the complete contraction. Crop both inputs before *all* encoders: individual residuals and boxes may change. Only their complete sum is divided coefficientwise by 4^L. The J^2 products, mask preparation, translations, Walsh transforms, and output summation cost polynomial factors in m per original coefficient. The bias-list construction is below the D input-formation budget. Sharing across outer bands is preserved. □

Lemma 46 (An exact half-weight source cap)**.** *Lemma 45 also holds with $\rho = 1/2$. An ordered list of $J_m = 64m^3$ translations can be constructed deterministically in $2^m \operatorname{poly}(m)$ bit operations so that, for every invertible $A \in \mathbb{F}_2^{m\times m}$ and $h \in \mathbb{F}_2^m$, some list index j has $\operatorname{wt}(A(h + \delta_j)) \leq \lfloor m/2 \rfloor$.*

Proof. Allow the bias to shrink with the tensor parameter m. Set

$$\epsilon_m = \frac{1}{2m}, \qquad r_m = 1 + \frac{1}{4m}, \qquad C_m = \frac{r_m + r_m^{-1}}{2}, \qquad J_m = 64m^3.$$

For $x = 1/(4m) \leq 1/4$ the elementary logarithm inequalities give

$$\ln r_m \geq x - x^2/2 \geq \frac{7}{32m}, \qquad \ln C_m \leq C_m - 1 = \frac{x^2}{2(1+x)} \leq \frac{1}{32m^2}.$$

Therefore

$$J_m(\epsilon_m \ln r_m - \ln C_m) \geq 5m > \ln(2(2^m - 1)).$$

The last inequality follows from $2(2^m - 1) < 4^m$ and $\ln 4 < 2$. The same two-sided character potential as in Lemma 45 has initial expectation strictly below $r_m^{\epsilon_m J_m}$. Exact conditional expectations produce a list with every nonzero character bias strictly below ϵ_m. Here common rational numerators and denominators have $O(J_m \log(m+1))$ bits, rather than the earlier $O(m)$ bound. There are $J_m m$ choices and $2^m - 1$ characters, so all bit costs are still $2^m \operatorname{poly}(m)$.

The rows of an invertible A are nonzero characters. Thus, uniformly in A, h,

$$J_m^{-1} \sum_j \operatorname{wt}(A(h + \delta_j)) < m/2 + \epsilon_m m/2 = m/2 + 1/4.$$

If every integer weight exceeded $\lfloor m/2 \rfloor$, their average would be at least $\lfloor m/2 \rfloor + 1$, a contradiction for either parity of m. Use the first valid *list index* in the two original-coordinate masks. The same translated-product decomposition, Walsh commutation, and same-degree argument permit cropping both opposite sources to $\lfloor m/2 \rfloor$ before both encoders. Complete outputs retain their normalization by 4^L. The $J_m^2 = O(m^6)$ products, their masks, and their output sums contribute only fixed polylogarithmic factors. The larger rational bit lengths are confined to the separately charged list construction; the matrix coefficients still have $O(\log n)$ bits. □

Lemma 47 (Grouped completion with asymmetric row chunks)**.** *Suppose B middle-block products have the same outer size N, inner size D, cyclic width q, good-set layout, folded locator, modulus, and padding.*

Suppose an exact rectangular algorithm multiplies dimensions $s^a \times s$ by $s \times s^\kappa$ in $O(s^w)$ operations, where $\kappa \geq a \geq 1$ and w includes a fixed positive allowance above a proved exponent bound. Write

$$\alpha = w - \kappa - 1 \geq 0, \qquad \beta = w - \kappa - a \geq 0, \qquad \gamma(r) = \frac{\mathcal{H}((\rho - r)/(c-1))}{\ln 4}.$$

Use either $\rho = 51/100$ from Lemma 45 or $\rho = 1/2$ from Lemma 46. Assume $c > 5$, $0 < 2\theta < \rho$, and

$$\zeta > 0, \quad \zeta < (c-1)\gamma(2\theta), \qquad \frac{2\ln((1-\theta)/\theta)}{\ln 4} + \frac{\zeta\gamma'(\theta)}{2\gamma(\theta)^2} > 0. \tag{L1}$$

Under the original geometry $N \geq KN_0$ and the strict batch fit $Bq \geq D^{\kappa\zeta+\xi}$ for some fixed $\xi > 0$, requested residual values for M_0 original output incidences across the common shape can be obtained with costs

$$\widetilde{O}\left(qBN(D^{E_{\mathrm{root}}} + D^{E_{\mathrm{res}}}) + BqM_0D^{q_1}\right), \tag{L2}$$

where

$$E_{\mathrm{root}} = 2C_K + (2c-1)\log_4 3 + Q - T,$$
$$E_{\mathrm{res}} = T + Q - \left(\frac{1}{2\gamma(\theta)} - 1 - \alpha\right)\zeta, \qquad q_1 = Q + \beta\zeta.$$

The incidence list may depend on the middle block if M_0 bounds its union over blocks. Duplicate requested rows may be shared.

Proof. For the shifted identity put $H(z) = -x_1 - x_2 + (1+z)p_1$. Then

$$P + zI = H(z)(2s) + zp_2w.$$

The residual remains (A10)–(A11), with the same physical channel set in both inputs and a single shift by its size. Work modulo z^t throughout.

Fix distinguished sets Q_n, Q_o of size m and channels $C_n \subseteq Q_n$, $C_o \subseteq Q_o$ with total size below t. Contract these positions first. Normal nonchannel forms have three raw supports x_1, x_2, p_1, normal channels are p_2, and opposite distinguished forms are the singletons $2s, w$. Every root has 3^{2k} remaining raw coordinates, each formed using at most 3^m explicit Walsh-source lookups. Across all groups, even zero ones, the cost per band per block is at most

$$\widetilde{O}\left(qK^2 \sum_{h<t} \binom{2m}{h} 3^{2k+m}\right).$$

There are $\widetilde{O}(N/D^T)$ bands. This proves the root term in (L2) and pays for all root writes and intermediate distinguished contractions.

The remaining normal rare count is at most $\lfloor \rho m \rfloor - |C_n|$, and the opposite rare count at most $\lfloor \rho m \rfloor - |C_o|$. For the opposite half use the forward integer evaluation map $(s, v, w) \mapsto (s+v, s-v, -w)$; after permuting ordinary labels its four forms have the normal pattern $(y_1 + w', y_1, y_2 - w', y_2)$. This map preserves the rare support. Every one of its 3^{2k}-entry layers is paid by the root bound; no inverse or division by two is used.

Choose the half with the larger channel count, breaking ties deterministically. For any suffix length $\ell \leq k$, a raw word with at most r_0 rare positions has a cyclic window of length ℓ containing at most $\lfloor r_0\ell/k \rfloor$ of them, by averaging the k windows. Assign the raw source coordinate to its first such window *before* transformation. This is a disjoint linear partition. Keep all k components, and sum their reconstructed values on each input before multiplying the two residual polynomials.

In each component, transform the other $2k-\ell$ coordinates to their four ordinary outputs. In the suffix keep raw coordinates with at most $\lfloor r_0\ell/k\rfloor$ rare letters. For each nonrare pair-bit string of length ℓ create a separate group. A raw suffix with j rare letters belongs to 2^j such strings; copy it to every one of those groups. The exact length of a group vector is

$$s_\ell = \sum_{j\le\lfloor r_0\ell/k\rfloor} \binom{\ell}{j}.$$

Thus each prefix has $2^\ell s_\ell$ explicitly stored entries, including all duplication. An ordinary suffix output chooses its pair-bit group and an activity set A; its selector is $\mathbf{1}\{J\subseteq A\}$ for each rare support J, with the fixed signs absorbed in the stored vector. The selector is independent of the middle block and of the formal and cyclic coefficient. To charge intermediate transforms explicitly, the raw suffix volume is $V_\ell=\sum_{j\le\lfloor r_0\ell/k\rfloor}\binom{\ell}{j}2^{\ell-j}\le 2^\ell s_\ell$. After d prefix transforms a full allocated layer has $4^d3^{2k-\ell-d}V_\ell$ entries, at most $4^{2k-\ell}2^\ell s_\ell$. Thus every layer, including its initialization and zeros, fits the final duplicated group volume. The source partition scans and the opposite forward map were already charged in root formation.

For channel fractions $x=|C_n|/m$, $y=|C_o|/m$, choose $\ell=\zeta m/\gamma(\max(x,y))+O(1)$. More precisely use the actual k and rare budget with a rational upper entropy enclosure of error $O(m^{-2})$, and round the quotient down. Binary logarithm enclosures compute this in polynomial time in m; the common padding dimension is $s_*=\mathrm{poly}(m)D^\zeta$. Condition (L1) ensures $\ell\le k$ with strict slack. The number of groups before band sharing is bounded by

$$\widetilde{O}\left(D^{2T+\max_{x+y\le 2\theta}\{(\mathcal{H}(x)+\mathcal{H}(y))/\ln 4-\zeta/(2\gamma(\max(x,y)))\}}\right). \tag{L3}$$

The factor $2^{-\ell}$ comes from replacing 4^ℓ suffix outputs by 2^ℓ pair-bit groups; the s_* entries per group are charged separately, not omitted.

At fixed $x+y=2r$, entropy concavity and decreasing γ make the expression in braces largest at $x=y=r$. Since γ is positive, decreasing, and concave, $1/\gamma$ is convex. Consequently $2\mathcal{H}(r)/\ln 4-\zeta/(2\gamma(r))$ is concave. Its positive endpoint derivative in (L1) makes its maximum on $[0,\theta]$ equal to $Q-\zeta/(2\gamma(\theta))$. Ceilings in L,t change these bounds only by polynomial factors.

For P_* groups and a total of M_* requested selector rows, form separately for each group an $r\times s_*$ selector and an $s_*\times Bq$ data matrix. Chunk rows into size s_*^a and columns into size s_*^κ. Even a short final row chunk is charged: there are $O((1+r/s_*^a)Bq/s_*^\kappa)$ products per group. Their total cost is

$$\widetilde{O}\big(Bq[P_*s_*^{1+\alpha}+M_*s_*^\beta]\big). \tag{L4}$$

This dominates data formation BqP_*s_*, selector formation M_*s_*, and every output write BqM_*. It never concatenates unrelated groups for free. Formal coefficients, degree runs, window components, and translations contribute polynomial factors. Each output incidence fixes its own two distinguished sets; hence $M_*\le\widetilde{O}(M_0D^Q)$, without another factor K^2. Substitution of (L3), the $\widetilde{O}(N/D^T)$ bands, and (L4) gives (L2). Complete cyclic vectors are recovered for both inputs before their residual products, so the factor q is retained. □

Lemma 48 (A separate jointly capped high-box encoder)**.** *For the shifted and cropped sources above, the high-box bases have encoder exponent $E_{\mathrm{box}}=T-G$. The stored-box preprocessing and query exponents remain G and q_2.*

Proof. Use private $H(0)=-x_1-x_2+p_1$ on the normal side and $2s$ on the opposite side. Retain at most m private-or-star positions per half and at most $2m-t$ jointly. Expanding a star into an ordinary

or private position preserves both caps, so all high-box states have the required base leaves. Their count is

$$S_{\mathrm{box}} = \sum_{\substack{j,l\le m\\ j+l\le 2m-t}} \binom{L}{j}\binom{L}{l} 4^{2L-j-l} = \widetilde{O}(D^{2T-G}).$$

Indeed $\binom{L}{j}4^{-j}$ increases up to m for $c > 5$ and is log-concave. For each sum $j + l$ the maximum is balanced; the largest sum is $2m - t$, so the exponent is $2c - 2 + 2\theta + 2c\mathcal{H}((1-\theta)/c)/\ln 4 = 2T - G$. All-ordinary completions pay for every intermediate array; the root volume 12^L is below the 16^L all-ordinary leaves already included. Sharing over $\widetilde{O}(N/D^T)$ bands gives $\widetilde{O}(qND^{T-G})$ per block. All bases and intermediate boxes store their full cyclic vectors. The unchanged high-box recurrence costs $\widetilde{O}(qN^2D^{-G})$ per block, and a requested coefficient uses $\widetilde{O}(D^{q_2})$ stored scalar reads. Cropping is applied before these encoders as required by Lemma 45. □

Lemma 49 (A rectangular endpoint for the grouped completion)**.** *There is a fixed exact characteristic-zero algorithm with exponent $w = 16.7220002$ for aspect $(a, 1, \kappa)$, where $a = 861/500$ and $\kappa = 15$. Thus Lemma 47 may use $\alpha = 0.7220002$ and $\beta = 0.0000002$.*

Proof. Specialize Proposition 5.1 and Theorem 3.2 of [3] to level one of the big Coppersmith–Winograd tensor with parameter $q_{\mathrm{CW}} = 16$. Put probabilities $(ap, p, \kappa p, d_0, e_0, f_0)$ on its six types (011), (101), (110), (002), (020), (200), where

$$p = \frac{4000}{75249}, \quad e_0 = \frac{1}{18}, \quad f_0 = \frac{646603}{10^9}, \quad d_0 = e_0 - p - f_0 > 0.$$

They are positive and sum to one. Their marginals are

$$\begin{aligned} X &= (ap + d_0 + e_0, (\kappa+1)p, f_0),\\ Y &= (p + d_0 + f_0, (\kappa+a)p, e_0),\\ Z &= (\kappa p + e_0 + f_0, (a+1)p, d_0). \end{aligned}$$

The index-two marginals determine the three corners; the index-one marginals uniquely determine the other three probabilities. Thus the compatibility entropy penalty is zero. Choose the theorem's $A_1 = 1$, $A_2 = A_3 = 0$; at level one the complete split distributions are point masses, so their penalty is zero as well. The extraction rate is $\min\{H(X), H(Y), H(Z)\}$, where H here is ternary entropy. The constituent dimensions are $\langle 16^{pN}, 16^{\kappa pN}, 16^{apN}\rangle$. Cyclic permutation and the asymptotic sum inequality give

$$\omega(a, 1, \kappa) \le \frac{\ln 18 - \min\{H(X), H(Y), H(Z)\}}{p \ln 16}.$$

For this rational witness $Y = (1, 16, 1)/18$, and $H(Y)+p(\kappa+a)\ln 16 = \ln 18$ exactly. Rational logarithm enclosures give $H(X), H(Z) > H(Y)$, so the infimum is at most $\kappa + a = 16.722$. The verifier separately checks all three marginal inequalities using $w - 10^{-7}$; the remaining fixed 10^{-7} allowance permits choosing a finite exact algorithm. This does not assert a finite algorithm attaining the infimum.

The characteristic-zero tensor construction uses rational degenerations, interpolation, and bilinear arithmetic. Choose its finite base size and interpolation once, within that fixed allowance, then recurse. Intermediate coefficient growth is at most a constant per recursion level; input integers of $O(\log n)$ bits therefore retain $O(\log n)$ bits. Clearing the fixed rational denominators and exact recovery cost polynomial logarithmic factors. The same argument applies to the scalar coefficient matrices in (L4); their complete cyclic and formal coefficient counts have already been charged. □

Corollary 50 (Deterministic APSP with light structured sources)**.** *For polynomially bounded integer weights, all-edges Exact Triangle, min-plus product, and APSP without negative cycles admit deterministic $O(n^{2.9926863})$ algorithms.*

Proof. Use the composite-incidence construction of Theorem 38 with

$$c = 10.422903143551736, \quad \theta = 0.0656840679765022, \quad b = 0.04884094,$$
$$R = 10^{14}, \quad S = 76777119351634, \quad \eta = b/R, \quad u = S\eta, \quad v = b - u, \quad e = b + u,$$
$$\rho = 51/100, \qquad \zeta = 1/4, \qquad \delta = 0.0073137.$$

These are exact rational constants. In the first phase let the folded exponent x range over $[0, u]$ with cyclic exponent $y = 0$; in the second fix $x = u$ and let y range over $[0, v]$. The common middle-block count, outer dimension, inner dimension, and cyclic width are, up to the already charged logarithmic factors,

$$B = n^{1-b}, \qquad N = n^{1+x/2}, \qquad D = n^{b+x}, \qquad q = n^{y}.$$

For each fixed candidate, mask, bit level and restart, the folded location and requested coefficient depend only on the original outer pair and the common modulus, not on the middle block. The union of requested original pairs has size at most n^2, so (L2) applies with $M_0 \le n^2$. Each such call is charged separately in the prime-stage bound. Residues are folded at their actual locations; no extra modulus factor is inserted in the query count.

Use Lemmas 45–49, with all three encoder charges $E_{\mathrm{root}}, E_{\mathrm{res}}, E_{\mathrm{box}}$ kept separately. After charging every candidate prime by n^{η}, the savings against cubic time are

$$\begin{array}{ll} \text{stored-box construction:} & b - x - y + (b+x)G - \eta, \\ \text{residual completion and products:} & b - y - (b+x)q_1 - \eta, \\ \text{stored coefficient queries:} & b - (b+x)q_2 - \eta, \\ \text{each encoder } E_i: & 1 + b - x/2 - y - (b+x)E_i - \eta, \\ \text{input and Walsh formation:} & 1 - 3x/2 - y - \eta, \\ \text{incidence bookkeeping:} & b - \eta. \end{array} \tag{L5}$$

The independent feasibility margins are

$$1 + x/2 - (b+x)T > 0, \qquad 1 - b + y - \kappa(b+x)\zeta > 0. \tag{L6}$$

The first retains the original full-grid exponent T, even though the new encoders are cheaper. The second supplies the strict batch fit for (L4); it is not implied by geometry.

The exact-rational verifier `verify_light_source_apsp.py` checks 61 strict inequalities and the exact small-bias potential inequality. These include all source feasibility conditions, (L1), the three rectangular marginal bounds with positive allowance, (L5)–(L6) at both endpoints of both phases, and the exact prime-stage identities. Every phase expression is affine, so its endpoint checks cover the full phase. For interpretation only, the exponents are approximately

$$\begin{array}{lll} T = 11.7993263169, & G = 0.08470894353, & q_1 = 0.34960703849, \\ E_{\mathrm{res}} = 11.6676126746, & E_{\mathrm{box}} = 11.7146173734, & E_{\mathrm{root}} = 9.0305563991, \\ q_2 = 0.48097512444. & & \end{array}$$

The minimum runtime margin above δ exceeds $3.25 \cdot 10^{-8}$, and the minimum geometric margin exceeds $1.38 \cdot 10^{-7}$. All cyclic coefficients are charged in formation, encoders, completion, and residual products; stored boxes alone permit single-coefficient reads. Walsh weights, fixed local maps, $O(m)$ formal degrees,

residual coefficients, and polynomially many translations preserve polynomial coefficient norms. Exact arithmetic therefore has the bit costs charged above.

As before, the fixed R requires sufficiently large inputs to supply all distinct candidate primes. Factors such as $(\log n)^{O(R)}$, bit scaling, restarts, APSP squaring, padding, and the deterministic translation construction fit within the strict runtime margin. The resulting pre-absorption bound is $\widetilde{O}(n^{2.9926863-\epsilon_*})$ for a fixed $\epsilon_* > 0$, hence the stated ordinary $O(n^{2.9926863})$ bound. Earlier certificates and their separately parameterized counting, clique, real-input, 3SUM, and subset-convolution consequences remain valid. This corollary does not substitute its witness into those reductions. □

Corollary 51 (Exact-half sources and a refined numerical witness)**.** *For polynomially bounded integer weights, all-edges Exact Triangle, min-plus product, and APSP without negative cycles admit deterministic $O(n^{2.9926862626})$ algorithms.*

Proof. Apply Lemma 46 and the same grouped completion, jointly capped boxes, and rectangular endpoint, with

$$c = 10.42290323194818, \qquad \theta = 0.06568406426999, \qquad b = 0.04884094420,$$
$$R = 10^{14}, \qquad S = 76777127384957, \qquad \eta = b/R, \quad u = S\eta, \quad v = b - u, \quad e = b + u,$$
$$\rho = 1/2, \qquad \zeta = 11/50, \qquad \delta = 0.0073137374.$$

Retain $a = 861/500$, $\kappa = 15$, $w = 16.7220002$ and the fixed 10^{-7} matrix-algorithm allowance of Lemma 49. In particular $\alpha = 0.7220002$ and $\beta = 0.0000002$ are unchanged. Only the source parameter in $\gamma(r) = \mathcal{H}((\rho - r)/(c - 1))/\ln 4$ and the explicit rational witness change. All three encoder terms and every complete cyclic coefficient remain charged.

The verifier `verify_half_source_apsp.py` checks 61 strict rational inequalities. It verifies (L1), suffix feasibility, the rectangular endpoint with its allowance, all three encoders, both queries, input formation, bookkeeping, and both endpoints of both phases in (L5)–(L6). The exact identity $\rho = 1/2$ and the positive uniform potential margin use Lemma 46, in place of the earlier fixed-bias condition $\rho > 1/2$. The minimum runtime margin above δ exceeds $8.43 \cdot 10^{-11}$, and the original geometry margin exceeds $2.19 \cdot 10^{-10}$. At the last cyclic endpoint the box, residual-query, and stored-query savings each exceed 0.0073137374843; the residual encoder's saving exceeds 0.0075728. The additional polynomial factors in m from the shrinking-bias list, together with all earlier rounding, exact bit arithmetic, prime stages and APSP squaring, fit within these fixed positive gaps.

This is a small numerical refinement of the preceding corollary, accompanied by a stronger source-support lemma. Much of the change from the displayed 2.9926863 is slack in that earlier target. The original G, q_2, T and $q_1 \geq Q$ remain in force; no escape from the geometry/query cost family's bottleneck is claimed. The earlier witnesses and separately parameterized downstream statements retain their own certificates. □

Two black-box uses of the five-term data structure

Lemma 31 is a statement about an arbitrary $N \times D$ by $D \times N$ integer product with $N \geq KN_0$. Two earlier reductions use the thin-product data structure only as such a black box, and both improve when the five-term lemma replaces Theorem 2. Neither uses composite certification, cyclic coefficient rings, or caches, so the exponent functions are exactly those of Lemma 31.

Corollary 52 (Five-term thin products in the source's regimes)**.** *Let $X \in \mathbb{Z}^{N\times D}$, $Y \in \mathbb{Z}^{D\times N}$ have entries of magnitude $N^{O(1)}$, and let W contain at most $N^2/\sqrt{D}$ positions. If $N \geq D^{18}$, the entries*

$(XY)[I,J]$, $(I,J)\in W$, *are computed deterministically in* $O(N^2/D^{0.0991})$ *time, and after that preprocessing any single entry is returned in* $O(D^{0.4009})$ *time. If* $N\ge D^{14}$, *the times are* $O(N^2/D^{0.0853})$ *and* $O(D^{0.4147})$.

Proof. Pad D to a power of four and apply Lemma 31 with

$$(c,\theta,r,y)=(16.2514,\ 0.05033,\ 0.04997,\ 0.02004)$$

for $N\ge D^{18}$ and with

$$(c,\theta,r,y)=(12.420851,\ 0.05478,\ 0.0150708,\ 0.01)$$

for $N\ge D^{14}$. The costs are $N^2D^{-G}+|W|D^{\max(q_1,q_2)}+ND^{E_{\mathrm{aug}}}+ND^{E_{\mathrm{box}}}$ with the tile condition $N\ge D^T$. The exact-rational certificate checks, for the first witness, $G=0.099159\ldots>0.0991$, $1/2-q_1=0.099179\ldots>0.0991$, $1/2-q_2>0.0991$, $q_1=0.400820\ldots<0.4009$, $T=17.9613\ldots<18$, and $E_{\mathrm{aug}}+0.0991<18$, so that both encoder families are below $N^2D^{-0.0991}$; and for the second witness $G=0.085309\ldots>0.0853$, $1/2-q_i>0.0853$, $q_2=0.414690\ldots<0.4147$, $T=13.9296\ldots<14$, and $E_{\mathrm{aug}}+0.0853<14$. The strict gaps absorb the padding and the polylogarithmic factors of the lemma. □

The source displays $O(N^2/D^{0.063})$ and $O(D^{0.437})$ at $N\ge D^{18}$; the ten-term analysis of Theorem 2 gives 0.07195 and 0.42805 there; Majdoddin [6] displays 0.092 and 0.403 at $N\ge D^{18}$ and 0.077 and 0.421 at $N\ge D^{14}$. The fixed-profile slots, the two separately charged encoders, and the prefix queries account for the difference from [6]; the tile condition is the same.

Corollary 53 (Real inputs with the five-term data structure)**.** *In the real RAM with comparisons, additions, and subtractions on real numbers, real 3SUM has a Las Vegas algorithm with expected time* $O(n^{1.993805})$, *and real min-plus product, APSP without negative cycles, and Exact Triangle have Las Vegas algorithms with expected time* $O(n^{2.993805})$.

Proof. Lemma 29 calls the thin-product data structure on explicit integer matrices with entries at most d and inner dimension $D=\Theta(d^2\log^5 n)$, rounded up to a power of four, and uses no other property of it. Replace Theorem 2 by Lemma 31 with

$$(c,\theta,r,y,e)=(10.7769,\ 0.0567,\ 0.03854,\ 0.02511,\ 0.08183),\qquad d=\max\{1,\lfloor n^{e/2}\rfloor\}.$$

The certified comparison counts, their complementary certification, the fresh primes for adaptive calls, and the expected-round argument are unchanged. For real 3SUM the expected cost is

$$\widetilde{O}\big(nD^{E_{\mathrm{aug}}}+nD^{E_{\mathrm{box}}}+n^2D^{-G}+n^2D^{\max(q_1,q_2)-1/2}+nD+n^2D^{-1/2}+nd\big),$$

and for real min-plus, APSP, and Exact Triangle it is

$$\widetilde{O}\big(n^2D^{E_{\mathrm{aug}}}+n^2D^{E_{\mathrm{box}}}+n^3D^{-G}+n^3D^{\max(q_1,q_2)-1/2}+n^2D+n^2d+n^3/d\big),$$

exactly as before with the two encoder families and the two query exponents of the five-term lemma. The exact-rational certificate gives the savings

$$eG=0.0061952\ldots,\qquad e(1/2-q_1)=0.0061954\ldots,$$
$$e(1/2-q_2)=0.0061952\ldots,\qquad 1-eE_{\mathrm{aug}}=0.0062665\ldots,$$

all above 0.006195, with $1-eE_{\mathrm{box}}>0.0096$, the remaining terms far from binding, and the unfolded tile condition $1-eT=0.003422\ldots>0$. The strict gaps absorb the rounding of D, the logarithms, and the expected number of certification rounds. Hence real 3SUM takes expected $O(n^{2-0.006195})=O(n^{1.993805})$ time and the three cubic problems expected $O(n^{2.993805})$ time. □

Majdoddin [6] notes that the real-valued bounds of the source follow by inserting his matrix theorem into the source's argument, which keeps a quarter of the saving, and leaves open whether real-input reductions can keep the whole saving. Lemma 29 keeps the whole unfolded saving with a Las Vegas guarantee, matching the guarantee type of the source's real-input reduction; the corollary above is its five-term instance. The deterministic integer bounds of this section are also Las Vegas bounds, so for integer inputs the Las Vegas rows of the summary table are superseded by 2.9926862626 and 1.9931055. For lopsided clique instances see Corollaries 55 and 56.

7 Further displayed bounds

The consequences below include both substitutions into source reductions and explicitly proved changes to their blocking, counting, or covering steps. Logarithmic factors are absorbed only after checking strict slack.

Weighted k-Clique

Use the standard weighted Nešetřil–Poljak reduction underlying Corollary 39 of the source. Group the k vertex positions into groups of a, a, m positions, with $k = 2a + m$. The tripartite graph has outer parts of size at most $R = n^a$ and a middle part of size at most $H = n^m$. Vertices represent valid internal cliques; edges require disjoint, compatible choices. Assign each original edge weight to exactly one tripartite edge, using the source's designated-edge-class convention for weights internal to one group. A triangle then has exactly the original k-clique weight. Missing edges can be filled with a sufficiently large polynomial sentinel. Construction takes $O(R^2 + RH)$ time for fixed k. Minimum weight is obtained by one rectangular min-plus product followed by adding the outer-edge weights and minimizing; maximum weight follows by negating all original weights.

Corollary 54 (Every fixed weighted clique size)**.** *For each fixed integer $k \geq 3$, Zero-, Min-, and Max-Weight k-Clique with polynomially bounded integer weights have Las Vegas expected time*

$$O\left(n^{k-0.004057\lfloor k/2.02914\rfloor}\right)$$

and deterministic time

$$O\left(n^{k-0.002125\lfloor k/2.09687\rfloor}\right).$$

Proof. For the Las Vegas bound, put $a = \lfloor k/2.02914\rfloor$ and $m = k - 2a$. Group the k vertex parts into groups of a, a, m parts, as in the lopsided reduction above. Then $m \geq 0.02914a$, so its middle part of n^m vertices is at least $\sqrt{D}$ for $D = \Theta(n^{0.05828a})$, with power-of-four rounding downward. Apply Lemma 12 with $R = \Theta(n^a)$ and the already certified folded parameters $(c, \theta, e) = (21.0477, 0.111971, 0.05828)$. Each of its four cost terms has strict saving greater than $0.004057a$ below n^{2a+m}, and the folded tile condition has positive slack. The graph construction costs $O(R^2 + RH)$: its R^2 term saves $m \geq 0.02914a > 0.004057a$, and its RH term saves a. The same bit-scaling argument gives the min-plus product used for minimum weight, and negating the input weights gives maximum weight. This removes the previous upper restriction on k by allowing the middle group to grow and by eliminating the successful-scan term.

For the deterministic bound put $a = \lfloor k/2.09687\rfloor$, $m = k-2a$, and retain the already certified parameters $(c, \theta, e, z) = (21.6844, 0.1163026, 0.0565835, 0.0021252)$, with $D = \Theta(n^{ae})$. Then $m \geq 0.09687a$. The cyclic-polynomial multiplication argument below computes prime-selection ring operations in quasilinear time in the modulus, giving a factor P^2 over all candidate primes, where $P = \Theta(n^{a(e+z)/2})$. Uniform

square blocking of an $R \times H$ by $H \times R$ product costs $\widetilde{O}(R^2 H / \min(R, H)^{3-\omega})$ ring operations: use blocks of order H when $H \le R$, and split the inner dimension into blocks of order R otherwise. This also covers $k = 4$ without a separate exception. Thus the saving for prime selection is at least

$$0.628 \min(a, m) - a(e + z) \ge a(0.628 \cdot 0.09687 - e - z) = 0.00212566a > 0.002125a.$$

Successful scans save $m - a(e - z)/2 \ge 0.06964085a$, and the other costs have the previously certified savings scaled by a. The middle part contains full blocks, and both tile and graph-construction costs have strict slack. The deterministic bit-scaling transfer and negation give the minimum- and maximum-weight versions. □

Corollary 23 above improves the deterministic bound to $O(n^{k-0.003739\lfloor k/3 \rfloor})$ for every fixed $k \ge 3$, with $O(n^{4.99276})$ for $k = 5$. The square-case five-term bound improves it further without any lopsided accounting.

Corollary 55 (Chunked clique bounds from the square case)**.** *For every fixed $k \ge 3$, Zero-, Min-, and Max-Weight k-Clique with polynomially bounded integer weights take $O(n^{k-0.00705208\lfloor k/3 \rfloor})$ deterministic time.*

Proof. Put $a = \lfloor k/3 \rfloor$ and $m = k - 2a \ge a$, and group the parts as (a, a, m) as in Corollary 28. Partition the middle part of n^m vertices into n^{m-a} chunks of n^a vertices. Each chunk, with the two outer parts, is a square all-edges Exact Triangle instance of size $N = n^a$ with weights polynomial in N, since a is fixed. Corollary 44 solves it deterministically in $O(N^{3-0.00705208})$ time; the zero-weight all-edges answers are the entrywise disjunction over chunks, and the min-plus product of the lopsided instance is the entrywise minimum of the chunk products, so the deterministic bit-scaling transfer gives the minimum- and maximum-weight versions. The total is $n^{m-a} N^{3-0.00705208} = O(n^{k-0.00705208a})$, and the graph construction $O(R^2 + RH) = O(n^{k-a})$ is smaller. □

Chunking needs $m \ge a$. The composite-incidence construction itself needs only one block to fit in the middle part, which allows a split with two nearly equal outer groups.

Corollary 56 (Lopsided five-term clique bounds)**.** *For every fixed $k \ge 3$, Zero-, Min-, and Max-Weight k-Clique with polynomially bounded integer weights take deterministic time*

$$O\Big(n^{\,k-0.00705208\lfloor k/2.046046346 \rfloor}\Big),$$

where $2.046046346 = 2 + b$ with the aligned-cache block exponent $b = 0.04604634600$. For example, $k = 5$ gives $O(n^{4.98589584})$ and $k = 7$ gives $O(n^{6.97884376})$.

Proof. Put $\alpha = \lfloor k/(2 + b) \rfloor \ge 1$ and $m = k - 2\alpha$, so that $m \ge b\alpha$, and group the parts as (α, α, m): outer parts $R = n^\alpha$, middle part $H = n^m = R^\mu$ with $\mu = m/\alpha \ge b$. Run the algorithm of Corollary 44 on this instance with every exponent measured in powers of R: blocks of $h = \Theta(R^b)$ middle vertices fit because $b \le \mu$; for a modulus $d = R^u$ each block is one folded product with $D = hd = R^{b+u}$ and outer dimension $R\lceil\sqrt{d}\rceil = R^{1+u/2}$ on both sides; the incidences (a, b, B) number at most $2R^2 H/h$. Lemma 33 and Theorem 38 use only h, the prime scale, the divisor bound, and sums of exact per-incidence counts, so their proofs apply with $2R^2 H/h$ incidences, $\lceil \log_2(2H) \rceil = O(\log n)$ halving stages, and a final scan of $O(R^2)$ pairs. Every per-block cost carries the factor H/h and every per-incidence cost the factor $R^2 H/h$, while the budget is $R^2 H = n^k$; hence the savings per unit α are exactly the entries of (A12)–(A13), all exceeding 0.00705208 at the certified endpoints, and the tile condition $1 + u/2 > (b + u)T$ is unchanged. Unlike Theorem 20, the selection is linear in the number of candidates and incidences, so no regime depends on H against R. The remaining costs are the graph construction $O(R^2 + RH)$ and the final

scan, with savings $\mu \geq b$ and 1 below $R^{2+\mu}$, both above 0.00705208. The zero-weight all-edges answers decide and locate zero cliques; the rectangular $R \times H \times R$ min-plus product follows by the deterministic bit-scaling transfer, and adding the outer edge weights and minimizing gives minimum weight, with negation for maximum weight. □

This contains Corollary 55 as the case $\alpha = \lfloor k/3 \rfloor$, and it supersedes the $k = 5$ bounds $O(n^{4.991886})$ and $O(n^{4.99276})$ above and, being deterministic, the Las Vegas clique bounds of Corollary 54 and of the $(2, 1, 1)$ split below for every $k \geq 3$. Majdoddin's Corollary 39 analogue [6] gives the saving $0.0055\lfloor k/3 \rfloor$. The split is explicit and feasible, not claimed optimal; the fixed-profile parameters of Corollary 35 give the weaker $O(n^{k-0.006324\lfloor k/2.0419064555 \rfloor})$ by the same argument.

Four vertices. For $k = 4$ the split $(2, 1, 1)$ beats $(1, 1, 2)$. Take outer parts $R_1 = n^2$ and $R_2 = n$ and middle part $H = n$, fold the whole residue onto the small side, so the rows are the n^2 vertices of R_1 and the columns are the pairs (b, ρ) with $\rho \in \mathbb{Z}_p$, and use the rectangular tiles of Lemma 69 in the clamped regime $K_1 = K$, $K_2 = 1$, which applies because $R_1/(R_2 p) = n^{1-d/2} \geq K$ for $D = n^d$ when $1 - d/2 - dC_K > 0$. Per block the boxes cost $R_1 R_2 p D^{-G}$, the queries $R_1 R_2 D^q$, and the encodings $R_1 N_0 + R_2 p K N_0 \leq R_1 D^{A_0} + \sqrt{R_1 R_2 p}\, D^E$; over the n/h blocks the savings below n^4 are dG, $d(1/2-q)$, $1-d(A_0-1/2)$, and $3/2-d(E-1/4)$, with the small side holding a tile when $1+d/2 > dA_0$. Take $(c, \theta, d) = (19.8418,\ 0.113322,\ 0.06896)$: the savings are $0.004514113\ldots$, $0.004514199\ldots$, $0.004784918\ldots$, and $0.388997\ldots$, with $1 - d/2 - dC_K = 0.768\ldots > 0$ and $1 + d/2 - dA_0 = 0.004784\ldots > 0$. Hence Zero-, Min-, and Max-Weight 4-Clique with integer weights take $O(n^{3.995486})$ Las Vegas expected time, against $O(n^{3.995943})$ from the balanced split; with the symmetric fold of Lemma 8 the same split gives $O(n^{3.995561})$.

Real weighted cliques for every fixed k

The rank-prefix oracle also permits a split that works for every fixed $k \geq 3$. Put

$$e_1 = 0.056846, \quad u_1 = e_1/2 = 0.028423, \quad a = \left\lfloor \frac{k}{2 + u_1} \right\rfloor, \qquad m = k - 2a.$$

Then $a \geq 1$ and $m \geq a u_1$. Group the k vertex parts into groups of a, a, m parts. The standard weighted Nešetřil–Poljak reduction has outer sizes $R = n^a$ and middle size $H = n^m$. Use the certified real parameters $(c, \theta) = (21.2733, 0.111731)$, list length $d = \max\{1, \lfloor R^{u_1} \rfloor\} \leq H$, and the actual rounded dimension $D = \Theta(R^{e_1} \log^5 R)$ of Lemma 29.

The source's Lemma 36(a) applies to these rectangular products by partitioning their H middle indices into blocks of at most d indices. There are $\widetilde{O}(H/d)$ counting calls, each with d comparison instances on R row and R column lists whose prescribed pair sets have total size R^2. Its additional list and minimum-selection costs are $\widetilde{O}(RHd + R^2 H/d)$. The resulting complete expected time is

$$\widetilde{O}\left(RHD^E + R^2 H D^{-G} + R^2 H D^{q-1/2} + RHD + RHd + R^2 H/d\right).$$

Relative to $R^2 H = n^k$, the first three savings are respectively $a(1 - e_1 E)$, $a e_1 G$, and $a e_1 (1/2 - q)$, each strictly greater than $0.003999a$. The remaining savings are $a(1 - e_1)$, $a(1 - u_1)$, and $a u_1$, and the strict tiling condition is $1 - e_1 T > 0$. All polylogarithmic factors are therefore absorbed. Constructing the tripartite graph takes $O(R^2 + RH)$ time for fixed k; this fits as well, because $m \geq a u_1 > 0.003999a$ and $a > 0.003999a$.

Exact Triangle tests the zero-weight case. For minimum weight, compute the rectangular min-plus product on the same three parts, add the edge between the two outer groups, and minimize over its R^2

entries; negating the weights handles maximum weight. Thus Zero-, Min-, and Max-Weight k-Clique with real weights have Las Vegas expected time

$$O\Big(n^{k-0.003999\lfloor k/2.028423\rfloor}\Big) \qquad \text{for every fixed } k \geq 3.$$

This is an algorithmic guarantee in the same real model as the source, not an assertion of literature priority. The choice of a is one explicit feasible split, not a claim that the split is optimal.

With the five-term data structure the same split improves. Take the parameters of Corollary 53, $e_1 = 0.08183$, $u_1 = e_1/2 = 0.040915$, $a = \lfloor k/(2+u_1)\rfloor$, and $m = k - 2a \geq au_1$; the lists have length $d = \max\{1, \lfloor R^{u_1}\rfloor\} \leq H$ and the rounded dimension is $D = \Theta(R^{e_1}\log^5 R)$. The cost has the two encoder terms $RHD^{E_{\text{aug}}} + RHD^{E_{\text{box}}}$ in place of RHD^E and the query exponent $\max(q_1, q_2)$, and otherwise is the display above. Relative to n^k the savings are $a(1-e_1E_{\text{aug}})$, $a(1-e_1E_{\text{box}})$, ae_1G, and $ae_1(1/2-q_i)$, each above $0.006195a$ by the certificate of Corollary 53, with the remaining savings $a(1-e_1)$, $a(1-u_1)$, au_1, the tiling condition $1 - e_1T > 0$, and graph construction fitting because $m \geq au_1 > 0.006195a$. Hence Zero-, Min-, and Max-Weight k-Clique with real weights have Las Vegas expected time

$$O\Big(n^{k-0.006195\lfloor k/2.040915\rfloor}\Big) \qquad \text{for every fixed } k \geq 3,$$

which is below the display above for every $k \geq 3$ and no worse than the chunked square bound $0.006195\lfloor k/3\rfloor$ for every k.

Problems solved by one product or one APSP

Negative Triangle and Min-Weight Triangle, also in their all-edges forms, Metricity, min-plus product verification, Diameter, Radius, Median, the Wiener index, and directed Minimum Weight Cycle with nonnegative weights (the minimum over edges (u, v) of $w(u, v) + d(v, u)$) each need a constant number of $n \times n$ integer min-plus products or one APSP plus $O(n^2)$ work. By Corollary 51 they inherit $O(n^{2.9926862626})$ deterministic time, which is also a Las Vegas bound, and with real weights $O(n^{2.993805})$ Las Vegas expected time; the source displays $O(n^{2.9995})$. Exact Triangle with real weights is covered by the source's Lemma 36(a) together with the min-plus product, with the same call counts, so it also takes $O(n^{2.993805})$ Las Vegas expected time.

Directed APSP with small integer weights

The source's Theorem 33 turns a bound $O(n^{2+\mu-\varepsilon_1})$ for one rectangular min-plus product into a bound $O(n^{2+\mu-\varepsilon''})$ for directed APSP with weights of magnitude polylog n, with ε'' left unspecified. We make the step quantitative. Throughout, $\bar{\mu} = 0.5275$ and $\bar{\omega} = 2.372$ are the source's displayed bounds: $\omega(1, \bar{\mu}, 1) \leq 1 + 2\bar{\mu}$ follows from $\mu < \bar{\mu}$ and the trivial $\omega(1, \kappa + t, 1) \leq \omega(1, \kappa, 1) + t$ (the table of [4] lists 2.054999 at $\bar{\mu}$), and $\omega \leq \bar{\omega}$. Zwick's algorithm runs in $\widetilde{O}(n^{2+\bar{\mu}})$.

Lemma 57 (Uniform lopsided product bound)**.** *Let $b = 0.04604634600$ and $\delta = 0.00705208$ be the block exponent and the saving of Corollary 44. For every fixed rational $\rho \in [b, 1]$, the min-plus product of an $n \times n^\rho$ by an $n^\rho \times n$ matrix with integer entries of magnitude $n^{O(1)}$ and ∞ entries is computed deterministically in $\widetilde{O}(n^{2+\rho-\delta})$ time, where the logarithmic power depends only on the magnitude exponent and the fixed parameters; a middle part of size $\widetilde{\Theta}(n^\rho)$ is allowed. Like Corollary 44, the bound is asymptotic and holds only for astronomically large n.*

Proof. Replace ∞ by a sentinel $S = 3W + 1$, where W bounds the finite magnitudes; a sum containing a sentinel exceeds $2W$, so outputs above $2W$ are ∞. Run the bit-scaling transfer of the subsection

on integer APSP: $O(\log n)$ levels, each with two all-edges Exact Triangle calls on the tripartite graph with outer parts of $R = C = n$ vertices and middle part $H = n^\rho$, all weights polynomial. Solve each call exactly as in Corollary 56 with $\alpha = 1$ and middle exponent $\mu = \rho \geq b$: blocks of $h = \Theta(n^b)$ middle vertices fit, every per-block cost carries the factor H/h, every per-incidence cost the factor $R^2H/h = n^{2+\rho-b}$, and the budget is $R^2H = n^{2+\rho}$; so the savings are exactly the entries of (A12)–(A13), all above δ at their certified endpoints, with the unchanged tile condition. The graph construction $O(R^2 + RH)$ and the final scan $O(R^2)$ save $\rho \geq b > \delta$ and 1. The strict gap absorbs the $O(\log n)$ levels. □

Theorem 58 (Directed APSP with small integer weights, explicit exponent)**.** *APSP in directed n-vertex graphs with integer weights of magnitude $(\log n)^{O(1)}$ and no negative cycles, in particular directed unweighted APSP, is solved deterministically in $O(n^{2.525754})$ time. More precisely, for every certified chord slope $m < 1$ with $\omega(1, \kappa, 1) \leq 1 + 2\bar{\mu} + m(\kappa - \bar{\mu})$ on $[\bar{\mu}, \kappa_1]$, and $c = 1 - m$, the exponent is $2 + \bar{\mu} - \varepsilon''$ for every*

$$\varepsilon'' < \frac{\delta c}{1+c}, \qquad \text{provided } \bar{\mu} + \frac{\delta}{1+c} < \kappa_1$$
$$\text{and } 1 - \kappa + \omega(1, \kappa, 1) < 2 + \bar{\mu} - \varepsilon'' \text{ at the certified nodes } \kappa \in [\kappa_1, 1].$$

With the chord from $(\bar{\mu}, 1 + 2\bar{\mu})$ to $(1, \bar{\omega})$ this is $\varepsilon'' < \delta(2 + \bar{\mu} - \bar{\omega})/(3 - \bar{\omega}) = 0.0017461\ldots$.

Proof. Run the deterministic reduction of the source's Theorem 33 with level ratio $R = \lceil n^\gamma \rceil$, $\gamma = 10^{-8}$. Its tables cost $\widetilde{O}(n^{\omega+\gamma})$ and its bridges $\widetilde{O}(n^{2+\gamma})$ per stage, with $O(1/\gamma)$ stages; both are below $n^{2+\bar{\mu}-\varepsilon''}$ since $\bar{\omega} + \gamma < 2 + \bar{\mu} - \varepsilon''$. The stage for $s = t_{i+1} = n^\sigma$ needs one min-plus product of an $n \times n^\rho$ by an $n^\rho \times n$ matrix with $\rho = 1 - \sigma + \gamma$ and finite entries of magnitude $\widetilde{O}(n^{1-\rho+\gamma})$, together with ∞ entries. We bound its exponent $\Phi(\rho)$ by the minimum of three deterministic methods. (i) Naive: $2 + \rho$. (ii) Lemma 57, for $\rho \geq b$: $2 + \rho - \delta$. (iii) Zwick's Lemma 3.3, cited by the source: $\widetilde{O}(Wn^{\omega(1,\rho,1)+\varepsilon})$ for entries of magnitude W, giving $g(\rho) = 1 - \rho + \gamma + \varepsilon + w(\rho)$ with $\varepsilon = 10^{-8}$ the matrix-multiplication exponent allowance.

The property of $\omega(1, \cdot, 1)$ used. Only this: if $\omega(1, \kappa_0, 1) \leq w_0$ and $\omega(1, \kappa_1, 1) \leq w_1$ are algorithmic bounds, then for rational $\lambda \in [0, 1]$, $\omega(1, (1-\lambda)\kappa_0 + \lambda\kappa_1, 1) \leq (1-\lambda)w_0 + \lambda w_1$, by tensoring the two algorithms in rational proportions, exactly as in the interpolation paragraph after Lemma 24. No global convexity, monotonicity, or lower bound $\omega(1, \kappa, 1) \geq 2$ is needed; the certified point at $\bar{\mu}$ follows from $\mu < \bar{\mu}$ by padding the middle dimension, since $\omega(1, \kappa + t, 1) \leq \omega(1, \kappa, 1) + t$. Hence $w(\rho) = 1 + 2\bar{\mu} + m(\rho - \bar{\mu})$ on $[\bar{\mu}, \kappa_1]$ with $m = (w_1 - 1 - 2\bar{\mu})/(\kappa_1 - \bar{\mu})$, and g decreases on this segment at rate $c = 1 - m > 0$ from $g(\bar{\mu}) = 2 + \bar{\mu} + \gamma + \varepsilon$.

Crossover. For $\rho < b$, $\Phi \leq 2 + b$. For $b \leq \rho \leq \rho^*$ use (ii), and for $\rho^* \leq \rho \leq \kappa_1$ use (iii), where $2 + \rho^* - \delta = g(\rho^*)$, that is $\rho^* = \bar{\mu} + (\delta + \gamma + \varepsilon)/(1 + c)$. Both bounds equal $2 + \bar{\mu} - \varepsilon_2$ there, with

$$\varepsilon_2 = \frac{\delta c - \gamma - \varepsilon}{1+c},$$

and (ii) increases while (iii) decreases, so $\max_{\rho \leq \kappa_1} \Phi(\rho) \leq 2 + \bar{\mu} - \varepsilon_2$. On $[\kappa_1, 1]$ the piecewise-linear g is checked at its nodes. With the single chord to $(1, \bar{\omega})$, $c = (2 + \bar{\mu} - \bar{\omega})/(1 - \bar{\mu})$ and $\varepsilon_2 \to \delta(2 + \bar{\mu} - \bar{\omega})/(3 - \bar{\omega})$ as $\gamma, \varepsilon \to 0$. The exact-rational certificate gives $\varepsilon_2 = 0.001746160812\ldots > 0.001746$ with $\rho^* = 0.5328059\ldots \in [b, 1]$, so every stage product costs $O(n^{2+\bar{\mu}-0.001746})$ and the strict gap absorbs the fixed number of stages, the $O(\log n)$ bit-scaling levels, and all polylogarithmic factors. Every step is deterministic. The exponents are certified in the script `verify_directed_apsp.py`. □

The reduction's own hypothesis form, with entries bounded by $n^{1-\mu}$ only and a saving ε_1 at $\rho = \bar{\mu}$ alone, gives instead $\varepsilon'' < \min\{\varepsilon_1(1 - \bar{\mu})/(3 - \varepsilon_1),\ \varepsilon_1(2 + \bar{\mu} - \bar{\omega})/(3 - \bar{\omega})\}$: for $\rho < \bar{\mu}$ the padded instance of

size $n^{(1-\rho)/(1-\bar{\mu})}$ meets the naive bound at $\bar{\mu}-\rho=\varepsilon_1(1-\bar{\mu})/(3-\varepsilon_1)$, and this side binds (0.0011133 at $\varepsilon_1=\delta$). The uniform lemma removes that side. Chunking the outer parts into $n^{\bar{\mu}}$-sized square products gives only $\varepsilon_1=\bar{\mu}\delta$. One further certified row, $\omega(1,0.6,1)\le 2.092351$ ([4], Table 1), gives the chord slope $m=0.5151\ldots$, $c=0.4848\ldots$, $\rho^*=0.5322\ldots<0.6$, and $\varepsilon_2>0.0023025$, that is $O(n^{2.5251975})$; the SODA 2024 row $\omega(1,0.55,1)\le 2.066134$ gives $\varepsilon_2>0.0023667$ and $O(n^{2.5251333})$. These two refinements use the additional external table rows of [4] and [3], recorded in the bibliography; the main statement uses only the two numbers the source displays. The Las Vegas route through Lemma 12 is not better: a deterministic bound is a Las Vegas bound, and no certified Las Vegas saving exceeds δ.

Hinted OMv

The source's Corollary 40 refutes the three hinted OMv conjectures of van den Brand, Nanongkai, and Saranurak for thin hints by preprocessing the hint with the data structure. With $N=n$ and $D=n^{\tau}$, Phase 2 costs $\widetilde{O}(n^{1+\tau E}+n^{2-\tau G})$, Phase 3 is n queries, $\widetilde{O}(n^{1+\tau q})$, and the tile condition is $\tau T<1$; for the uMv conjecture, n/t_2 products with $N=t_2$ and $D=t_1$ give the same conditions with $\tau=t_1/t_2$. Two observations follow.

1. For $0<\tau\le 1/18$, the parameters of the first corollary give Phase 2 in $O(n^{2-0.07195\tau})$ and Phase 3 in $O(n^{1+0.42805\tau})$, since $G>0.07195$, $q<0.42805$, and $18-E>0.07195$; the source displays 0.063 and 0.437. Correspondingly Conjecture 5.12 of that paper fails for $\tau_1\le\tau_2/18$ with Phase 3 in $O(n^{1+\tau_2-0.07195\tau_1})$ and Phase 4 in $O(n^{\tau_2+0.42805\tau_1})$. Preprocessing U times $N_{I,J}$ as one $n\times t_1$ by $t_1\times t_2$ product with the rectangular tiles of Lemma 69, instead of n/t_2 square products, costs $\widetilde{O}(n^{1+\tau_1 A_0}+n^{(1+\tau_2)/2+\tau_1 E}+n^{1+\tau_2-\tau_1 G})$, so the same Phase 3 and Phase 4 bounds hold for all $\tau_1\le\min\{\tau_2/16.54,\,(1+\tau_2)/36\}$, since $16.54>A_0+0.07195$ and $36>2(E+0.07195)$ at these parameters.

2. The conjectured Phase 3 bound is $n^{1+\tau-\varepsilon}$, so any query exponent $q<1$ refutes it, not only $q<1/2$. The branch $c<10$ of Theorem 2 then reaches beyond the source's limit $\varepsilon^*=0.1204\ldots$: for $0<\tau\le 0.14$ take $c=8.58$ and $\theta=0.335$, where $G=0.00971\ldots>0.0097$, $q=0.99093\ldots<0.991$, $1/0.14-E=0.01282\ldots>0.0097$, and $1/0.14>T$, giving Phase 2 in $O(n^{2-0.0097\tau})$ and Phase 3 in $O(n^{1+0.991\tau})$; for $0<\tau\le 0.145$ take $c=8.29$ and $\theta=0.3385$, where $G>0.001$, $q<0.999$, $1/0.145-E>0.001$, and $1/0.145>T$, giving $O(n^{2-0.001\tau})$ and $O(n^{1+0.999\tau})$. Conjecture 5.12 fails accordingly for $\tau_1\le 0.145\,\tau_2$ with square products, and with one clamped rectangular product for all $\tau_1\le\min\{0.1726\,\tau_2,\,(1+\tau_2)/13.8\}$, using the $c<10$ encoding of Lemma 69: at $c=8.29$, $\theta=0.3385$ one has $1-0.1726(A_{\mathrm{cl}}+0.001)>0$, $1-0.1726A_0>0$, and $1-2(E+0.001)/13.8>0$. The limits of these families, about 0.1457 and 0.1736, are not claimed; the savings vanish there.

All of these are deterministic, as in the source.

Rectangular min-plus convolution

The square block size in the usual min-plus-convolution reduction loses nearly half the available saving. A rectangular block choice retains much more. For any integer $1\le s\le n$, put $r=\lceil n/s\rceil$ and form

$$Q[u,v]=b[su+v],\qquad P_z[d,v]=a[sd+z-v],$$

where $0\le u<r$, $0\le v,z<s$, and $0\le d\le r$. Replace absent entries by a sentinel $S>3W$, where W bounds all input magnitudes. For $F_z=P_z\star Q^{\mathsf{T}}$, every output satisfies

$$c[sw+z]=\min_{\substack{0\le u<r\\ 0\le w-u\le r}} F_z[w-u,u],\qquad 0\le sw+z\le 2n-2.$$

Indeed, writing the index of the b entry as $j = su + v$ forces the index of the a entry to be $i = s(w-u) + z - v$; every genuine pair occurs, and values of $d = w - u$ outside $[0, r]$ cannot give a valid a index. The sentinel rule prevents absent entries from beating a genuine sum, also for negative inputs. Both boundary rows are retained and all $2n-1$ outputs are produced.

There are s products of an $(r+1) \times s$ by $s \times r$ matrix. Padding the outer dimensions to $r+1$ changes only constants. Forming all inputs costs $O(ns + s^2)$ and reading all product entries and taking their diagonal minima costs $O(sr^2)$. Take

$$s = \lceil n^{\beta} \rceil, \quad r = \Theta(n^{\alpha}), \quad \alpha = 1 - \beta, \quad D = 4^{\lfloor \log_4(r^e) \rfloor} = \Theta(r^e).$$

Multiplying (5) by the s products gives the four savings below n^2

$$\alpha\big(1 - e(E - 1/4)\big), \quad \alpha e G, \quad \alpha e(1/2 - q), \quad \alpha(1 - 3e/4). \tag{7}$$

The folded tile condition has positive gap $\alpha(1 - e(T - 1/4))$, and the middle-size condition holds when $\beta - \alpha e/2 > 0$. Formation and extraction save $1 - \beta$ and β, respectively. Choose

$$(c, \theta, e, \beta) = (21.487,\ 0.111507,\ 0.057093,\ 0.027755).$$

Then $\beta - \alpha e/2 = 0.0000008081075 > 0$. The box, query, and encoding savings are respectively

$$0.003943484351370\ldots, \quad 0.003943490849588\ldots, \quad 0.003953476962084\ldots.$$

All terms have strict saving greater than 0.003943. Thus integer min-plus convolution takes Las Vegas expected

$$\boxed{O(n^{1.996057})}.$$

This improves the $O(n^{1.997972})$ square-block consequence.

The deterministic version also benefits from rectangular blocks. Use the explicit folded parameters

$$(c, \theta, e, z) = (23.1026,\ 0.1151488,\ 0.053132,\ 0.00212056),$$

and write $A = (e + z)/2$, $B = (e - z)/2$. For each product choose a prime of scale r^A minimizing its modular triangle count, and use middle blocks of scale r^B. The same averaging and distinct-false-triple charging as in the deterministic reduction give, after all s products, savings

$$\alpha(eG - z), \quad \alpha\big(e(1/2 - q) - z/2\big), \quad \alpha z, \quad \alpha\big(1 - e(E - 1/4) - 3z/4\big) \tag{8}$$

for boxes, queries, failed scans, and encodings. Construction and successful scans save $\alpha(1 - 3A/2)$ and $\beta - \alpha B$. The tile condition has gap $\alpha(1 + A/2 - eT) > 0$.

Prime selection adds a constraint, but its ring operations admit a useful improvement. Degree-p cyclic-polynomial multiplication takes $\widetilde{O}(p)$ word operations, rather than quadratic time, when coefficients have $O(\log n)$ bits. To justify that bit bound within the matrix algorithm, fix an exact rational bilinear matrix-multiplication algorithm of constant block size and exponent less than 2.372, using the same matrix-multiplication bound as the source. Its recursion has depth $O(\log s)$. Each input linear form and output reconstruction uses fixed rational coefficients; their numerators, denominators, and the numbers of terms grow at most as a fixed constant to this depth. Clearing the predetermined common denominator therefore uses $O(\log n)$ bits. The bilinear leaf products and all reconstructed coefficients also have $O(\log n)$ bits, since the degree p and all dimensions are polynomial in n. The bilinear identities remain valid over the commutative $\mathbb{Q}$-algebra $\mathbb{Q}[x]/(x^p - 1)$.

For a concrete multiplication, split signed integer coefficient arrays into positive and negative parts, use a power-of-two Kronecker base larger than any coefficient of their ordinary convolution, and multiply the

resulting $O(p \log n)$-bit integers exactly. Fast integer multiplication, coefficient unpacking, and folding degrees modulo p cost $\widetilde{O}(p)$ word operations; the fixed common-denominator bookkeeping does not change this bound. The final triangle counts are exact integers. Consequently $O(P)$ candidate primes of size $\Theta(P)$ cost a factor $\widetilde{O}(P^2)$ in total.

Since $s \le r$, split the $r \times s$ by $s \times r$ matrix product into $O((r/s)^2)$ square products of order $O(s)$. The resulting prime-selection cost is $\widetilde{O}(r^2 s^{0.372} r^{2A})$ per rectangular product. Over all s products it saves

$$0.628\beta - 2\alpha A$$

below n^2. With $\beta = 0.083712$, this saving is 0.00194387830272. The box, query, failed-scan, and encoding savings in (8) are respectively

$$0.001943037983720\ldots, \quad 0.001943040668583\ldots, \quad 0.001943043681280, \quad 0.001945978936429\ldots.$$

All conditions and auxiliary costs have strict slack above saving 0.001943. Hence the deterministic bound is

$$\boxed{O(n^{1.998057})},$$

improving the square-block $O(n^{1.998938})$ consequence of the single-prime product bound. Polynomial input weights remain polynomial in the dimensions of each product, and all rounding, bit-scaling, and amplification logarithms are absorbed by the strict gaps before the final exponents are rounded. The same square-block reduction, $\sqrt{n}$ min-plus products of $\sqrt{n} \times \sqrt{n}$ matrices with weights polynomial in $\sqrt{n}$, applied to the deterministic product bound of Corollary 44, gives $2 - 0.00705208/2 = 1.99647396$, so deterministic integer min-plus convolution takes $O(n^{1.996474})$ time; the rectangular Las Vegas bound $O(n^{1.996057})$ remains smaller.

Corollary 59 (Five-term rectangular min-plus convolution)**.** *Integer min-plus convolution of two length-n sequences with entries of magnitude $n^{O(1)}$ has a Las Vegas algorithm with expected time $O(n^{1.993927})$ and a deterministic algorithm with running time $O(n^{1.993927})$.*

Proof. Use the block reduction of this subsection unchanged, writing $\bar{r}$ for its row count $r = \lceil n/s \rceil$: $s = \lceil n^\beta \rceil$, $\bar{r} = \Theta(n^\alpha)$ with $\alpha = 1 - \beta$, both boundary rows $0 \le d \le \bar{r}$, the sentinel $S > 3W$, and all $2n - 1$ outputs. There are s min-plus products with outer dimensions $R = C = \bar{r} + 1$ and middle dimension $H = s$, whose entries are polynomial in $\bar{r}$ because α is fixed. Forming them costs $O(ns + s^2)$ and reading all product entries and taking the diagonal minima costs $O(s\bar{r}^2)$, with savings $1 - \beta$ and β below n^2. Each product is computed by the priority-preserving bit-scaling transfer through $O(\log n)$ all-edges Exact Triangle calls on the parts R, H, C with matrix-local shifts. Write ϱ for the prefix parameter r of Lemma 31, fix

$$(c, \theta, \varrho, y, e, \beta) = (11.0588596,\ 0.0563850479,\ 0.0383287249,\ 0.0249694743,\ 0.0814836742,\ 0.03914792),$$

and put $D = 4^{\lfloor \log_4 \bar{r}^{\,e} \rfloor} = \Theta(\bar{r}^{\,e})$ and $b = e/2$. The exact-rational certificate gives

$$\begin{aligned} T &= 12.480128412\ldots, & G &= 0.077579669\ldots, \\ E_{\mathrm{aug}} &= 12.444817353\ldots, & E_{\mathrm{box}} &= 12.402548742\ldots, \\ q_1 &= 0.422420330988\ldots, & q_2 &= 0.422420331044\ldots, \end{aligned}$$

the tile margins $1 - e(T - 1/4) > 0.003444$ and $1 - bT > 0.49$, and the middle-size margin $\beta - \alpha e/2 > 1.04 \cdot 10^{-6}$, so $H = s \ge \sqrt{D}$ for all sufficiently large n.

Las Vegas. Apply Lemma 12 to every all-edges call with $R = \bar{r} + 1$ and $H = s$, using Lemma 32 with $R = C = (\bar{r} + 1)\lceil \sqrt{p} \rceil$ in place of Theorem 2. The lemma uses the data structure only through

its preprocessing bound, now $\widetilde{O}(ND^{E_{\mathrm{aug}}} + ND^{E_{\mathrm{box}}} + N^2D^{-G})$, its query bound $\widetilde{O}(D^{\max(q_1,q_2)})$, and its tile condition $N \geq D^T$ polylog; no relation between E and T is used in its proof, so the five-term exponents $E_{\mathrm{aug}}, E_{\mathrm{box}} < T$ are admissible. The folded instances have $N = \Theta(\bar{r}D^{1/4})$ up to logarithms, which satisfies the tile condition by the first margin, and there are $\widetilde{O}(HD^{-1/2})$ blocks by the middle-size margin. Multiplying (5), with the two encoder terms, by the s products gives the savings

$$\alpha eG, \quad \alpha e(1/2 - q_1), \quad \alpha e(1/2 - q_2), \quad \alpha\big(1 - e(E_{\mathrm{aug}} - 1/4)\big), \quad \alpha\big(1 - e(E_{\mathrm{box}} - 1/4)\big), \quad \alpha(1 - 3e/4)$$

for boxes, residual queries, stored queries, the two encoder families, and folded matrix formation; the certificate bounds them below by

$$0.0060740038, \quad 0.0060740038, \quad 0.0060740038, \quad 0.0060740032, \quad 0.00938, \quad 0.90,$$

while $1 - \beta > 0.96$ and $\beta > 0.039$. The isolation trials, label decoding, exact checks, and fallback scans are charged inside (5). All savings exceed $\delta = 0.006073$ with a strict gap above $1.0 \cdot 10^{-6}$, which absorbs the logarithmic factors of amplification, isolation, bit scaling, and the rounding of D and s. Fresh randomness in every adaptive call preserves the expected bound.

Deterministic. Apply Lemma 34 with $\Lambda = 100000$, $h = \Theta(\bar{r}^{\,b})$, and the oracle obtained by folding the composite modulus into the outer indices, as in the paragraph following Lemma 33, and preprocessing with Lemma 32. For a modulus $d = \bar{r}^{\,u}$, $0 \leq u \leq b$ up to polylogarithms, there are $O(H/h)$ folded products with $N = (\bar{r} + 1)\lceil\sqrt{d}\rceil$ and inner dimension $hd = \bar{r}^{\,e}\,\mathrm{polylog}(n)$ at the completed modulus, padded to a power of four with the polylogarithmic factor absorbed as in Corollary 35, and at most $O(\bar{r}^2H/h)$ requested incidences. Charging $\widetilde{O}(\bar{r}^{\,\eta})$ candidates per round with $\eta = b/\Lambda$ and $O(\Lambda \log n)$ rounds, and multiplying by the s products, the savings below n^2 are

$$\alpha\big(b - u + (b+u)G - \eta\big), \quad \alpha\big(b - (b+u)q_i - \eta\big), \quad \alpha\big(1 + b - u/2 - (b+u)E_j - \eta\big), \quad \alpha(1 - 3u/2 - \eta), \quad \alpha(b - \eta)$$

for boxes, the two query terms, the two encoder families, folded matrix formation, and incidence bookkeeping, with tile margin $\alpha(1 + u/2 - (b+u)T)$. All decrease in u, so the endpoint $u = b$ is binding, where they equal the Las Vegas expressions minus $\alpha\eta$; the tile margin is certified at both endpoints. Computing the residues $w(a, b) \bmod d$ saves $\beta - \alpha\eta$ and the final scans cost $O(RC)$ per product, saving β. The certificate gives $\eta = 4.07 \cdot 10^{-7}$ and endpoint savings above

$$0.0060736123, \quad 0.0060736123, \quad 0.0060736123, \quad 0.0060736117, \quad 0.00938, \quad 0.90, \quad 0.039,$$

all above $\delta = 0.006073$ with a strict gap above $6.1 \cdot 10^{-7}$. This absorbs the fixed-Λ logarithmic factors, including the $(\log n)^{\Theta(\Lambda)}$ in the completed modulus, bit scaling, and rounding; as in Corollary 35, the bound is asymptotic and no practical speedup is asserted. Both guarantees therefore give $O(n^{2-0.006073}) = O(n^{1.993927})$, improving the rectangular Las Vegas $O(n^{1.996057})$, the deterministic $O(n^{1.998057})$, and the square-block consequence $O(n^{1.996474})$ of Corollary 44. The unrounded maximal cost exponents are $1.9939259967\ldots$ and $1.9939263882\ldots$; the floating-point suprema of this family, about 0.0060740 and 0.0060736, are not claimed. The exponents are certified in the script `verify_five_term_convolution.py`. □

MonoConvolution

The source quotes, from Lincoln, Polak, and Vassilevska Williams, that an $O(n^{2-\varepsilon})$-time 3SUM algorithm gives MonoConvolution in $\widetilde{O}(n^{3/2-\varepsilon/(8-2\varepsilon)})$ time. With the deterministic $\varepsilon = 0.0068945$ proved in Theorem 64, the inherited exponent is

$$\frac{3}{2} - \frac{0.0068945}{8 - 2(0.0068945)} = \frac{11972422}{7986211} = 1.49913669949\ldots < 1.4991368.$$

Hence MonoConvolution takes deterministic $O(n^{1.4991368})$ time through the quoted reduction. That reduction loses a factor two in the saving because its light-value step calls All-Integers 3SUM through the reduction of Vassilevska Williams and Williams, which turns an $O(n^{2-\varepsilon})$ decision algorithm into an $O(n^{2-\varepsilon/2})$ All-Integers algorithm. Corollary 65 gives All-Integers 3SUM directly in $O(n^{2-\varepsilon})$ time with $\varepsilon = 0.0068945$. Rerunning the same argument, set $a = (1-\varepsilon)/(2-\varepsilon) > 1/3$ and $t = \lceil n^a \rceil$. The t most frequent values are handled by t Boolean convolutions in $\widetilde{O}(tn)$ time. For each remaining value i, let A_i, B_i, C_i be its three sets of occurrence indices, with total size $n_i \le 3n/t$; over all such values, $\sum_i n_i \le 3n$. When $n_i \le t$, enumerate $A_i \times B_i$ and look up each sum in sorted C_i. These small classes cost $\widetilde{O}(\sum_{n_i \le t} n_i^2) \le \widetilde{O}(3nt)$ in total. For every remaining class $n_i > t \ge n^a$, its indices are at most $n < n_i^{1/a} < n_i^3$, so the polynomial-magnitude hypothesis of Corollary 65 holds with a fixed exponent. Applying that corollary only to these classes costs at most $\widetilde{O}(\sum_{n_i > t} n_i^{2-\varepsilon}) \le \widetilde{O}(3n(3n/t)^{1-\varepsilon})$. Sorting and writing all answers cost $\widetilde{O}(n)$ and are covered as well. The choice of t balances tn and $n(n/t)^{1-\varepsilon}$ up to constant factors, giving total time

$$\widetilde{O}\Big(n^{(3-2\varepsilon)/(2-\varepsilon)}\Big), \qquad \frac{3-2\varepsilon}{2-\varepsilon} = 1.498270412680\ldots < 1.498271,$$

so MonoConvolution takes deterministic $O(n^{1.498271})$ time. The exponent is certified in the script `verify_five_term_realinput.py`. The earlier Las Vegas $\varepsilon = 0.003999$ also retains its valid $O(n^{1.4995})$ expected-time consequence, now dominated by the deterministic bound. These use the source's reduction and its assumptions unchanged; the strict numerical gap absorbs its logarithmic factors.

Deterministic exact witness counting

Lemma 60 (Purifying a window with known genuine multiplicity)**.** *Let A, B, T be integer matrices with polynomially bounded entries, and partition the middle labels into blocks B_0 of actual sizes at most $h = \Theta(n^b)$, where $0 < b < 1$. Write $s_k(a,b) = A_{ak} + B_{kb} - T_{ab}$. Suppose the exact counts*

$$G_\ell(a,b,B_0) = \#\{k \in B_0 : s_k(a,b) = \ell\}, \qquad \ell \in \{-1,0,1\},$$

are already known. Given any two entry masks $\alpha_{ak}, \beta_{kb} \in \{0,1\}$, the twelve counts

$$N_{\ell,i,j}(a,b,B_0) = \#\{k \in B_0 : s_k(a,b) = \ell,\ \alpha_{ak} = i,\ \beta_{kb} = j\}$$

can be recovered deterministically using the same asymptotic number of modular counting calls and the same modulus scale as composite incidence certification. In particular, a counting oracle for all three shifted targets at moduli $d \le h\,\mathrm{polylog}(n)$, applied to the entry-masked input matrices, gives the twelve counts in $\widetilde{O}(n^{3-s+b/R} + n^3/h)$ time if each oracle call costs $\widetilde{O}(n^{3-s})$ and $R \ge 1$ is a fixed integer. The polylogarithmic power may depend on R and the input-magnitude bound.

Proof. For an incidence $x = (a,b,B_0)$ put $G(x) = G_{-1}(x) + G_0(x) + G_1(x)$. An incidence with $G(x) = 0$ has all twelve desired counts zero and can be finished immediately. For any modulus d divisible by 3 define

$$C_d(x) = \sum_{\ell=-1}^{1} \#\{k \in B_0 : s_k(a,b) \equiv \ell \pmod d\}.$$

The three target residues are distinct. Each original middle label is counted at most once, every genuine window label is counted, and hence $C_d(x) = G(x) + F_d(x)$ with $F_d(x)$ a nonnegative integer. Initially $d = 3$, so $C_3(x) = |B_0|$ exactly, without an oracle call. Dummy labels introduced by rectangular padding are absent from every count.

Fix the active incidence set $\mathcal{I}$ for one purification stage. If a false label survives a modulus divisible by 3, its target $\ell \in \{-1, 0, 1\}$ is uniquely determined by its residue modulo 3. Thus the target does not change when further prime factors are appended. For this false label $(s_k - \ell)/3$ is a nonzero integer. Let $U' \geq 1$ bound the magnitude of every such nonzero integer and put $\zeta = \max(1, \ln U')$. Choose

$$P \geq 4\zeta(4h)^{1/R}, \qquad P \geq 8\zeta,$$

within a constant factor of these bounds, and sufficiently large. Enumerate the primes in $[P, 2P)$, excluding 3 and all primes previously selected in this stage. For all sufficiently large n, the number M of remaining candidates is at least $P/(4 \ln P)$, even after these constant-in-n exclusions. Each false label has at most $\lambda = \lfloor \ln U' / \ln P \rfloor$ candidate prime divisors. Consequently $\lambda/M \leq 4\zeta/P =: a \leq 1/2$.

For each of all R rounds, compute $C_{dp}(x)$ for every remaining candidate prime p and every $x \in \mathcal{I}$. Choose a single common prime minimizing $\sum_{x \in \mathcal{I}} C_{dp}(x)$ and set $d \leftarrow dp$. The true mass $\sum G(x)$ is independent of the candidate. Averaging over candidates therefore contracts the total false mass by at most a in every round. Starting from $F_3(x) \leq h$, after all R rounds the chosen modulus satisfies

$$\sum_{x \in \mathcal{I}} F_d(x) \leq h|\mathcal{I}|a^R \leq |\mathcal{I}|/4.$$

At this same chosen modulus, mark every incidence with $C_d(x) = G(x)$. At least three quarters of the active incidences are marked, since every unmarked incidence contributes an integer at least one to the false mass. Equality proves that every modular survivor in the union of the three residue classes is a genuine window label. In particular, it proves the absence of false labels separately in each target class.

Now form the four masked products, one for each $(i, j) \in \{0, 1\}^2$, by keeping an A-entry only when $\alpha = i$ and a B-entry only when $\beta = j$. An excluded entry is zero in the folded or cyclic input array (an absent incidence), not an edge of numerical weight zero. Query the three target residues at the same chosen modulus for every marked incidence. Their modular counts are exactly $N_{\ell,i,j}$: masking removes labels and introduces none. These four products are made only after the common modulus is chosen. Remove the marked incidences, restart with $d = 3$, and continue until none remain. Since initially there are $O(n^3/h)$ incidences, $O(\log n)$ stages suffice. No fallback scans and no whole-pair discarding are needed.

Each candidate modulus is 3 times a product of at most R distinct primes, so it is at most $3(2P)^R = h \operatorname{polylog}(n)$. There are $\widetilde{O}(n^{b/R})$ candidates and $O(R \log n)$ rounds in total. The sieve, exact comparisons, candidate summation, survivor bookkeeping, and the last prime-selection round are charged. Counts, moduli, and their sums have $O(\log n)$ bits. Three target queries and four masked products change only constant factors. The finitely many small input sizes can be handled by direct enumeration. □

Theorem 61 (Deterministic exact all-edges witness counting)**.** *For polynomially bounded integer matrices A, B, T, all the counts*

$$Z_{ab} = \#\{k : A_{ak} + B_{kb} = T_{ab}\}$$

can be computed deterministically in $O(n^{2.99294792})$ time, using the aligned five-term cyclic counting oracle and its certified parameters. These counts retain multiplicities of original middle labels.

Proof. Choose nonnegative integer shifts a_0, b_0 such that $A + a_0$, $B + b_0$, and $T + a_0 + b_0$ are entrywise nonnegative. This is possible with polynomially bounded shifts, and preserves every sum $A_{ak} + B_{kb} - T_{ab}$. Denote the shifted matrices by $\widehat{A}, \widehat{B}, \widehat{T}$. Use a common number $L = O(\log n)$ of bits, so their prefixes obtained by dividing by 2^L and taking floors are all zero. At this base level initialize $G_0(a, b, B_0) = |B_0|$ and $G_{-1} = G_1 = 0$ for every actual block.

Suppose the current prefix matrices are A', B', T' and the next prefixes are

$$A^+ = 2A' + \alpha, \qquad B^+ = 2B' + \beta, \qquad T^+ = 2T' + \gamma, \qquad \alpha, \beta, \gamma \in \{0,1\} \text{ entrywise.}$$

If a label has old signed gap $\ell = A'_{ak} + B'_{kb} - T'_{ab}$, its new signed gap is

$$2\ell + \alpha_{ak} + \beta_{kb} - \gamma_{ab}.$$

The residual lies in $[-1, 2]$. An old gap at least 2 therefore becomes at least 3, and an old gap at most -2 becomes at most -2. Thus a label in the new window $\{-1, 0, 1\}$ necessarily belonged to the old window. This nesting holds for arbitrary target matrices; no minimum or nonnegative-gap promise is used.

Apply Lemma 60 to the known old-window counts, using the next input bits α, β as its masks. The new exact per-block counts are

$$G_r^+(a, b, B_0) = \sum_{\substack{\ell \in \{-1,0,1\},\ i,j \in \{0,1\} \\ 2\ell + i + j - \gamma_{ab} = r}} N_{\ell,i,j}(a, b, B_0), \qquad r \in \{-1, 0, 1\}.$$

The target bit γ_{ab} is used only in this scalar update; it introduces no extra matrix dimension or mask family. In particular, an incidence with zero old-window count remains zero and needs no oracle call. Induction over the L bit levels proves every stored count exact. At the last level return $Z_{ab} = \sum_{B_0} G_0(a, b, B_0)$.

For completeness, the oracle implementation is unchanged except for constant factors. For one candidate modulus it uses the same input weights A', B' and three targets $T' - 1, T', T' + 1$. Each target changes only the requested folded position and cyclic coefficient, and is queried once per incidence. The four post-selection input masks give four more products of exactly the same dimensions. Use $h = n^b$ with

$$b = 0.04604634600, \qquad R = 1000000000, \qquad S = 768979045.$$

In the direct phase fold the modulus 3 times the selected prime product. At the split freeze 3 times the first S selected primes as the physical modulus and place the remaining selected factors in the cyclic coefficient ring. Excluding prime 3 and choosing all other factors distinctly preserves coprimality. The constant factor 3 changes no dimension exponent, and every direct or cyclic phase endpoint is the same as in the aligned certificate. In particular,

$$u = bS/R = 0.03540867517281957, \quad v = b - u = 0.01063767082718043, \quad e = b + u,$$

and the candidate charge is still b/R. Every selected and tested modulus, including those in the final greedy round, is charged. Complete cyclic coefficient vectors are charged in preprocessing and residual contractions; a stored box only permits a scalar coefficient read because its full polynomial has already been stored. As in the aligned oracle, divide the complete residual-plus-box numerator by $4^{L_{\text{tens}}}$, where L_{tens} is that oracle's tensor depth, coefficientwise before comparing its exact integer counts; individual terms need not be divisible.

The extra count tables, their updates, and the output accumulation use $O(n^3/h)$ entries and work per bit level. This is already within the certified incidence-bookkeeping cost. The constant number of targets and masks, $O(\log n)$ purification stages, $O(\log n)$ bit levels, and fixed-R polynomial logarithmic and exact-bit costs are absorbed by the same strict slack in the 47-inequality aligned certificate. No new numerical witness or changed exponent is needed. □

Corollary 62 (Exact min-plus multiplicities)**.** *The minimum values and the exact numbers of minimizing original middle labels for integer min-plus product can be computed deterministically in $O(n^{2.99294792})$ time under the same polynomial-magnitude assumption.*

Proof. First compute the min-plus product C using the existing deterministic algorithm. Then apply Theorem 61 with target $T = C$. The two calls have the same exponent. This counts minimizing middle labels in one matrix product, not shortest paths or their decompositions across successive APSP squarings. □

Earlier counting construction with one-sided error

Running the amplified reduction without any scans gives counts. For each sampled prime p_i and block B, the folded product returns $\mathrm{Count}_i(a,b,B) = Z(a,b,B) + F_i(a,b,B)$, the number of $c \in B$ with $S(a,b,c) \equiv 0 \pmod{p_i}$, where Z counts exact zeros and $F_i \geq 0$ counts false positives. Output $\widetilde{Z}(a,b) = \sum_B \min_i \mathrm{Count}_i(a,b,B)$. Always $\widetilde{Z} \geq Z$, with equality unless some pair/block has $F_i > 0$ for every i, which by the block-level bound of Theorem 9 has probability at most 2^{-k}; with $k = \lceil(\gamma+3)\log_2 n\rceil$ primes a union bound over the $O(n^3/h)$ pair/blocks gives error probability at most $n^{-\gamma}$. The cost is that of Theorem 9 without the scan terms, and it is a worst-case bound, since only the scans had random cost. So the number of zero triangles through every outer pair, hence the total number of zero triangles, is computed in worst-case $O(n^{2.995943})$ time by a Monte Carlo algorithm that never undercounts and is exact with probability $1 - n^{-\gamma}$. Counting the witnesses of every entry of an integer min-plus product first needs the product itself from the Las Vegas algorithm above, whose time is only expected. Restart it whenever a run exceeds twice a uniform upper bound on its expected time, for at most $\lceil(\gamma+1)\log_2 n\rceil$ independent attempts; by Markov's inequality all attempts fail with probability at most $n^{-\gamma-1}$, and the strict gap absorbs the logarithmic factor. If every attempt times out, output n for every witness count; this cannot undercount. Otherwise run the counting algorithm on the product with error parameter $\gamma + 1$. For $n \geq 2$, the total error probability is at most $2n^{-\gamma-1} \leq n^{-\gamma}$. Witness counting therefore has the same worst-case time and error probability. This earlier construction has no exactness certificate and retains its one-sided Monte Carlo guarantee. The separate signed-window construction in Theorem 61 supersedes this integer all-edges and min-plus witness-counting bound with deterministic exact counts; its certificate uses genuine window counts inherited from the preceding bit level. The following direct reduction gives a Monte Carlo bound for integer #3SUM.

7.1 Deterministic exact 3SUM counts with fixed buckets

Lemma 63 (Fixed-bucket cyclic count lifting)**.** *Let A, B, C be sets of at most n polynomially bounded integers. Set $s = \lceil n^b \rceil$, where $0 < b < 1$ is fixed. There is a fixed list of $\widetilde{O}(n^2/s)$ sub-bucket incidences, each containing at most $h = s^2$ original pairs, on which the signed-window count refinement can be implemented by exact cyclic thin-product queries. At physical modulus d and cyclic modulus q, with $\gcd(d,q) = 1$, these products have dimensions*

$$D = \Theta(d), \qquad N = O((n/s)\lceil\sqrt{d}\rceil).$$

There are a constant number of target queries per incidence and a constant number of input parity masks. Formation of each product, including all coefficient arrays, costs $O(n + qND)$.

Proof. Choose a prime $r \in [\lceil n/s\rceil, 2\lceil n/s\rceil]$ minimizing $\Phi(r) = \sum_i(|A_i|^2 + |B_i|^2)$, where these are buckets of the *original full values*. The same prime-divisor averaging as in Theorem 14 gives $\Phi(r) = O(ns\log n)$ and deterministic selection cost $\widetilde{O}(n^2/s)$. Here distinctness of the full values is used; their bit prefixes need not be distinct. Cut every nonempty bucket into sub-buckets of size at most s. On either side there are at most $n/s + r = O(n/s)$ sub-buckets.

Keep the incidences (α, β, z) with $z \in C$, $\alpha \subseteq A_i$, and $\beta \subseteq B_{z-i \bmod r}$. Every exact solution belongs to exactly one such incidence. Their number is at most

$$\frac{n\Phi(r)}{2s^2} + \frac{2n^2}{s} + nr = \widetilde{O}(n^2/s).$$

Enumerating the corresponding lists of sub-bucket identifiers achieves this bound. Keep these memberships and incidences fixed throughout all bit levels. A prefix sum need not respect the original bucket residues: the old residue condition only specifies this fixed query list, and is never used as a prefix congruence test.

Choose polynomially bounded shifts a_0, b_0 making $x + a_0, y + b_0, z + a_0 + b_0$ nonnegative for all inputs. They preserve $x + y - z$. Start at a common all-zero prefix above the largest bit. For every incidence initialize

$$G_{-1} = G_1 = 0, \qquad G_0 = |\alpha|\,|\beta|.$$

These are counts of original pairs; different original values that coalesce at a prefix retain their multiplicities. If the current prefix gap is $\ell = x' + y' - z'$, the next gap is $2\ell + \epsilon_x + \epsilon_y - \epsilon_z$, with each ϵ a bit. Thus only old gaps in $\{-1, 0, 1\}$ can enter that window at the next level, exactly as in Theorem 61.

Apply Lemma 60 with original pairs $(x, y) \in \alpha \times \beta$ as its labels and $h = s^2$. Start each purification attempt at modulus 3, exclude 3, r, and previously selected primes, and use all R greedy prime stages. The initial union count is $|\alpha||\beta|$. The targets $z' - 1, z', z' + 1$ are distinct modulo every selected modulus, which is divisible by 3. Each false original pair therefore has a unique fixed target offset, and the lemma's same-label nonzero-divisor argument applies without change. Minimizing the total modular union count minimizes false mass because the exact true mass $G_{-1} + G_0 + G_1$ is known. At the completed modulus, equality of these two counts certifies all three classes for that incidence. Four parity-masked products at that same modulus yield the twelve exact counts $N_{\ell,i,j}$ and the update

$$G_t^+ = \sum_{\substack{\ell \in \{-1,0,1\},\ i,j \in \{0,1\} \\ 2\ell + i + j - \epsilon_z = t}} N_{\ell,i,j}, \qquad t \in \{-1, 0, 1\}.$$

At least three quarters of the active incidences are certified per purification attempt; $O(\log n)$ attempts suffice. Zero-window incidences remain zero. There are $O(\log n)$ bit levels, and summing the final G_0 over the fixed incidence list counts all exact triples. No witness scans or randomized decisions occur.

Here is the actual cyclic count oracle. Write the current modulus as dq, keep the factor 3 in d, and put $d_1 = \lceil \sqrt{d} \rceil$, $d_2 = \lceil d/d_1 \rceil$. Over $\mathcal{R}_q = \mathbb{Z}[x]/(x^q - 1)$, with middle coordinate $\sigma \in \mathbb{Z}_d$, define the histogram matrices

$$X[(\alpha, \rho_1), \sigma] = \sum_{\substack{a \in \alpha \\ a' - \rho_1 \equiv \sigma \pmod d}} x^{a' \bmod q}, \qquad 0 \le \rho_1 < d_1,$$

$$Y[\sigma, (\beta, \rho_2)] = \sum_{\substack{b \in \beta \\ d_1\rho_2 - b' \equiv \sigma \pmod d}} x^{b' \bmod q}, \qquad 0 \le \rho_2 < d_2.$$

For a requested target $v = z' + \ell$, write $v \bmod d = \rho_1 + d_1\rho_2$. The coefficient of $x^{v \bmod q}$ in the corresponding product entry counts precisely the original pairs with $a' + b' \equiv v \pmod{dq}$. Indeed each such pair uses one middle residue if and only if its sum is correct modulo d; its cyclic monomial tests the same pair modulo q. Coprimality combines the two tests. These entries are multiplicity histograms, not Boolean presence indicators. A mask removes its excluded inputs from these sums, rather than changing their numerical weights to zero.

For formation, first build the two-dimensional residue histogram ($a' \bmod d, a' \bmod q$) of each sub-bucket in $O(n + (n/s)dq)$ time and storage. Shift-copy these histograms into the folded rows and columns in $O(qND)$ time, including zero initialization. This avoids reinserting each input separately for every outer residue. Pad the two outer dimensions to the stated common N and the inner dimension to the next power of four; all added entries are zero. Padding changes dimensions by constant factors. A target offset changes only its folded position and coefficient; it causes no extra factor d or q in the number of queries. Distinct incidences whose prefix targets coincide remain separate in the count tables, every greedy candidate-score sum, and the final sum, even when an oracle answer can be shared. Reusing a queried entry does not deduplicate the incidences in the objective or its contraction argument. □

Theorem 64 (Deterministic integer 3SUM and exact counts)**.** *For sets of polynomially bounded integers, 3SUM and exact #3SUM admit deterministic $O(n^{1.9931055})$ algorithms. The one-set form can require three distinct elements, with either ordered or unordered exact counts.*

Proof. Use Lemma 63 and the aligned cache of Lemma 43, with

$$c = 10.83107856, \qquad \theta = 0.06504251269, \qquad b = 0.04436367207,$$
$$R = 1000000000, \qquad S = 884889170, \qquad \delta = 0.0068945.$$

Here $s = \lceil n^b \rceil$ and the per-incidence label bound is $h = s^2$. Each of the R prime factors comes from a pool of scale $\widetilde{O}(h^{1/R})$, so enumerating all candidates charges the exponent

$$\eta = \frac{2b}{R}, \qquad u = S\eta = 0.0785138659123489638, \qquad v = (R-S)\eta = 0.0102134782276510362.$$

In each purification attempt, fold 3 and the first S selected factors physically, then freeze their product d and put subsequent factors into the cyclic ring length q. These are actual integer stage counts, so $u + v = 2b$ holds exactly. Distinct selected primes, excluding 3 and r, ensure $\gcd(d, q) = 1$.

Write

$$T = c - 1 + \frac{c\mathcal{H}(1/c)}{\ln 4}, \qquad G = \frac{c\mathcal{H}(1/c) - c\mathcal{H}((1-\theta)/c) - \theta \ln 4}{\ln 2}, \qquad q_1 = \frac{2\mathcal{H}(\theta)}{\ln 4},$$

and $q_2 = q_1 + 2\theta$, $E = T + q_1$. The source cache, boxes, formation, and all requested coefficient queries cost, respectively,

$$\widetilde{O}(qND^E), \quad \widetilde{O}(qN^2D^{-G}), \quad O(n + qND), \quad \widetilde{O}\big((n^2/s)(qD^{q_1} + D^{q_2})\big).$$

The separate box encoder exponent $T - G$ is smaller than E and is included. Original full-grid tiling requires $N \geq KN_0$; its exponent condition uses T, not the cache exponent E.

For the scalar phase let $x \in [0, u]$ denote the current physical modulus exponent. Then $N = n^{1-b+x/2}$, $D = n^x$ up to fixed polylogarithmic factors. After charging all candidate primes, the savings below exponent two are

boxes:	$2b - x(1-G) - \eta$,
residual queries:	$b - xq_1 - \eta$,
stored queries:	$b - xq_2 - \eta$,
polynomial cache:	$1 + b - x(E + 1/2) - \eta$,
dense histogram formation:	$1 + b - 3x/2 - \eta$.

The tile margin is $1 - b + x/2 - xT$. In the cyclic phase, with d frozen and current ring exponent $w \in [0, v]$, the savings are

boxes:	$2b - u + uG - w - \eta$,
residual queries:	$b - uq_1 - w - \eta$,
stored queries:	$b - uq_2 - \eta$,
polynomial cache:	$1 + b - u(E + 1/2) - w - \eta$,
dense histogram formation:	$1 + b - 3u/2 - w - \eta$.

The tile margin is now $1 - b + u/2 - uT$. Input reading separately has saving $1 - \eta$. Incidence enumeration, count updates, candidate summation and survivor bookkeeping have saving $b - \eta$; the original bucket-prime selection has the still larger saving b.

All these expressions are affine in the stage exponent. The exact rational certificate checks both endpoints of both phases, the parameter and tail regimes, and the exact stage identities. At the final cyclic endpoint $w = v = 2b - u$, the five displayed runtime savings exceed, respectively,

$$0.00689451426923, \qquad 0.00689451427575, \qquad 0.00689451426351,$$
$$0.00689451423864, \qquad 0.91637939488509.$$

The original tile margin exceeds 0.03415019380523, and the minimum runtime slack above δ exceeds $1.4238 \cdot 10^{-8}$.

Every R, S here is fixed independently of n. For sufficiently large n the prime pool contains enough primes after all exclusions, and the completed modulus is at most h times a fixed polynomial in $\log n$, with constants allowed to depend on R and the input magnitude exponent. Its bit length is $O(\log n)$, including the fixed-R logarithmic factors. Histograms, exact counts, and sums over incidences have polynomial magnitude. The aligned oracle's signed linear encodings, auxiliary polynomial caches and bilinear boxes have polynomial coefficient norms by its existing proof. Exact cyclic arithmetic therefore has the stated $\widetilde{O}(q)$ bit cost. Divide the *complete* residual-plus-box numerator coefficientwise by $4^{L_{\mathrm{tens}}}$ before using a count in a certificate; this is exact integer division and requires no inverse modulo q.

The constant number of masks and offsets, all candidate calls, $O(\log n)$ purification attempts per bit level, $O(\log n)$ bit levels, padding, sieving and fixed-parameter logarithmic overhead fit within the strict slack. Small input sizes are handled by direct enumeration. This proves the colored exact count and hence decision.

For the one-set form use $A = B = S_0$, $C = -S_0$. If Z is the resulting ordered count, put $t = |\{(x, z) \in S_0^2 : 2x + z = 0\}|$ and $z_0 = \mathbf{1}_{0 \in S_0}$. Inclusion–exclusion on equality of the three labels gives the distinct-element ordered count $Z - 3t + 2z_0$. Sorted lookup computes t in $O(n \log n)$ time. Divide this count by six for unordered triples, or test positivity for decision. No correction is needed for the colored sets problem. □

Corollary 65 (All-Integers 3SUM and per-target counts)**.** *For sets A, B, C of at most n polynomially bounded integers, the counts $\#\{(a, b) \in A \times B : a+b = c\}$ for every $c \in C$ are computed deterministically in $O(n^{1.9931055})$ time. In particular All-Integers 3SUM, which asks for every $c \in C$ whether some $a + b = c$, has the same deterministic bound.*

Proof. Lemma 63 fixes the incidence list (α, β, z) with $z \in C$, and every solution $a+b = z$ lies in exactly one incidence. The proof of Theorem 64 ends with the exact window count $G_0(\alpha, \beta, z) = \#\{(x, y) \in \alpha \times \beta : x + y = z\}$ for every incidence at the last bit level. Summing these $\widetilde{O}(n^2/s)$ exact counts by target z costs $\widetilde{O}(n^2/s)$ and gives every per-target count; the theorem only adds them once more. □

7.2 Deterministic 3XOR and exact counts with fixed linear buckets

The source's Remark 23 sketches deterministic 3XOR through linear hash functions over $\mathbb{F}_2$ with rows chosen from an ε-biased set by conditional expectations, and Majdoddin [6] (Section 18, Lemmas 18.7–18.9, 18.11–18.12, 18.16) proves $O(n^{1.99451})$ with cut classes, the carry-free fold of a hash into the outer indices, halving stages, and a minimum-over-rows certificate. The construction below keeps his grouping, fold, and greedy rows, and replaces the stages by the fixed-bucket exact-count lifting of Theorem 64.

Vectors of $\mathbb{F}_2^d$, $d \le c_0 \log_2 n$, occupy $O(1)$ words and an inner product $\langle u, x\rangle$ costs $O(\log d)$ word operations. For a linear map $\Lambda : \mathbb{F}_2^d \to \mathbb{F}_2^r$ and an incidence $x = (\alpha, \beta, z)$ put $\mathrm{cnt}_\Lambda(x) = \#\{(a, b) \in \alpha \times \beta : \Lambda(a \oplus b \oplus z) = 0\}$. A multiset $E \subseteq \mathbb{F}_2^d$ is β-biased if $\left||E|^{-1} \sum_{u\in E}(-1)^{\langle u,w\rangle}\right| \le \beta$ for every $w \ne 0$. The explicit construction of Alon, Goldreich, Håstad and Peralta gives a $(1/r)$-biased multiset $E_{d,r}$ with $|E_{d,r}| \le 4(dr)^2$ in $\mathrm{poly}(dr)$ time; its projection to any coordinate subspace is again $(1/r)$-biased, since characters of the subspace are characters of $\mathbb{F}_2^d$. For a nonzero w at most a $(1+\beta)/2$ fraction of E annihilates w, so rows chosen one at a time from $E_{d,r}$, each minimizing the surviving weight of a family of nonzero vectors, leave at most $e2^{-t}$ of the weight after $t \le r$ rows. This replaces the prime-divisor bound: one row doubles the modulus and halves the false mass up to the factor $1 + 1/r$, so the loss over all rows is e and no fixed stage count or polynomial candidate pool is needed.

Lemma 66 (Group-algebra coefficient ring)**.** *Lemmas 36, 42 (whose coefficient-ring clause is coefficientwise), and 43 hold verbatim with $\mathcal{R}_q = \mathbb{Z}[x]/(x^q - 1)$ replaced by $\mathbb{Z}[\mathbb{F}_2^m] = \mathbb{Z}[x_1, \dots, x_m]/(x_1^2 - 1, \dots, x_m^2 - 1)$, $q = 2^m$.*

Proof. Their proofs use only that the coefficient ring is commutative and free of rank q over $\mathbb{Z}$, that linear operations cost $O(q)$ and products $\widetilde{O}(q)$ word operations, that one product coefficient is a length-q dot product, and that coefficient ℓ_1 norms are submultiplicative. Store an element as its q integer coefficients indexed by $\mathbb{F}_2^m$. The product is $(fg)(w) = \sum_v f(v)g(v \oplus w)$: coefficient w is a dot product with a permuted copy of g, and the whole product is obtained exactly by two unnormalized integer Walsh–Hadamard transforms, a pointwise product, a third transform and exact division by 2^m, in $O(mq)$ operations on $O(\log n)$-bit integers. Norms satisfy $\|fg\|_1 \le \|f\|_1\|g\|_1$. The complete degree-summed numerator equals 4^L times the product entry coefficientwise, so the final division is exact integer division per coefficient. □

Lemma 67 (Fixed linear buckets and prefix-exact count lifting)**.** *Let $S_1, S_2, S_3 \subseteq \mathbb{F}_2^d$ be sets (no repeated vectors) of at most $n \le 2^d$ vectors, $d \le c_0 \log_2 n$, $s = \lceil n^b \rceil$, $h = s^2$, $k = \lceil \log_2(n/s) \rceil$, $l = \lceil \log_2(4eh) \rceil$, and fix $0 \le l_1 \le l$. There is a fixed list of at most $8n^2/s$ incidences (α, β, z), computable in $\widetilde{O}(n^2/s)$ time, such that every solution $a \oplus b \oplus z = 0$ lies in exactly one of them, and the exact counts $G(x) = \mathrm{cnt}_{\mathrm{Id}}(x)$ of all incidences follow from $O(d \log n \cdot l \cdot |E_{d,l}|) = O(\log^7 n)$ counting calls. Each call takes a linear map (H_1, H_2) with $t_1 \le l_1$ rows H_1 and $t_2 \le l - l_1 + 1$ rows H_2 and a set $\mathcal{I}$ of incidences, and returns $\mathrm{cnt}_{(H_1,H_2)}(x)$ for all $x \in \mathcal{I}$ as one coefficient of one entry of an $N \times D$ by $D \times N$ product over $\mathbb{Z}[\mathbb{F}_2^{t_2}]$ with $D = 2^{t_1}$, $N = O((n/s)\sqrt{D})$, formed in $O(n + 2^{t_2}ND)$ time, whose entries are multiplicity histograms of magnitude at most s.*

Proof. Buckets. Choose rows $w_1, \dots, w_k \in E_{d,k}$ one at a time, each minimizing $\Phi(\pi) = \sum_{\gamma\in\mathbb{F}_2^k}(|A_\gamma|^2 + |B_\gamma|^2)$ for the map π with the rows so far followed by the candidate, where $A_\gamma = \{a \in S_1 : \pi(a) = \gamma\}$ and B_γ likewise. Here $\Phi(\pi) = |S_1| + |S_2| + P_1 + P_2$, where P_i counts the pairs $(x, x') \in S_i^2$ with $x \ne x'$ and $\pi(x \oplus x') = 0$, and $x \oplus x' \ne 0$ by distinctness, so the greedy bound gives $\Phi(\pi) \le 2n + e2^{-k}(|S_1|^2 + |S_2|^2) \le 2n + 2ens$. Each evaluation costs $O(nk \log d + 2^k)$, so selection costs $\widetilde{O}(n)$. Cut every nonempty A_γ, B_γ into sub-buckets of at most s vectors; there are at most $n/s + 2^k \le 3n/s$ on each side. The incidences are (α, β, z) with $z \in S_3$, $\alpha \subseteq A_i$, $\beta \subseteq B_{i\oplus\pi(z)}$. By linearity a solution has $\pi(b) = \pi(a) \oplus \pi(z)$, so it lies in exactly one incidence. Using $\lceil p/s \rceil \lceil p'/s \rceil \le pp'/s^2 + p/s + p'/s + 1$ and $|A_i||B_j| \le (|A_i|^2 + |B_j|^2)/2$, the number of incidences is at most $n\Phi(\pi)/(2s^2) + 2n^2/s + n2^k \le 8n^2/s$. Keep this list fixed.

Levels. For $0 \le \ell \le d$ let Π_ℓ project to the first ℓ coordinates; any complete flag of functionals serves. Maintain $G^{(\ell)}(x) = \mathrm{cnt}_{\Pi_\ell}(x)$ for every incidence, with $G^{(0)} = |\alpha||\beta|$ and $G^{(d)} = G$. Since $\Pi_{\ell+1}s = 0$ implies $\Pi_\ell s = 0$, an incidence with $G^{(\ell)} = 0$ stays zero. There is a single window $\{0\}$ and no signed targets, because XOR has no carries.

One attempt at level ℓ. Let $\mathcal{I}$ be the active incidences with $G^{(\ell)} > 0$ and $E = E_{d,l}$. Start with no rows; l times, for every $u \in E$ evaluate $\Sigma(H,u) = \sum_{x\in\mathcal{I}} \mathrm{cnt}_{(H,u)\Pi_\ell}(x)$ by one counting call and append a minimizer. The first l_1 rows form H_1 and the rest H_2. A false pair (a,b) of x, one with $\Pi_\ell(a\oplus b\oplus z) \neq 0$, is killed by a row u exactly when $\langle u, \Pi_\ell(a\oplus b\oplus z)\rangle = 1$; genuine pairs survive every row. The true mass $\sum_{x\in\mathcal{I}} G^{(\ell)}(x)$ does not depend on the rows, so the greedy minimizes the false mass $F = \sum_x(\mathrm{cnt} - G^{(\ell)})$, and the projected E is $(1/l)$-biased on $\mathbb{F}_2^\ell$, so after l rows $F \le e2^{-l}\sum_x F_\emptyset(x) \le e2^{-l}h|\mathcal{I}| \le |\mathcal{I}|/4$. Certify every x with $\mathrm{cnt}_{H\Pi_\ell}(x) = G^{(\ell)}(x)$; each uncertified incidence contributes at least one to F, so at least three quarters are certified. For a certified x the pairs with $H\Pi_\ell(a\oplus b\oplus z) = 0$ are exactly its pairs with $\Pi_\ell(a\oplus b\oplus z) = 0$. Hence one further call with the coordinate functional $e_{\ell+1}$ appended as a group-algebra row returns $\mathrm{cnt}_{(H\Pi_\ell, e_{\ell+1})}(x) = \mathrm{cnt}_{\Pi_{\ell+1}}(x) = G^{(\ell+1)}(x)$ for every certified x. The target bit enters only through the queried position and coefficient; no input masks are needed. Remove the certified incidences and repeat; $O(\log n)$ attempts finish the level and d levels give G.

The counting call. Write $H_1 = (H_1', H_1'')$ with $\lceil t_1/2\rceil$ and $\lfloor t_1/2\rfloor$ rows, $\sigma \in \mathbb{F}_2^{t_1}$, and over $\mathbb{Z}[\mathbb{F}_2^{t_2}]$

$$X[(\alpha,\rho_1),\sigma] = \sum_{a\in\alpha:\ H_1 a\oplus(\rho_1,0)=\sigma} x^{H_2 a}, \qquad \rho_1 \in \mathbb{F}_2^{\lceil t_1/2\rceil},$$

$$Y[\sigma,(\beta,\rho_2)] = \sum_{b\in\beta:\ H_1 b\oplus(0,\rho_2)=\sigma} x^{H_2 b}, \qquad \rho_2 \in \mathbb{F}_2^{\lfloor t_1/2\rfloor}.$$

At position $(\rho_1,\rho_2) = H_1 z$, coefficient $x^{H_2 z}$ of the product entry counts the pairs with $H_1(a\oplus b) = H_1 z$ and $H_2(a\oplus b) = H_2 z$, that is $(H_1,H_2)(a\oplus b\oplus z) = 0$; the two tests are independent linear conditions and need no coprimality. The split of $H_1 z$ is exact, without carries. Build the histogram $(H_1 a, H_2 a)$ of each sub-bucket in $O(n + (n/s)2^{t_1+t_2})$ time and shift-copy it into the folded rows and columns in $O(2^{t_2}ND)$ time; pad to common dimensions with zero entries. One coefficient is read per incidence; incidences sharing a position and coefficient share the answer but remain separate in every sum. □

Theorem 68 (Deterministic 3XOR and exact counts)**.** *For three lists of at most n vectors of $\mathbb{F}_2^d$ with $d = O(\log n)$, 3XOR, exact #3XOR, and the per-target counts $\#\{(a,b) : a\oplus b = z\}$ for every $z \in S_3$ admit deterministic $O(n^{1.9931055})$ algorithms.*

Proof. For $n \le 1$ use direct enumeration. Otherwise retain the original ambient size bound n and append zero coordinates to every vector until $d' = \max\{d, \lceil\log_2 n\rceil\}$. This preserves XOR equations and multiplicities, gives $n \le 2^{d'}$, and keeps $d' = O(\log n)$; write d for this padded dimension below. Remove repeated vectors in $O(n\log n)$ time, keeping multiplicities. Use Lemma 67 with the aligned cache of Lemmas 43 and 66 and

$$c = 10.83107856, \quad \theta = 0.06504251269, \quad b = 0.04436367207, \quad u = 0.0785138659, \quad \delta = 0.0068945,$$

with $l_1 = \lfloor u\log_2 n\rfloor$, so $D \le n^u$ and $2^{t_2} \le 2^{l-l_1+1} = O(n^{2b-u})$. With T, G, q_1, q_2, E as in Theorem 64, a call with $D = n^x$, $0 \le x \le u$, and ring length $q = n^w$, $0 \le w \le 2b-u$, where $w = 0$ while $x < u$, has $N = n^{1-b+x/2}$, costs $\widetilde{O}(qND^E + qN^2D^{-G} + qND)$ to preprocess and $\widetilde{O}(qD^{q_1} + D^{q_2})$ per incidence, for $O(n^{2-b})$ incidences. The savings below exponent two are those displayed in Theorem 64 with $\eta = 0$: the pool $E_{d,l}$ has $O(\log^4 n)$ rows, each level has $O(\log n)$ attempts of $l+1 = O(\log n)$ rows, and there are $d = O(\log n)$ levels, so all $O(\log^7 n)$ calls are charged by a fixed polylogarithmic factor. Bucket selection, incidence enumeration, count tables, and the final summation cost $\widetilde{O}(n^{2-b})$. The exact rational certificate `verify_cyclic_3xor.py` checks 46 strict inequalities at both endpoints of both phases; at the final group-algebra endpoint the box, residual, stored and cache savings exceed 0.0068945143, the tile margin exceeds 0.03415, and the minimum slack above δ exceeds $1.4\cdot10^{-8}$, which absorbs the polylogarithmic factors for large n; small n use direct enumeration. The sum $\sum_x G(x)$ is the exact number of solutions over the deduplicated sets and summing by z gives the per-target

counts. For lists with multiplicities, at every incidence certified in some attempt the hash $(H\Pi_{d-1}, e_d)$ of that attempt is exact, so one weighted histogram product per attempt, with entries $\sum m(a)x^{H_2 a}$ of $O(\log n)$ bits, returns $\sum m(a)m(b)$ over its genuine pairs; multiplying by $m(z)$ gives the multiset count. For a single set S of $N = |S|$ distinct values, let Z be its ordered count with repetition allowed. Requiring three distinct values gives $Z - 3N\mathbf{1}_{0\in S} + 2\mathbf{1}_{0\in S}$. For a single list of length N, let Z instead be the weighted ordered count and let $m(0)$ be the number of zero occurrences. Requiring three distinct occurrence indices gives $Z - 3Nm(0) + 2m(0)$. Indeed, each of the three index-equality events has $Nm(0)$ solutions, since the third value must be zero, and each intersection of two or three such events has $m(0)$ solutions. Inclusion–exclusion proves this correction and its set specialization. Decision follows from the count. □

Earlier integer 3SUM counting with one-sided error

The direct bucket-pair reduction also gives a counting bound. This is an additional consequence of the present reductions, not a claim of literature priority. Let A, B, C be sets of polynomially bounded integers as in Theorem 14. In Lemma 13, replace Boolean membership by multiplicity:

$$X[(\alpha, \rho_1), \sigma] = \#\{x \in \alpha : x - \rho_1 \equiv \sigma \pmod p\},$$
$$Y[\sigma, (\beta, \rho_2)] = \#\{y \in \beta : p_1\rho_2 - y \equiv \sigma \pmod p\}.$$

These entries are at most s and have logarithmic bit length. At the folded position for z, their product counts the pairs $(x, y) \in \alpha \times \beta$ with $x + y \equiv z \pmod p$, including distinct pairs that collide at a residue. Thus for each compatible sub-query $Q = (z, \alpha, \beta)$ and sampled prime p_i its answer is $C_i(Q) = Z(Q) + F_i(Q)$, where $Z(Q)$ counts the exact integer equalities and $F_i(Q) \geq 0$ counts the false pairs. Return

$$\widehat{Z} = \sum_Q \min_i C_i(Q).$$

Every exact triple belongs to exactly one compatible sub-query, so $\widehat{Z} \geq \#\{(x, y, z) \in A \times B \times C : x + y = z\}$ always. For a fixed Q, the union bound over its at most s^2 nonzero differences $x + y - z$ gives $\Pr[F_i(Q) > 0] \leq 1/2$, with precisely the same sub-bucket size and prime-divisor estimate as in Theorem 14. This holds even if Q has exact solutions. Independent samples imply that its minimum is inexact with probability at most 2^{-k}. There are at most n^3 sub-queries, since there are at most n nonempty sub-buckets on each side and n choices of z. Hence $k = \lceil(\gamma + 3)\log_2 n\rceil$ ensures that all minima are exact with probability at least $1 - n^{-\gamma}$ for every fixed $\gamma > 0$ and $n \geq 2$. Finitely many small sizes can be counted directly.

There are no scans. All preprocessing, query enumeration, multiplicity construction, and queries have the same deterministic cost bounds as in Theorem 14, with its scan costs omitted. The certified unfolded parameters therefore give *worst-case* $O(n^{1.996001})$ time for integer #3SUM, by a Monte Carlo algorithm that never undercounts and has error probability at most $n^{-\gamma}$. This is not a Las Vegas counting algorithm; the folded Exact Triangle saving 0.004057 does not apply to this reduction.

For a single set S, run the algorithm with $A = B = S$ and $C = -S$. Write $t = \#\{x \in S : -2x \in S\}$ and $z_0 = \mathbf{1}_{\{0\in S\}}$. Among the ordered zero-sum triples, exactly $R = 3t - 2z_0$ use a repeated element: each of the three equality events has t triples, and every intersection is the all-zero triple when zero is present. Sorting computes R in $O(n \log n)$ time. Thus

$$\left\lceil \frac{\widehat{Z} - R}{6} \right\rceil$$

never undercounts the unordered zero-sum triples of three distinct elements and is exact whenever $\widehat{Z}$ is exact. This counting statement is for sets; no multiset bucket-balance guarantee is being asserted.

Min-sum subset convolution through rectangular restrictions

Let $f, g : 2^{[n]} \to \mathbb{Z} \cup \{+\infty\}$ be explicit tables with finite values of at most β bits, and define $h(S) = \min_{A \cup B = S}(f(A) + g(B))$. We use the restriction products and local-cover organization of Mihail Stoian, *Truly Sub-3^n Min-Sum Subset Convolution and Join Ordering*, arXiv:2610.10282v1 [7], Sections 2–4. The changed ingredient is an asymmetric restriction cost from rectangular tiles. This improves even the direct substitution of our square min-plus exponent into that paper's transfer theorem.

Lemma 69 (Rectangular tiles, $c > 10$)**.** *Put $A_0 = (c-1)\ln 3/\ln 4$, so $N_0 = \widetilde{\Theta}(D^{A_0})$, and retain $M = KN_0^2 = \widetilde{\Theta}(D^{2T})$ and $E = T$. For outer dimensions R, C satisfying $R, C \geq N_0$ and $RC \geq M$, the refined data structure preprocesses an $R \times D$ by $D \times C$ product in*

$$\widetilde{O}\left(\sqrt{RC}\, D^E + (R+C)D^{A_0} + RCD^{-G}\right)$$

time, with individual query time $\widetilde{O}(D^q)$.

Proof. Use a $K_1 \times K_2$ grid of block products per tile, with $K_1 K_2 = \Theta(K)$ and $K_1 K_2 \leq K$. Assign distinct inner subsets to grid positions and zero the unused products. The source's independent batch identity applies unchanged; a left input depends only on its row band and a right input only on its column band. The same boxes suffice.

Choose K_1 within a constant factor of $\min\{K, \max\{1, \sqrt{KR/C}\}\}$ and $K_2 = \lfloor K/K_1 \rfloor$, with harmless constant-factor padding. In the unclamped case the number of bands on each side is $O(\sqrt{RC/M})$, and their encodings cost $O(\sqrt{RCM})$. If $R/C \geq K$, take $K_1 = K, K_2 = 1$: the encoding cost is $O(RN_0 + CKN_0) = O(RN_0)$. The opposite case is symmetric. The assumptions ensure both bands fit up to constants. Ceiling terms cost $O(M) \leq O(\sqrt{RCM})$. There are $O(RC/M)$ tiles; the unchanged box bound therefore gives $\widetilde{O}(RCD^{-G})$. □

For $1 < c < 10$ the same layout applies with each band encoding costing $10^L \operatorname{poly}(m)$ instead of $O(M)$, so in the clamped regime $K_1 = K$, $K_2 = 1$ the encodings cost $RD^{A_{\mathrm{cl}}} + CD^{A_{\mathrm{cl}}+C_K}$ with $A_{\mathrm{cl}} = c\ln 10/\ln 4 - C_K - A_0$, and in the unclamped regime $\sqrt{RC}\, D^E$ with the $c < 10$ branch of E; the box term is unchanged.

Write the original outer dimensions of an all-edges Exact Triangle instance as $2^y, 2^z$ and its middle dimension as 2^x, and put $a = (y+z)/2$. Applying residue folding, amplification, and Lemma 69 gives the following complete cost list:

$$\begin{aligned}
\text{encodings: } & 2^{a+x}D^{E-1/4} + (2^y + 2^z)2^x D^{A_0 - 1/4}, \\
\text{boxes: } & 2^{x+y+z}D^{-G}, \qquad & \text{queries: } & 2^{x+y+z}D^{q-1/2}, \\
\text{writing: } & (2^y + 2^z)2^x D^{3/4}, & \text{successful scans: } & 2^{y+z}D^{1/2}.
\end{aligned}$$

These expressions suppress polynomial factors in the logarithms of the dimensions. Indeed, each folded instance has outer dimensions $\widetilde{\Theta}(2^y D^{1/4}), \widetilde{\Theta}(2^z D^{1/4})$, and there are $\widetilde{O}(2^x D^{-1/2})$ instances. Every outer pair is queried once per prime and block, as before. Taking $O(x+y+z+1)$ independent primes makes expected failed-scan work negligible; original sums still certify acceptance, and successful scans stop separately for each pair. The tiling requirements are

$$a > (T - 1/4)\log_2 D, \qquad \min\{y, z\} > (A_0 - 1/4)\log_2 D,$$

with sufficient slack for the suppressed logarithms.

Use the fixed rational constants

$$(c,\theta,e,\delta)=(21.0477,0.111971,0.05828,0.004057),\quad \lambda=0.916,\quad \kappa=0.034,$$

and choose

$$b=\min\left\{\frac{y+z}{2},\frac{y}{\lambda},\frac{z}{\lambda},\frac{x}{\kappa}\right\},\qquad D=4^{\lfloor eb/2\rfloor}.$$

Every listed cost is $\widetilde{O}(2^{x+y+z-\delta b})$. The respective savings, in units of b, are bounded below by

$$1-e(E-1/4),\quad \lambda-e(A_0-1/4),\quad eG,\quad e(1/2-q),\quad \lambda-3e/4,\quad \kappa-e/2,$$

all strictly exceeding δ. The same inequalities secure the tile and block-size requirements. Bounded b, or a failure due only to polylogarithmic padding when $b=O(\log(n+1))$, can use direct enumeration with a polynomial factor in $n=x+y+z$.

Arbitrary input bit length. The integer magnitude assumption need not be imposed here. Put $Q=(\beta+n+1)(n+1)^2$, $P=\lceil\sqrt{D}\,Q\rceil$, and $h=\lfloor D/(2P)\rfloor$. A nonzero β-bit sum has $O((\beta+1)/\log P)$ prime divisors in $[P,2P]$, so the probability that a zero-free block passes one prime is $O(h(\beta+1)/P)=O(D(\beta+1)/P^2)$, below $1/2$ for sufficiently large n. When $h\ge 1$, the new block counts and folded dimensions increase each of the costs above by only $\mathrm{poly}(n,\beta)$; the lower-bound tiling conditions become easier. If $h=0$, then $\sqrt{D}<4Q$ for $D\ge 4$. Direct enumeration loses at most $2^{\delta b}=O(D^{\delta/e})=\mathrm{poly}(n,\beta)$. Exact arithmetic on original weights has polynomial bit cost; the folded matrices contain Boolean entries and their encoded counts have polynomial bit length in n. The priority-preserving min-plus bit-scaling reduction takes $O(\beta)$ levels and therefore gives the same rectangular min-plus bound times $\mathrm{poly}(n,\beta)$. Address operations are charged by their $O(n+\log(\beta+1))$ bit length. For absent entries use $4W+1$, where $W\ge 1$ bounds finite magnitudes; a sum involving this sentinel exceeds $2W$ and is mapped back to infinity.

Theorem 70 (Subset convolution)**.** *Min-sum subset convolution on arbitrary β-bit integer tables has a Las Vegas algorithm with expected running time $O(2.997190^n\,\mathrm{poly}(n,\beta))$.*

Proof. For a restriction $[n]=X\,\dot\cup\,Y\,\dot\cup\,Z$, use rows $U\subseteq Z$, middle indices $T\subseteq X$, and columns $V\subseteq Y$, with matrices

$$F[U,T]=f(U\cup T),\qquad G[T,V]=g((X\setminus T)\cup V).$$

Each product entry updates $h(X\cup U\cup V)$. It enumerates exactly the disjoint pairs compatible with this restriction: X forbids outside membership, Y forbids left membership, and Z forbids right membership. Put $x=|X|,y=|Y|,z=|Z|$. The rectangular bound above is at most $2^n w(X,Y,Z)\,\mathrm{poly}(n,\beta)$, where

$$w=2^{-\delta(y+z)/2}+2^{-\delta y/\lambda}+2^{-\delta z/\lambda}+2^{-\delta x/\kappa}.$$

Indeed $2^{-\delta b}$ is the maximum of these four terms. Forming the matrices and performing the output updates fit the same bound, since $x\ge\kappa b$ and $y,z\ge\lambda b$, with $\kappa,\lambda>\delta$.

For a block of r coordinates, summing w over all restrictions gives

$$B_1^r+2B_2^r+B_3^r\le 4B^r,\quad B_1=1+2^{1-\delta/2},\quad B_2=2+2^{-\delta/\lambda},\quad B_3=2+2^{-\delta/\kappa},\quad B=\max_i B_i.$$

Each candidate ternary membership string belongs to exactly 2^r compatible restrictions. Round each of the four terms upward to a multiple of 2^{-r}. Every term is at least 2^{-r}, so rounding inflates total weight by at most two. Weighted greedy set cover now produces a complete local family of weight at most

$$8H_{3^r}(B/2)^r.$$

Here $H_t = \sum_{j=1}^{t} 1/j$. For completeness, if u candidates remain, the sum of the numbers covered by all restrictions is $u2^r$. Some restriction therefore has rounded weight per newly covered candidate at most $8B^r/(u2^r)$. Charging successive removals and summing $1/u$ gives the displayed harmonic bound. Explicit construction takes $2^{O(r)}$ time. Since the exponents are fixed rationals, upward rounding uses integer root comparisons on $O(r)$-bit integers and is exact.

Partition the universe into $\lceil n/\lceil\sqrt{n}\rceil\rceil = O(\sqrt{n})$ blocks, all but the last of size $\lceil\sqrt{n}\rceil$, and concatenate choices from the corresponding local families. Compatibility is preserved and every candidate is covered. Each of the four summands in w is separately multiplicative under concatenation. Consequently, writing W_j for the total local weight,

$$\sum_{\sigma\in\prod_j \mathcal{F}_{r_j}} w(\sigma) \le 4\prod_j W_j,$$

and the complete expected time is

$$B^n\, 2^{O(\sqrt{n}\log(n+1))}\,\mathrm{poly}(n,\beta).$$

This includes empty and arbitrarily unbalanced restriction parts. Exact certificates give $B_i < 2.997190$ for each i, so the strict gap absorbs the subexponential factor. Every matrix call is Las Vegas, and linearity of expectation gives the claimed total bound. □

Lemma 71 (Rectangular isolated recovery with square five-term tiles)**.** *Let the two outer parts have R and C vertices, the middle part H vertices, and let the finite weights have at most β bits; put $\Lambda = \log_2(R+C+H)+\log_2(\beta+1)$. Fix parameters of Lemma 31 and a power of four D with $H \ge \sqrt{D}$ and $\min\{R,C\}D^{1/4} \ge D^{T+\eta}$ for a fixed $\eta > 0$. Then all-edges Exact Triangle has an always-correct algorithm with expected time*

$$\mathrm{poly}(\Lambda,\beta)\big((R{+}C)H(D^{E_{\mathrm{aug}}-1/4}{+}D^{E_{\mathrm{box}}-1/4}){+}RCH\,D^{-G}{+}RCH\,D^{\max(q_1,q_2)-1/2}{+}(R{+}C)HD^{3/4}{+}RC\big),$$

and the $R\times H$ by $H\times C$ min-plus product, with $+\infty$ entries allowed, has the same bound.

Proof. Put $Q = (\beta+\Lambda+1)(\Lambda+1)^2$, $P = \lceil\sqrt{D}\,Q\rceil$, and $h = \lfloor D/(2P)\rfloor$. If $h = 0$ then $\sqrt{D} < 4Q$ for $D \ge 4$; enumerate all RCH triples directly, which costs $RCH\beta \le \beta(16Q^2)^G \cdot RCH\,D^{-G}$ and is within the bound. Otherwise $h \le \sqrt{D}/2 \le H$, so the middle part splits into $O(HQ/\sqrt{D})$ blocks of at most h vertices. Run the algorithm of Lemma 12 with Lemma 32 as the data structure on the folded instances of Lemma 8: $\lceil L\Lambda\rceil$ independent modular filters per block, at most $\Lambda+1$ dyadic mask levels with $C_0\Lambda$ trials each, a fresh uniform prime from $[P,2P]$ per trial, one count product and one label-sum product per trial, exact integer checks of decoded witnesses, and fallback scans of every unsolved pair/block that passed all filters. A nonzero sum has at most $\beta+2$ bits and hence $O((\beta+\Lambda)/\log P)$ prime divisors in $[P,2P]$, so a zero-free pair/block passes one filter with probability $O(h(\beta+\Lambda)/P) = O((\beta+\Lambda)/Q^2) \le 1/2$, and conditional on any mask retaining exactly one genuine witness a fresh prime isolates it with probability at least $1/2$; the arguments of Lemma 12 then give failure probability at most $2^{-L\Lambda}$ per pair/block and expected fallback work $RCH\beta\,2^{-L\Lambda} = o(1)$ for a sufficiently large fixed L. Correctness holds for every random outcome.

Each folded instance has outer dimensions $R' = R\lceil\sqrt{p}\rceil = \Theta(RD^{1/4}\sqrt{Q})$ and $C' = \Theta(CD^{1/4}\sqrt{Q})$ and inner dimension at most $2Ph \le D$; the hypothesis $\min\{R,C\}D^{1/4} \ge D^{T+\eta}$ gives $\min\{R',C'\} \ge KN_0\,\mathrm{poly}(\log D)$ for all sufficiently large D. There are $O(\Lambda^2HQ/\sqrt{D})$ instances. Lemma 32 charges their encodings $(R'+C')(D^{E_{\mathrm{aug}}}+D^{E_{\mathrm{box}}})$, their boxes $R'C'D^{-G}$, and one query per outer pair at exponent $\max(q_1,q_2)$; writing their explicit matrices costs $(R'+C')D$. Multiplying by the instance count gives the first four displayed terms. Residues $w(a,b) \bmod p$, label decoding, and exact checks cost $RC\,\mathrm{poly}(\Lambda,\beta)$ per instance, which the query term dominates since $D^{\max(q_1,q_2)} \ge 1$. Labels are

block-local, in $\{1,\dots,h\}$, and counts are at most h, so all entries are at most $h^2 \le D$, polynomial in the dimensions as Lemma 32 requires. For min-plus, shift each product's two finite matrices to be nonnegative, use the sentinel $4W+1$ for absent entries, where $W \ge 1$ bounds finite magnitudes, run the bit-scaling recurrence with its priority rule for $O(\beta)$ levels of two all-edges calls each, remove the local shifts, and map sums above $2W$ back to $+\infty$. Fresh randomness in each adaptive call preserves the expected bound. □

Corollary 72 (Subset convolution through square five-term tiles)**.** *Min-sum subset convolution on arbitrary β-bit integer tables has a Las Vegas algorithm with expected running time $O(2.995627^n \operatorname{poly}(n,\beta))$.*

Proof. Use the restriction products of Theorem 70: for $[n] = X \dot\cup Y \dot\cup Z$ with sizes x, y, z, rows $U \subseteq Z$, middle indices $T \subseteq X$, columns $V \subseteq Y$, and $F[U,T] = f(U \cup T)$, $G[T,V] = g((X \setminus T) \cup V)$, a $2^z \times 2^x$ by $2^x \times 2^y$ min-plus product whose entries update $h(X \cup U \cup V)$. Apply Lemma 71 with the parameters of Corollary 35,

$$(c, \theta, r, y, e) = (10.74132662,\ 0.056740653,\ 0.038571087,\ 0.025146880,\ 0.083812911),$$

and the fixed rationals $\delta = 0.006324$, $\lambda = 1$, $\kappa = 0.0425$,

$$b = \min\{y/\lambda,\ z/\lambda,\ x/\kappa\}, \qquad D = 4^{\lfloor eb/2 \rfloor}.$$

Here $\Lambda = O(n + \log(\beta+1))$, so every suppressed factor is $\operatorname{poly}(n,\beta)$. Relative to 2^{x+y+z}, the costs of the lemma save, in units of b, at least

$$eG, \quad e(1/2-q_1), \quad e(1/2-q_2), \quad \lambda - e(E_{\text{aug}} - 1/4), \quad \lambda - e(E_{\text{box}} - 1/4), \quad \lambda - 3e/4, \quad \lambda, \quad \kappa$$

for boxes, residual queries, stored queries, the two encoder families, folded matrix writing, input formation, and residues with output updates. The exact-rational certificate gives

$$eG = 0.0063251295\ldots, \qquad e(1/2-q_1) = 0.0063251295\ldots,$$
$$e(1/2-q_2) = 0.0063251295\ldots, \qquad 1 - e(E_{\text{aug}} - 1/4) = 0.0063251406\ldots,$$

all above δ, with $1 - e(E_{\text{box}} - 1/4) > 0.0097$ and the rest far from binding. The short side holds a square tile since $\min\{y,z\} \ge b$ and $1 - e(T - 1/4) = 0.0034181\ldots > 0$, and the middle part holds a block since $2^x \ge 2^{\kappa b} \ge 2^{eb/2} \ge \sqrt{D}$, as $\kappa - e/2 = 0.0005935\ldots > 0$. Thus every restriction product costs $2^{x+y+z-\delta b} \operatorname{poly}(n,\beta) \le 2^n w(X,Y,Z) \operatorname{poly}(n,\beta)$ with the three-term weight

$$w = 2^{-\delta y/\lambda} + 2^{-\delta z/\lambda} + 2^{-\delta x/\kappa},$$

whose maximum term is $2^{-\delta b}$. The geometric term of Theorem 70 is absent because square tiles charge encodings to $R' + C'$, and the clamped term is absent because Lemma 32 has no clamped regime: the fit is on the two outer sides $\min\{y,z\}$ only.

For a block of r coordinates, summing w over all restrictions gives exactly

$$2B_2^r + B_3^r \le 3B^r, \qquad B_2 = 2 + 2^{-\delta/\lambda}, \quad B_3 = 2 + 2^{-\delta/\kappa}, \quad B = \max\{B_2, B_3\}.$$

Each candidate ternary membership string belongs to exactly 2^r compatible restrictions. Since $\delta/\lambda < 1$ and $\delta/\kappa < 1$, every term is at least 2^{-r}, so rounding each term upward to a multiple of 2^{-r} inflates the total weight by at most two; the exponents are fixed rationals, so the rounding uses exact integer root comparisons on $O(r)$-bit integers. Weighted greedy set cover produces a complete local family of weight at most $6H_{3^r}(B/2)^r$: with u candidates uncovered the restrictions cover $u2^r$ candidate incidences, so some restriction has rounded weight per newly covered candidate at most $6B^r/(u2^r)$, and summing

$1/u$ gives the harmonic factor. Partition $[n]$ into $O(\sqrt{n})$ blocks of size at most $\lceil\sqrt{n}\rceil$ and concatenate local choices; compatibility is preserved, every candidate is covered, and each of the three terms is multiplicative under concatenation, so $\sum_{\sigma\in\prod_j \mathcal{F}_{r_j}} w(\sigma) \le 3\prod_j W_j$. The complete expected time is $B^n 2^{O(\sqrt{n}\log(n+1))}\,\mathrm{poly}(n,\beta)$, including empty and arbitrarily unbalanced parts. When b is bounded, when the tile condition fails only through its $\mathrm{poly}(n)$ padding because $b = O(\log n)$, or when $h = 0$ in the lemma, direct enumeration loses at most $2^{\delta b} \le (4D)^{\delta/e} = \mathrm{poly}(n,\beta)$. The exact certificate gives $B_2 = 2.9956261\ldots$ and $B_3 = 2.9020004\ldots$, both below 2.995627, and the strict gap absorbs the subexponential cover factor. Every matrix call is Las Vegas, and linearity of expectation gives the bound. □

Writing $g = \min\{G, 1/2 - q_1, 1/2 - q_2\}$, balancing just the box/query, augmented-encoder, and tile constraints gives the candidate ratio $r_* = g/\max\{T - 1/4,\ E_{\mathrm{aug}} - 1/4 + g\}$. The chosen certified ratio $\delta/\lambda = 0.006324$ is strictly below r_*, as the exact-rational certificate checks; the other encoder, formation, and middle-size constraints are checked separately above. The same rounded saving $\delta = 0.006324$ is feasible for the square folded amplified reduction with Lemma 31. Since $\lambda = 1$, the base $2 + 2^{-\delta/\lambda}$ coincides with the transfer $2 + 2^{-\delta}$ of that certified square saving. Thus the displayed rectangular witness adds no improvement beyond this square saving; this comparison asserts no optimality of r_*. Stoian displays $O^*(2.9987^n)$ Las Vegas and $O^*(2.9997^n)$ deterministic. Substituting the deterministic square saving 0.00705208 of Corollary 44 into Stoian's transfer theorem gives deterministic $O^*((2 + 2^{-0.00705208})^n) = O^*(2.995124^n)$ for tables whose finite entries have magnitude $n^{O(1)}$: restrictions whose block dimension is below a fixed power of n are enumerated directly at polynomial loss, and the larger products have entries polynomial in their dimension, as that corollary requires. For arbitrary β-bit tables this would need the composite certification with divisor bound $\Theta(\beta)$ in place of $\Theta(\log n)$, which is not done here; the Las Vegas bound above has no such restriction. The separate script `verify_five_term_subset.py` certifies 34 strict rational inequalities for this corollary and the comparison.

The leading candidate base is $B_1 = 2.997189877936\ldots$; direct substitution of the same square saving gives $2 + 2^{-0.004057} = 2.997191852132\ldots$. The source's Section 5 join-ordering reduction uses polynomially many subset-convolution calls and only polynomial bit growth, so its C_{out} application inherits $O^*(2.997190^n)$, where O^* suppresses factors polynomial in n, β, under its stated cardinality-estimate assumptions, including its undirected-query-graph convention for forbidding cross products.

The separate script `verify_subset_convolution.py` certifies thirteen strict rational inequalities, and `check_subset_convolution.py` checks finite weighted covers and 120 signed or infinite input pairs against direct subset convolution; these checks do not replace the proof.

8 Verification and limits of the claims

The accompanying `verify_exponents.py` uses exact rational interval arithmetic. It bounds logarithms via

$$\ln x = 2\sum_{j=0}^{r-1} \frac{z^{2j+1}}{2j+1} + R_r, \qquad z = \frac{x-1}{x+1}, \qquad 0 \le R_r \le \frac{2z^{2r+1}}{(2r+1)(1-z^2)}$$

after rescaling to $1 \le x \le 2$, and uses $r = 64$. All asserted exponent inequalities compare rational interval endpoints. Decimal values displayed in this note are not used as certificates. The main run certifies 172 strict inequalities, covering historical and current witnesses for the thin products, integer and real 3SUM, the unbatched and batched APSP reductions, deterministic all-edges Exact Triangle by per-pair moments and its clique instances, rectangular convolution, the lopsided 4-Clique split, hinted OMv and uMv, MonoConvolution, and all-fixed-k clique splits. Its output is

in `certificate-results.txt`. The subset-convolution verifier adds 13 strict inequalities, recorded in `subset-certificate-results.txt`. The five-term black-box certificates add 89 inequalities (thin products in the source's regimes, real inputs, the MonoConvolution transfer, and the lopsided clique split), 44 (min-plus convolution), and 34 (subset convolution); their recorded outputs are the three files named `five-term-*-certificate-results.txt` with the corresponding middle words. The explicit directed APSP exponent adds 31 inequalities and the 3XOR construction 46, in `directed-apsp-` and `cyclic-3xor-certificate-results.txt`.

The script `check_fixed_linear_bucket_counting.py` checks the 3XOR construction on finite instances: the group-algebra products against naive XOR convolution, the small-bias families and their prefix projections, the bucket second moment and incidence bounds, the folded products against brute force, the level certification with a fixture that forces restarts, and end-to-end exact counts for sets, multisets, and the one-list correction, in twelve tests. The separate `verify_deterministic_apsp.py` adds 45 strict inequalities for the two deterministic cyclic-encoding witnesses, recorded in `deterministic-apsp-certificate-results.txt`. The two external rectangular-multiplication theorems are explicit dependencies, not proved by these numerical certificates.

`check_pruned_encoding.py` performs 3,232 exact integer checks: full versus restricted encodings, retained-leaf counts, the prefix work bound, and decoded matrix entries for small recursion depths. These support the implementation of the encoding lemma; they do not replace the asymptotic proof, implement the complete large-instance algorithms, or benchmark practical performance.

`check_reductions.py` adds twelve small-instance correctness tests, including 240 signed modular-filter instances, 202 folded residue-label instances whose product entries are compared with direct modular counts, 152 bucket-pair 3SUM instances checked against direct modular pair sums, 60 deterministic modulus selections checked against brute-force false-positive counts, and the block reduction of min-plus convolution on all outputs for lengths up to 100, besides 400 signed min-plus instances. Its oracle is deliberately brute force: it checks the reductions, including different false witnesses for different primes, rather than claiming to implement the asymptotic data structure. All twelve tests pass.

The additional finite checkers exercise conditional modular moments (`check_modular_moments.py`, eight tests, including per-pair certificates on 300 planted all-edges instances) and certified prefix counts (`check_prefix_counts.py`, five tests). Rectangular convolution and singleton recovery are covered by `check_rectangular_reductions.py` in seven tests, including 4,160 signed convolution instances, and one-sided 3SUM counting by `check_3sum_counts.py` in six tests. The subset checker additionally verifies weighted covers, their Cartesian composition, and 120 full convolution tables.

`check_direct_sum_tiling.py` places independent signed matrices in fixed subset slots, reuses each band encoding across tiles, and decodes 5,079 entries against direct products, including partial packs and tiles. The script `check_aggregated_queries.py` checks the prefix-contracted query identity, the shrinking transforms, and the signed binomial aggregation against brute-force leaf sums in five tests. `check_lazy_encoding.py` adds five tests covering 9,217 cyclic windows, 600 sparse contractions, 1,936 retained-array lookups, 146 complete free-pruned queries, and 146 complete lazy-prefix queries. It counts sparse input reads and checks full intermediate-array storage, as well as values against direct products. The unified command `python3 scripts/verify.py` reruns every checker and compares all fifteen certificate outputs with the recorded files. These are finite oracle checks and numerical certificates, not a proof assistant or a full fast implementation.

The additional `verify_five_term_apsp.py` certifies 29 strict inequalities for Corollary 35, with saved output in `five-term-apsp-certificate-results.txt`. The five-term checks validate retained internal arrays, balanced residual queries, complete box closure, and an independent original-array oracle. The additional `check_five_term_profile.py` validates 216 complete fixed-profile queries, both actual retained encoders, truncated-transform predecessors, and the fixed-order box closure against an

original-array oracle. `check_five_term_profile_maps.py` checks exact conditional expectations and bijections for five profile-map fixtures. Separate composite checks compare residue-folded counts and incidence-halving decisions against brute force, including true incidences that require a restart. These checks do not independently prove the external squared embedding.

The additional `verify_five_term_ring_apsp.py` certifies 48 strict inequalities for Corollary 39, including both endpoints of each modulus phase and the exact integer prime-factor split. Its saved output is `five-term-ring-apsp-certificate-results.txt`. The finite `check_five_term_ring.py` compares folded cyclic counts with original signed sums, checks complete coefficient queries against original-array products, and exercises greedy prime selection, incidence certification, restarts, and fallback scans. Stored-box queries read only the target coefficient; residual queries retain their full cyclic factor. Schoolbook ring multiplication in these tests is an oracle, not an implementation of the asymptotic polynomial-arithmetic bound.

The partial-cache `verify_five_term_cache_apsp.py` certifies 58 strict inequalities for Corollary 41; its saved output is `five-term-cache-apsp-certificate-results.txt`. It charges the augmented encoding exponent E, retains T in geometry, checks both residual branches, and covers both endpoints of each modulus phase. The finite `check_five_term_cache.py` constructs the augmented encoders from signed source arrays, checks exact leaf and intermediate counts, verifies every truncated-transform predecessor and the disjoint lexicographic groups, and compares complete queries with original matrix products. Opposite low/high classifications on the two input sides are included. Its fixtures cover 544 complete integer outputs and 96 requested coefficients from 24 complete cyclic-ring outputs, using independent original-matrix products as oracles. These tests supplement the general cache proof; they do not independently prove the source embedding or an asymptotic exponent.

The aligned-cache `verify_five_term_aligned_apsp.py` adds 47 strict inequalities for Corollary 44, with recorded output in `five-term-aligned-apsp-certificate-results.txt`. It charges $E = T + q_1$, retains the original T geometry, and checks both endpoints of both prime phases. The accompanying `check_five_term_aligned_embedding.py` checks 352 original signed matrix outputs, including nontrivial transpose locators. `check_five_term_aligned_cache.py` checks 34 complete arbitrary source-array queries and 20 compact encoders, with 423,918 retained leaves and 1,088,138 total retained/intermediate entries. It verifies every residual coefficient and the constant-coefficient box bases. `check_five_term_aligned_integration.py` composes those steps and checks 530 normalized original-product coefficients, including every target in a signed cyclic-ring fixture. Its 68 encoders account for 2,011,348 retained and intermediate entries. The finite checks supplement the Walsh embedding and polynomial-cache proofs; they do not replace those proofs or independently establish the asymptotic bound.

The light-source `verify_light_source_apsp.py` checks 61 strict rational inequalities for Corollary 50 and the exact rational small-bias potential inequality. Its recorded output is `light-source-apsp-certificate-results.txt`. It retains all three encoder charges, both phase endpoints, original geometry, the independent batch fit, source-type derivative guard, suffix fit, and the positive rectangular-algorithm allowance. In the full research checkout, `scripts/verify.py` also reproduces the seven original research suites under `work/`: shifted sources, structured completion, adaptive lengths, cropped sources, jointly capped boxes, light-source translations, and the next-pass appendix. These include complete integer and cyclic products, explicit duplicated selector vectors, row/column chunk padding, exact conditional-expectation construction, and a strict backend that reconstructs omitted leaves. The results-only checkout retains the supporting checkers and their import dependencies, run by `scripts/verify_structured_sources.py`. Recorded outputs are compared byte for byte. Scoped ceiling checks in those suites describe only their stated cost families. These finite tests and rational certificates supplement the proofs in Section 6; they do not establish the external matrix-multiplication theorem or formally verify the asymptotic argument.

The exact-half refinement has its own 61-inequality certificate, `verify_half_source_apsp.py`, with recorded output in `half-source-apsp-certificate-results.txt`. In the full research checkout, `work/half-source-improvement/verify.py` checks 22,656 exact conditional choices for $m = 1, \ldots, 4$ and 2,922 affine half-weight coverage cases. Its embedding checker uses the actually constructed ordered lists and first valid indices, and verifies 416 complete normalized signed cyclic coefficients. It omits only translation indices whose masks are identically zero on the fixture. The results-only checkout runs these two finite checkers through `scripts/verify_structured_sources.py`. The earlier strict grouped-residual and joint-box checks already use the finite cap $\lfloor m/2 \rfloor$ and remain in the standard suite. These checks supplement the all-m potential argument in Lemma 46.

The signed-window reduction has two additional finite checkers. `check_exact_counting.py` runs ten tests using actual residue-folded cyclic products as the modular oracle, checking common-prime selection, restarts, four parity masks, signed original matrices, arbitrary targets, duplicate middle labels, and uneven or padded blocks. Its companion `check_signed_window_counts.py` independently checks 600 signed matrix instances through 4,997 prefix updates and 6,065,880 certified masked counts, plus exact contraction and restart fixtures. These check Theorem 61's reduction; they do not implement the fast tensor oracle. Its exponent uses the unchanged 47-inequality aligned certificate.

The separate `verify_cyclic_3sum.py` supplies 48 strict rational checks for Theorem 64, including both endpoints of both modulus phases, all candidate costs, the original full-grid geometry, and the inherited MonoConvolution exponent. Its saved output is `cyclic-3sum-certificate-results.txt`. The numerical certificate relies on the separately proved fixed-bucket count-lifting and cyclic histogram lemmas. The additional `check_fixed_bucket_counting.py` runs six finite tests with actual folded cyclic histograms, including 50 signed set instances, a constructed purification restart, prefix coalescence, and an objective-multiplicity regression for shared query answers. Its occurrence-list fixtures check a separate extension retained in the research notes; the stated theorem is for sets. All these finite checks supplement the asymptotic proof.

A floating-point exploration of the generalized Schönhage identities $\langle k, 1, n\rangle \oplus \langle 1, (k-1)(n-1), 1\rangle$ with the same box and encoding analysis, recorded in `work/tensor/shapes.py` in the full research repository, indicates that the source's 3×3 shape maximizes the integer savings in this framework, with or without folding. That is a numerical observation about this parameter family, not a theorem or an optimality claim.

The paper explicitly says some better exponents were omitted, and already notes the tight all-edges APSP transfer. Accordingly these results improve its displayed guarantees; no claim is made that the ideas are unknown to the authors or new to the literature. The derivations have been checked algebraically and computationally during this work, but have not been independently peer reviewed or formalized in Lean.

References

[1] Josh Alman and Virginia Vassilevska Williams. *Truly Subquadratic 3SUM and Truly Subcubic APSP via Triangles in Sparse Lopsided Graphs*. arXiv:2610.06783v1, October 5, 2026. https://arxiv.org/abs/2610.06783. Source licensed CC BY 4.0. References to numbered lemmas, theorems, and footnote 10 refer to this version.

[2] Tsvi Kopelowitz, Seth Pettie, and Ely Porat. *Higher Lower Bounds from the 3SUM Conjecture*. SODA 2016; arXiv:1407.6756. Its Theorem 1.4 and Section 2 give the reduction from 3SUM to offline set disjointness whose digit split is used above.

[3] Virginia Vassilevska Williams, Yinzhan Xu, Zixuan Xu, and Renfei Zhou. *New Bounds for Matrix Multiplication: from Alpha to Omega*. SODA 2024; arXiv:2307.07970v2. Table 1 supplies the earlier rectangular endpoints; Proposition 5.1 and Theorem 3.2 supply the level-one specialization in Lemma 49. https://arxiv.org/abs/2307.07970v2.

[4] Josh Alman, Ran Duan, Virginia Vassilevska Williams, Yinzhan Xu, Zixuan Xu, and Renfei Zhou. *More Asymmetry Yields Faster Matrix Multiplication*. arXiv:2404.16349v2. Its Table 1 supplies $\omega(1, 0.5275, 1) \leq 2.054999$, $\omega(1, 0.6, 1) \leq 2.092351$, $\omega \leq 2.371339$, and $\mu < 0.5275$; only the directed APSP refinement uses rows other than the source's displayed $\mu < 0.5275$ and $\omega < 2.372$.

[5] Przemysław Uznański. Rectangular matrix multiplication from polynomial products. Unrefereed preprint, arXiv:2610.10494v2. Its Corollary 1.2 and Lemma 6.1 are additional dependencies only for Corollary 27. https://arxiv.org/abs/2610.10494v2.

[6] Ruhollah Majdoddin. *Thin matrix products up to $D \leq N^{0.172}$ and lossless reductions for 3SUM, Exact Triangle and APSP, with 3SUM over abelian groups and a dichotomy for coordinatewise problems*. arXiv:2610.17420v1, October 12, 2026. Sections 3–7 supply the five-term identity and squared embedding; Section 15.4 supplies the composite incidence certification. https://arxiv.org/abs/2610.17420v1.

[7] Mihail Stoian. *Truly Sub-3^n Min-Sum Subset Convolution and Join Ordering*. arXiv:2610.10282v1. Sections 2–4 supply the restriction-product and local-cover framework; the join-ordering transfer uses its Section 5 assumptions. https://arxiv.org/abs/2610.10282v1.